OECD Economic Surveys

France

ECONOMICS

OECD

November 2001

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OECD ECONOMIC SURVEYS

2000-2001

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Table of contents

Assessment and recommendations 9

I. A phase of cyclical slowdown 23

Growth performance was favourable from 1998 to 2000 23
The output gap has closed and strains have appeared 33
In 2001 growth slowed distinctly and core inflation picked up 34
Activity should pick up in the second half of 2002, but strong risks remain 41

II. Fiscal policy issues 47

Fiscal policy in the short term 47
The draft budget for 2002 49
Structural fiscal issues 55

III. Structural reforms to raise potential growth 73

Towards full employment? 73
How can network industries be opened to competition? 92

IV. Towards ecologically sustainable growth 113

Introduction 113
The institutional setting 114
Ecotaxes and the greenhouse effect 116
Urban air quality 128
Water quality 135
Conclusions and recommendations 145

Notes 153

Bibliography 169

Annexes

I. Are the national accounts forgetting the new economy? 177
II. A breakdown of growth factors 179
III. Follow-up to the recommendations on policies concerning the health sector 181
IV. The reduction in working time: a post-mortem on the 1999 Interlink simulations 186
V. Calendar of main economic events 191

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Boxes

1. Why do French businesses invest so much abroad? 29
2. Can the French economy grow sustainably at 3 per cent a year? 37
3. Is the changeover to the euro neutral for the economy? 44
4. Tax reform 57
5. Privatisation: the current situation 107
6. The retail distribution sector also has competition problems 110
7. The advantages of economic instruments 114
8. Waste taxes: not much incentive 121
9. Voluntary agreements: an assessment 127
10. The Dutch system of taxes on excess nitrogen 143
11. Principal recommendations 146

Tables

1. Summary of short-term outturns and projections 43
2. General government expenditure 48
3. Main components of public finance, 2001-2002 50
4. Trend of central government expenditure in 10 categories 54
5. Overview of the progress in structural reform since 2000 74
6. Main characteristics of basic income support 78
7. Main public or partially public enterprises and main public corporations 109

Annex

A1. Average annual growth of activity and the factors of production 180

Figures

1. Labour utilisation and output per head 24
2. Interest rates and monetary conditions 26
3. Credit 27
4. Foreign trade 28
5. Employment and unemployment 32
6. Strains 33
7. The new economy 35
8. Cyclical indicators 36
9. Supply potential 38
10. Inflation 40
11. Wages, labour costs and competitiveness 42
12. Trend of public debt by government sector 48
13. Fiscal policy 52
14. The Pension Reserve Fund 62
15. Total health expenditure 67
16. Working time and participation rates 81
17. Unskilled jobs 87
18. Wage ratios 89
19. Rates for long-distance calls 94
20. Rates for local calls and fixed charges 95
21. Air fares from and to Paris 97

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22. Fares of low-cost airlines 98
23. Electricity tariffs for industrial users 99
24. Relative electricity prices by type of user 99
25. Taxes on the carbon content of fuels 117
26. Cost of air pollution by private cars 118
27. Diesel and gasoline taxes 119
28. Breakdown of the stock of motor cars by type of fuel used 120
29. Carbon dioxide emission intensities 122
30. Road-freight traffic and emissions 130
31. Marginal external cost of various transport modes in the Ile-de-France region 134
32. Water prices in selected OECD countries 141

Annex

A1. Simulation of the effects of the reduction in the statutory working week 189

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BASIC STATISTICS OF FRANCE

THE LAND

Area (1 000 km^2), 1998	632.8
Agricultural area, excl. overseas departments (1 000 km^2), 1998	300.0
Major cities (thousand inhabitants), 1999:	
Paris	2 125
Marseille	798
Lyon	445

THE PEOPLE (2000)

Population (thousands)	60 431
Number of inhabitants per km^2	95
Average annual increase (thousands), 1990-2000	240
Total labour force, excl. overseas departments (thousands)	26 466
Percentage of employment in:	
Agriculture	4.0
Industry and construction	24.4
Services	71.6

PRODUCTION (2000)

Gross domestic product at market prices (FF billion)	9 227.2
Gross domestic product per capita ($)	21 452
Gross fixed investment as a per cent of GDP (current prices)	19.6
Gross value-added by activity, at basic prices (per cent):	
Agriculture	2.9
Industry	21.5
Construction	4.7
Services (excl. FISIM)	70.9

GENERAL GOVERNMENT (2000)

ESA95 concept, as per cent of GDP:	
Current expenditure	47.4
Current revenue	49.6
Gross fixed investment	3.0

FOREIGN TRADE (2000)

Exports of goods and services (% of GDP)	28.7
Main exports as a percentage of total exports (SITC):	
Food, beverages and tobacco (0 + 1)	12.6
Chemical products (5)	12.0
Manufactured products (6 + 8 + 9)	25.5
Machinery and transport equipment (7)	45.0
Imports of goods and services (% of GDP)	27.2
Main imports as a percentage of total imports (SITC):	
Food, beverages and tobacco (0 + 1)	7.6
Chemical products (5)	11.2
Manufactured products (6 + 8 + 9)	29.2
Machinery and transport equipment (7)	39.2

THE CURRENCY

Monetary unit: franc	
Currency unit per euro, since 31 December 1998	6.55957
Currency unit per $, average of daily figures:	
Year 2000	7.12
October 2001	7.24

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of Member countries.

•

The economic situation and policies of France were reviewed by the Committee on 8 October 2001. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 13 November 2001.

•

The Secretariat's draft report was prepared for the Committee by Patrick Lenain, Rauf Gönenç and Ann Vourc'h under the supervision of Yutaka Imai.

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The previous Survey of France was issued in July 2000.

Assessment and recommendations

In recent years, France has experienced strong growth of activity and employment, although there are doubts about its sustainability

From 1998 to 2000 France recorded strong GDP and employment growth, and low inflation. The output gap that had accumulated during the 1990s was closed and the unemployment rate declined, reaching its lowest level in more than fifteen years at the beginning of 2001. The payoffs from growth were shared by most economic agents: households saw an increase in their purchasing power, the general government collected substantial tax revenues and corporate profits improved. Inflation was one of the lowest in the euro area, thanks to moderate increases in labour costs and increased competition. Growth was largely attributable to the strong expansion of world trade and was facilitated by particularly accommodating monetary conditions in Europe. But it also reflected the efforts made since the early 1990s to consolidate the budget, reduce inflation and implement structural reforms. In particular, labour market reforms accompanied by moderate wage growth contributed to the strong expansion of employment in recent years. However, the severe pressure that emerged in 2000 on the utilisation of production factors suggest that the output gap accumulated over the past decade has been closed. In order to reach the authorities' goal of full employment and for living standards to rise, the rate of growth of potential output will need to be raised above its current level. This requires continuing to implement structural policies to increase factor productivity gains, the employment rate and the investment ratio. Indeed, despite recent progress the latter two indicators still lag behind those of other OECD countries.

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A perceptible slowdown has occurred since the beginning of 2001

The economy entered a phase of marked cyclical slowdown at the beginning of 2001. GDP growth eased, business expectations deteriorated and unemployment has been rising since June. This cyclical shift can be largely explained by the sharp slowing of the American economy and the dip in the Japanese economy, which impacted on Europe via world trade. Other international transmission mechanisms, like parallel swings in stock markets or the reactions of multinational firms, have probably amplified the slowdown. The turnaround in the new technologies sector has also affected France, even though this sector's share in output is limited. In this depressed environment, French exports have fallen back, businesses have reduced their investment and inventories have been run down. In addition, rising inflation at the start of 2001 curbed household purchasing power in France. The risk of second-round effects in the euro-zone, among other factors, prevented the European Central Bank from significantly lowering its interest rates.

Growth should pick up next year, but the risk of a subdued economic environment remains,...

Since the terrorists attacks on 11 September in the United States, the international environment has become much less favourable. The OECD's new projections envisage a sharp slowdown in activity in the United States and to a lesser degree in Europe. As regards France, although private consumption growth should slow, it will continue to bolster activity via real wage growth, direct tax cuts and, to a lesser extent, net job creation. However, the world slowdown and uncertainty will continue to bear down on exports and investment in coming months and the first half of 2002. As a result, GDP is set to grow by only 2 per cent this year and by about 1.6 per cent next year. Accelerating from the second half of 2002, growth should approach 3 per cent in 2003. These projections are subject to exceptional uncertainties particularly as regards business and consumer confidence, which could worsen more than projected (both domestically and externally), or, conversely, improve more quickly than anticipated. Bearing in mind the decline in exports, investment and private consumption before the events in September, even an optimistic scenario for confidence would see the slowdown of activity continuing into the first months of 2002.

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... which would justify allowing automatic stabilisers to operate fully on the revenue side

The impact of the cyclical slowdown on public finances is already perceptible. Tax receipts are lower than forecast and unlikely to show much increase this year or next. The budget deficit will consequently be higher than was projected in the 2002-2004 multiyear programme of public finance. This deterioration, due to the play of built-in revenue stabilisers, is temporary and does not necessitate corrective measures even if the slowdown and widening of the cyclical deficit worsens. Nevertheless, while consolidation efforts did make possible substantial tax cuts, they were not large enough to bring down the structural deficit during these years of fast growth. Fiscal policy should, therefore, continue to target bringing the structural deficit into balance by 2004. Indeed, the public debt is large, and borrowing requirements related to population ageing will be substantial in the long term. Over and above cyclical fluctuations, priority should thus continue to be given to reducing the structural deficit of the general government. The government's decision, within the context of the Act on Public Finances, to keep the structural deficit unchanged in 2001 and 2002 will make achieving structural balance more difficult. This being so, a plan to revive domestic demand by amplifying tax cuts or by increasing public expenditure, including expenditures that may be associated with the introduction of the 35-hour work week among small and medium enterprises, would be dangerous and should be avoided. Furthermore, in order to permit a return to structural balance, the next stability programme should incorporate a tightening of public expenditure so as to achieve a substantial adjustment in 2003 and 2004.

Fiscal consolidation requires tighter control of public spending, especially on healthcare and staffing

Control of public expenditure will thus be the cornerstone of macroeconomic policy for many years still. The authorities have already put mechanisms in place to control the growth of general government expenditure. Spending increase norms are set in the multiyear programmes of public finance, then written into the Budget Laws and the Social Security Financing Laws. They have helped to curb central government expenditure for some years now but have proved much less effective with health insurance and local government spending. The present norms are unlikely to be stringent enough to consolidate public finances in the

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future, especially if economic growth falls short of the authorities' hopes. More ambitious and more tightly enforced norms should, therefore, be envisaged. Healthcare spending could be curbed by means of appropriate micro-economic reforms, as suggested in the previous S*urvey*, but little has been done along those lines recently. The growth of staffing expenditure could be slowed by not systematically replacing the large numbers of public servants who will be retiring in the near future. Finally, expenditure related to employment policy has not been reduced, despite lower unemployment. In any event, the present efforts to make public finance more transparent, especially the adoption of the organic law on public finance, are useful in that they focus on the effectiveness of public policies, rather than on the amount of financial resources allocated to them. In this regard, the some 600 *fonds de concours*, which are administered by ministries outside of ordinary budgetary procedures, should be abolished.

Moreover, the tax reform recently begun should continue

Better control of public expenditure will also be essential in order to permit a sustainable easing of the tax burden in France, the level of which is one of the highest in the OECD area. For highly qualified individuals – and therefore the highest paid – the average income tax rate can be up to 58 per cent of income. In an environment of increased tax competition, and given the major tax reforms undertaken elsewhere in the European Union, failure to reduce taxes could result in losses of human and financial capital. The French authorities have thus initiated a programme to ease tax pressure. A start has been made on reducing the marginal income tax rate and this reduction should, if possible, be amplified and extended in coverage. Across-the-board cuts that benefit all taxpayers are preferable to cuts targeted at certain professions (football players, performing artists, brokers). Measures to simplify the tax system and make it more efficient are also necessary. Here some taxes that were costly to collect and had a low yield, like the road tax, have been abolished. Other taxes might also be done away with (television licence fee or quasi-fiscal taxes).

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Other structural reforms are more modest, notably in regard to pensions

In other areas, structural reform has advanced more modestly during the past two years. The brighter economic outlook may have made some analysts believe, wrongly, that growth would itself resolve the structural problems and that reforms would not be essential. Such an illusion may have extended to the implications of the ageing of the French population, a process that will accelerate as from 2005. The authorities do not share this view. Before embarking on major reforms, they have decided to set up a *Conseil d'orientation des retraites* (COR, pensions advisory board) to reflect on the issue and to facilitate the emergence of a consensus between social partners on future reforms. In the immediate term, the authorities have decided to endow the *Fonds de Réserve des Retraites* (pensions reserve fund) with FF 1 000 billion between now and 2020 in order to smooth the financial burden created by population ageing. Public finances will have to show surpluses in the future to supply this fund. Receipts from privatisation should also be paid into it rather than being used to subsidise loss-making public enterprises, as at present. The authorities have also expanded existing employee saving schemes. But these measures cannot be a substitute for a reform of the different pension schemes. On equity and sustainability grounds, the main parameters for the public sector pension (minimum period of contribution, reference salary, mode of indexation) should be brought more into line with those of the private sector. According to the COR, alignment of the respective contribution periods would reduce the borrowing requirements for the pensions of central-government employees by one-third in 2040. In addition, free choice of the date of retirement could be introduced and pension amounts determined on an actuarially neutral basis. This would reduce incentives to early retirement.

A return to full employment in the medium term is possible

The government is aiming at a return to full employment by 2010, which is often defined in the public debate as a 5 per cent unemployment rate with higher labour force participation. Achievement of this full employment objective would make it possible not only to combat poverty and exclusion but also to increase the economy 's supply potential. However, it would necessitate a steep fall in the structural unemployment rate, which most studies estimate to be around 9 per cent. It is, therefore, a goal requiring ambitious measures on the side of both labour supply and demand.

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The authorities have already reduced wage costs for low-skilled labour…

Since the start of the 1990s, the authorities have made a big effort to reduce the *cost of unskilled labour*. This policy, which consisted mainly of substantial reductions in employer social security contributions, helped moderate the growth of labour costs and was bolstered by slower growth of individual net wages. The beneficial impact of this reduction on employment is now well established. It helped to stimulate corporate demand for labour and to stabilise the share of low skill-workers in total employment for the first time since the 1980s.

… and introduced measures to stimulate labour supply

The authorities have also recently sought to implement supply-side measures to reduce the "inactivity traps" produced by the interaction of various social minima and the tax system. Recent analysis shows that net income from work is not necessarily higher than basic income support, which may discourage a return to employment. Indeed, hiring difficulties for low-skilled jobs have become more marked in recent years, attaining historically high levels. Increasing the incentives to work for the least well-paid is not an easy task, especially when the authorities wish, among other things, to preserve social minima. Initially, the authorities undertook to rationalise the tax-benefit system, notably by allowing individuals to combine temporarily basic-income support with income from work, and by reforming housing benefits and the *taxe d'habitation*. However, within the context of a complicated system, selective measures of this kind do not provide adequate incentives. A more significant step forward was taken this year with the introduction of the "employment bonus", a tax credit that will benefit some 8 million low-income households. However, in its present form, the bonus primarily benefits those working full time and is less advantageous to part-time workers. The employment bonus should, therefore, be amended so as to provide more incentive for part-time working, which is often an effective bridge towards full-time employment.

Further employment policy measures are now necessary

Despite these important reforms, the hard core of unemployment still consists essentially of persons without qualifications or without experience. For these individuals, the challenge will be to stimulate labour demand by further

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reducing labour costs without affecting work incentives at the same time. To do so, the authorities should exploit the interaction of the three instruments at their disposal: the employment bonus to encourage the labour supply; cuts in employer social security contributions to reduce the cost of low-skilled labour, though the scope for further cuts is now limited, and the minimum wage, increases in which should remain moderate in order to safeguard recent progress. In this connection, a wider role for the employment bonus could be envisaged. It has the advantage of strengthening the incentive to work and permitting moderate wage growth, while protecting the purchasing power of workers who receive it. Corrective measures also need to be taken rapidly to prevent a sharp acceleration of the hourly minimum wage and the monthly guarantee as a result of the changeover to the 35-hour week. Otherwise, low skilled employment, especially in small firms, is likely to be penalised.

The gradual nature of the changeover to the 35-hour workweek needs to be reinforced,…

The *reduction of working time* has been considered by the authorities to be the cornerstone of their plan for reabsorbing unemployment. About half of all employees in the market sector are now working 35 or fewer hours per week. The reduction in working time was accompanied by a range of state-financed aids designed to reduce its impact on labour costs. These aids, combined with the scope for work reorganisation offered by the changeover, explain why some businesses were keen to make the change. Yet there are a number of challenges to be faced as the process continues. First, even when possible feedback effects are taken into account, the changeover is proving to be costly for the government and the sources of future financing are not fully identified. Moreover, the forthcoming switchover of the civil service to the 35-hour work week is likely to result in numerous new hires that will strain public finances still further. In addition, small businesses will experience difficulties when they move over to the shorter workweek on 1 January 2002. They will have to meet high fixed costs and, unlike larger businesses, will not necessarily be able to reap the benefits of work reorganisation. In anticipation of this, the authorities have decided to ease the arrangements for the 4½ million employees concerned.

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... the match between labour supply and demand needs to be improved...

It is also recognised today that the *functioning of the French labour market* lacks fluidity, as evidenced by the coexistence of mass underemployment and severe recruitment difficulties for employers. The mismatch between labour supply and demand is making the economy more vulnerable to recurrent shocks and slowing the return to equilibrium following those shocks. Even though they represent only a small proportion of total employment, fixed-term work contracts have increased and become more common in certain limited segments of the labour market. Also, in order to bring more young people into the labour market, the authorities created the "*emplois jeunes*" (youth employment) scheme, which provides five-year contracts in the public-sector. About 15 per cent of the young people on the scheme have managed to find dependent employment before the end of their contract. Nevertheless, it is not clear that the scheme helps young persons to find steady employment in the business sector when their initial contracts come to an end. As a result, the scheme risks increasing the public-sector workforce. Renewal of the initial contracts for another period of five years should be accompanied by targeted and effective training as a bridge to jobs in the private sector. In this regard, the social partners and the authorities have decided to set up schemes to assist persons wanting to rejoin the workforce by providing them with custom-made re-entry strategies that include further training. International experience suggests that an initiative of this kind can be effective. However, the concomitant abolition of the degressive scale for unemployment insurance benefits is likely to reduce further labour market flexibility. To prevent this from happening, it is important that the authorities apply more strictly the regulations resulting in the removal from the unemployment register of jobseekers who refuse jobs matching their qualifications, without a legitimate reason.

... and labour regulations need to be simplified

Another generally accepted fact is that *French labour legislation* is extremely complex. This applies in particular to redundancy legislation and the formalities an ailing business must comply with in order to reduce its workforce. The government has proposed in the draft Social Modernisation Act to make redundancy regulations even more restrictive. If

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the bill goes through, it will increase the formalities, lengthen the periods of redundancy notice and increase dismissal costs. Employee consultation procedures will be more complicated, the departmental préfet is given a formal role and the employees concerned will be able to appeal against dismissal to the courts. International experience indicates that legislative initiatives of this type are liable to reduce hiring flows to the detriment of the jobless. Indeed, there is nothing to prove that employment protection policies are an effective means of combating unemployment. If businesses are unable to lay off employees, they tend to hire less and make more use of temporary workers or workers on fixed-term contacts.

A number of network industries have been successfully opened up to competition

One area in which structural reform has progressed has been the opening of several network industries to competition, pursuant to EU directives. This major development has had positive effects on the economy as a whole: lower prices and better-quality services for consumers and higher incomes for the industries' employees. In the *telecommunications* sector, competition has developed strongly, though unevenly across the market. It established itself early-on for long distance and international telephone calls, but much later and only partially for local calls. The introduction of competition rapidly brought new investors into the sector and has generated entrepreneurial excitement, which has stimulated activity and employment. Similarly, in *air transport* the big European and non-European airlines have increased their flights to and from France, obliging Air France to squeeze its prices. The fragility of "low cost" airlines is nevertheless still a weakness of the French market. The opening of the *electricity* sector to competition is more recent. The creation of the increasingly independent Electricity Transmission Network (RTE) business unit within EDF was an important step towards separating the generation and transmission of electrical power in France. This process should be completed by fully separating the two entities, which would facilitate the further opening of the sector to competition. Although this liberalisation is only partial, it, nevertheless, represents a step forward in this traditionally monopolistic sector.

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Extension to other sectors is being held back by several concerns to which responses exist: competition need not compromise universal service,...

On the other hand, little progress has been made in opening up the *rail transport*, *natural gas* and *postal services* sectors. True, some moves have been made in this direction, but here the traditional monopolies remain essentially unchallenged. Certain fears are holding back reform in these sectors. The *first* is that competition will undermine the "public service" tradition, which in France is understood to mean that network industries should provide their services over the whole territory at identical prices that are affordable for the whole population. There is strong popular support for this tradition and further reform will not be possible unless the fear is quelled. But what has been achieved so far shows that maintenance of "public service" is not incompatible with competition. The reform of the telecommunications sector and the ongoing reform of the electricity sector have maintained universal access to network industry services, including for the poorest and most isolated segments of the population. The social mission of "public service" can therefore be preserved – even if certain practices lacking in transparency like cross-subsidisation have to be abandoned.

... the employees of operators need not lose their social benefits,...

The *second* concern is for the future of labour relations within the incumbent operators, whether in respect of work practices, fringe benefits or retirement arrangements. The experience of the telecommunications and air transport sectors offers useful lessons on how to make employees active participants in the process of change, even if this experience cannot be transposed in identical form. These experiences have shown that labour unrest can be resolved with less difficulty if employees are associated with the commercial prospects opened up by the firms' changed status. The transformation of the capital structure could provide an opportunity for their employees to acquire equity stakes and a share of future profits. Moreover, at a time when the entry of certain state-controlled network companies into foreign markets has prompted reservations about the conditions in which that entry might develop, this trend will help to create conditions of competition that are more transparent and, in some respects, probably more favourable to French firms themselves.

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... and security of supply and stability of services need not be compromised

The *third* fear is that enterprises exposed to competition will be disinclined to undertake the very long-term investments needed to ensure security of supply, whereas public corporations are less constrained by profitability requirements and have a different approach to long-term risk. Appropriate regulation could put this fear to rest. The regulatory authorities can, at the time of licensing and with the necessary incentives, get private operators to maintain a level of investment sufficient to ensure security of supply and stability of services over the long run. Although the misgivings of public opinion are real, they should not be allowed to justify a halt in reforms. Solutions exist that can dispel the fears expressed both by service users and by service employees.

Environmental policies still lack overall coherence and...

Like most other OECD countries, France has been endeavouring for some years to make its growth environmentally sustainable. These efforts, which are reviewed in the special chapter of this *Survey*, have been based on the traditional instruments of environmental protection-regulatory standards in particular. However, this approach is not very economically efficient because it does not allow efforts to be concentrated on pollution sources with the lowest abatement costs. Environmental taxes and charges have been used but more with a view to financing expenditure on pollution control than providing adequate micro-economic incentives. Maintenance of this financing has also reduced the economic and environmental efficiency of the aid distributed. Aware of these inadequacies, the authorities have adopted a new approach that involves more extensive and more efficient use of economic analysis and instruments. However, recent action, notably that concerning petrol taxation, has been a step backward if anything and the withdrawal of a proposed tax on intermediate energy consumption attests to the difficulty of framing a coherent environmental strategy. This is especially the case with greenhouse gas emissions, water pollution and emissions in urban areas.

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... economic instruments are required to reduce greenhouse gas emissions efficiently

France has pledged to reduce its greenhouse gas emissions under the Kyoto Protocol. For that purpose, the government planned to introduce a tax on the intermediate consumption of energy by enterprises. However, the Constitutional Council overruled this project in December 2000, essentially on the grounds that its provisions for tax allowances and exemptions treated taxpayers unequally. Yet some inequality is difficult to avoid if the aim is to provide the same marginal incentive to all sectors, while moderating the losses of competitiveness due to the lack of international co-ordination in this area. Even if the tax project was not perfect, alternative solutions could have been found to introduce an effective instrument for curbing greenhouse gas emissions. The aim was not to increase tax pressure but to bring taxation to bear more on scarce resources (the environment) than on abundant resources (*e.g.* labour supply). The government is currently studying alternative solutions to the ecotax overruled by the Constitutional Council, in particular the possibility of negotiating voluntary agreements to reduce greenhouse gas emissions with industrial enterprises. However, international experience has shown that such agreements often result in unambitious objectives and that they are unlikely to result in a significant and low cost reduction in these emissions. The possibility for firms to trade emission reductions would reduce the cost for them but would not modify the inherent problem of voluntary agreements. Only a real market in emission permits with a cap on the overall level of emissions and rules for allocating permits, or an appropriate tax, would provide the necessary incentives. A more coherent approach to fuel taxation is also necessary; even though they are consistent with Community law, the exemptions and excise duty reductions appear inappropriate, in both their environmental and economic impacts. Similarly, the present tax differential between diesel oil and gasoline is sending out an inappropriate signal, both from the standpoint of greenhouse gas emissions and from that of emissions contributing to urban air pollution.

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Improvement of urban air quality is being hampered by many obstacles and…

Air pollution due to road traffic in cities poses a serious health hazard for urban dwellers. Admittedly, air policy, which has rested principally on enforcement of emission standards and development of public transport, has made it possible to reduce the emissions of some local pollutants but it has not prevented a steep growth of motor traffic in urban areas. While there are no miracle solutions to this complex problem, the following approaches might enable some progress to be made. Parking charges should be revised so as to discourage the use of motor vehicles. The introduction of urban tolls in big cities should be studied. And incentives for car pooling should be provided. For these measures to work, there would have to be better co-ordination of local stakeholders (municipalities, public transport operators, associations, employers). In particular, joint targets for improving air quality and reducing traffic would have to be adopted.

… the reduction of water pollution, particularly of agricultural origin, necessitates an extensive reform

The quality of ground water and water in rivers, lakes and estuaries is deteriorating owing to pollution of agricultural origin, particularly the presence of nitrates connected with intensive livestock farming, fertilisation and irrigation. The proposed new legislation on water provides for the introduction of a tax on excess nitrogen. This is a first step towards internalising the cost of agricultural pollution, which up to now has been virtually exempt from all payment. However, to reduce agricultural pollution of water it would also be necessary to reform the domestic and European systems of agricultural aid, which encourages intensive farming and hence pollution. To improve the efficiency of water resource management, the charges paid for water consumption and pollution should no longer be earmarked to the budgets of water agencies. They should be more closely aligned with the cost of existing externalities and aid allocation procedures should be reformed so as to increase their environmental and economic efficiency. Finally, preferential charge rates for certain sectors or users should be avoided.

To sum up

After years of economic stagnation and unemployment, France has experienced a period of job-rich growth. The efforts made to consolidate the budget situation, bring down inflation, reduce labour costs, open markets to

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competition and privatise public enterprises have finally paid off. However, the rapidity with which strains on the factors of production have emerged shows that progress still needs to be made in boosting productive capacity. The French economy is on track but a substantial improvement in living standards and employment is still possible. Since the beginning of the year, activity has slowed appreciably and unemployment is rising again. For the time being, private consumption is underpinning growth, which should continue to run at close to potential. But the picture in the near term is cloudy and the economic environment may remain subdued. Fiscal policy, while continuing to exercise control over public expenditure, should therefore not attempt to offset the revenue shortfalls linked with the cyclical slowdown. Although this will mean a slightly larger budget deficit in the short term, the authorities should maintain their objective of a balanced budget in 2004. The growth slowdown usefully underscores the importance of structural reforms. These remain essential in order to deal with the problems regarding pensions and to achieve the goal of full employment so desired by public opinion. Here there is a need for additional measures to bring down structural unemployment, such as extending the employment tax credit to part-time working, introducing a new approach to setting the minimum wage, reducing labour-market skills mismatch, increasing gradualism in the phasing-in of the 35-hour work week and simplifying labour regulations. Also, further opening of network industries to competition would create new entrepreneurial opportunities. Finally, a new approach to environmental policy is needed in order to provide more microeconomic incentives to reduce polluting emissions. Pursuit of such policies with respect to public finance, employment, competition and the environment should enable France to, once again, achieve sustainable non-inflationary growth.

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I. A phase of cyclical slowdown

Like most other OECD economies, France went through a favourable period until 2000. But at the beginning of 2001 economic conditions began to deteriorate. Exports softened, business activity slowed, investment weakened considerably and, since May, unemployment has been rising. At the same time, inflation in the euro area accelerated temporarily, making it more difficult for the European Central Bank to ease monetary policy to any significant extent. The terrorist attacks of September 11 have almost certainly accentuated the process of slowdown already begun. Its scale and duration will largely depend on the impact these attacks have on the confidence of consumers and investors (in France and the rest of Europe as well as in the United States). But other influential factors will be the decisions taken by the French authorities, particularly in regard to fiscal policy (Chapter II) and structural policies (Chapter III). For the longer term, a more environment-oriented approach will also be necessary in order to make growth sustainable (Chapter IV).

Growth performance was favourable from 1998 to 2000

During the period from 1998 to 2000 real GDP growth averaged 3.3 per cent annually, a marked improvement on the 1.2 per cent average rate recorded from 1991 to 1997. Whereas fiscal consolidation in preparation for the European Economic Union held back growth in the first half of the 1990s, the last three years of the decade were marked by a number of favourable circumstances. Monetary conditions supported the economy. World trade expanded briskly. The euro's weakness against the dollar and the yen gave a fillip to export performance. And activity in the information and communication technologies sector grew substantially. These factors helped to improve business confidence and investment picked up. Households became optimistic again, partly because of declining unemployment, and consumption was reactivated. The French economy also benefited from the positive effects of the structural reforms carried out over a period of almost ten years, notably with respect to the labour and goods markets, and the wage moderation than began towards the beginning of the decade which allowed inflation be contained.

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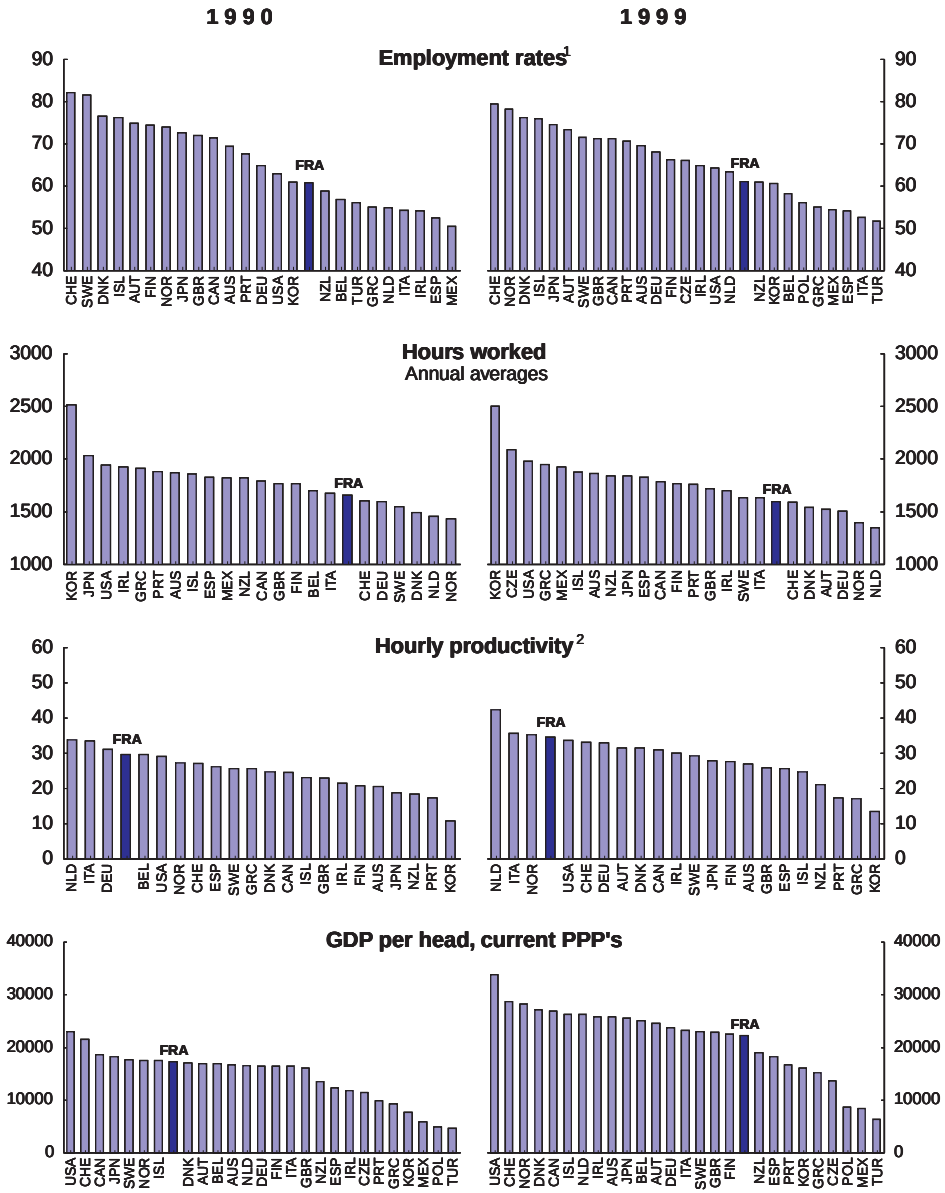
Figure 1. **Labour utilisation and output per head**



1. Total employment as per cent of population aged 15 to 64 years.
2. Real GDP in PPP's divided by total hours worked.

Source: OECD.

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Nevertheless, these three years of rapid expansion were not enough to close the growth gap with other industrialised countries, a gap has even tended to widen because of the brisk acceleration of growth in many countries. Although the level of French measured hourly labour productivity is still one of the highest in the OECD area, the combination of a falling employment rate and a decrease in hours worked is impacting on average income per inhabitant, as indicated by the comparison of per capita GDP levels at purchasing power parity in OECD countries (Figure 1). Also, the speed with which production capacity has become strained serves as a reminder that, to achieve strong sustainable growth in the years ahead, the French economy will need to need to mobilise its resources better. The fact that French employment and investment ratios are comparatively low shows that there is still room for growth. To harness this potential, structural reform of the labour and goods markets will need to be taken further. Not only will it be necessary not only to ensure that more intensive use is made of the nation's resources, but also to introduce policies to stimulate productivity growth will also have to be introduced, as indicated in the OECD report on growth (OECD, 2001).

Output increased thanks to accommodating monetary conditions,...

The upturn in growth was facilitated, as in the other euro zone countries, by particularly favourable monetary conditions (OECD, 2001*a*). Until 1999, short-term interest rates were at historically low levels (2.5 per cent at year-end). The European Central Bank subsequently raised its intervention rate in several stages to 4.75 per cent, but long-term rates remained low (Figure 2). The rapid expansion of credit, both to households and to businesses, reflects the favourable slant of monetary conditions until the end of 2000 (Figure 3). Furthermore, the depreciation of the euro on a real effective exchange rate basis also contributed to the accommodating stance of monetary conditions and made French exports more price-competitive in the short term.

... the strong growth of world trade,...

French exports grew briskly until the end of 2000. Sales were greatly helped by the buoyancy of world trade, the marked weakness of the euro, and the good performance of unit labour costs (Figure 4). There was a steep increase in exports to the most dynamic areas of the world economy – North America and the euro zone – and to the OPEC countries, but a more moderate advance in sales to Asia, Japan and Africa. This export growth concerned all products, not only those for which France has always held a comparative advantage (agro-food, luxury goods, motor vehicles, shipbuilding, aerospace), but also new technologies (mobile phones, in particular).

... a need to renew production capacity, and...

Businesses sharply increased their investments, largely in response to mounting strains on production capacity – the capacity utilisation rate peaked in

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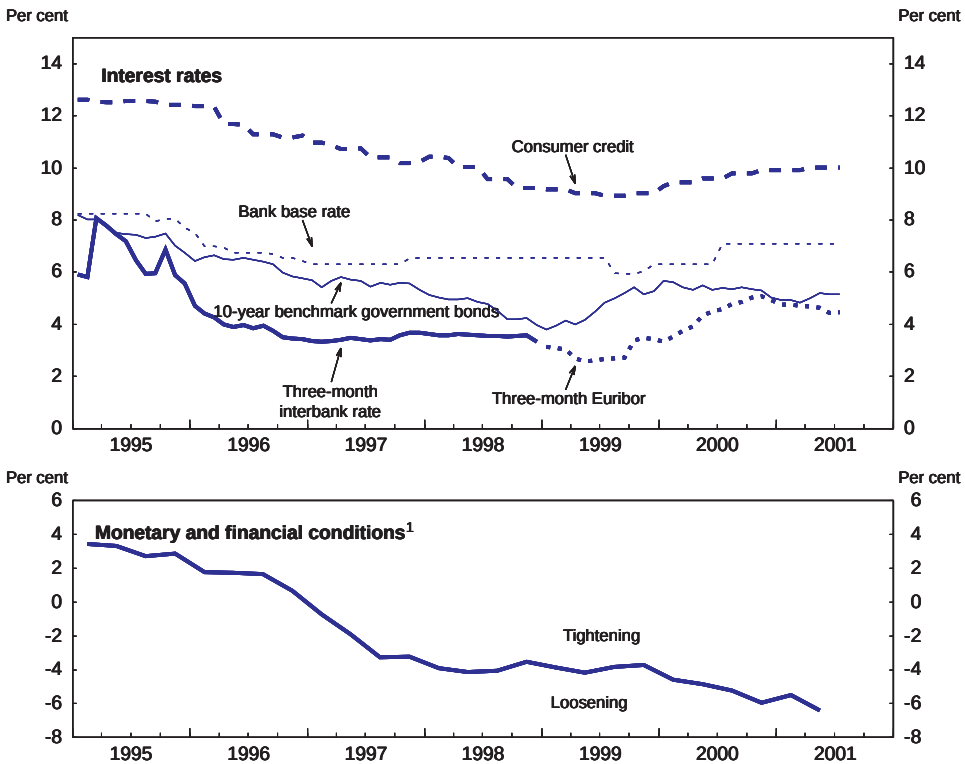
Figure 2. **Interest rates and monetary conditions**



1. Reflecting short and long rates (respective weights .35 and .65), calculated with the real effective exchange rate *vis-à-vis* forty-one countries based on unit labour costs expressed in common currency.

Source: Banque de France and OECD.

the second half of 2000 at close to its 1990 level. Investments were concentrated in the traditional categories (vehicles, machine tools and buildings). Investment in new information and communication technologies (ICT) also grew strongly until 1999, when it represented about 18 per cent of total corporate investment. Despite this remarkable growth, France falls short of the United States in respect of ICT investment, notably as regards computer hardware and software (Mairesse, Cette and Kocoglu, 2000).[1] Moreover, purchases of ICT products slowed towards the end of the period, as the Y2K-related renewal of microcomputer equipment come to an end. In addition to the post-Y2K slowdown, the weaker euro made these products more expensive and reduced demand for them.

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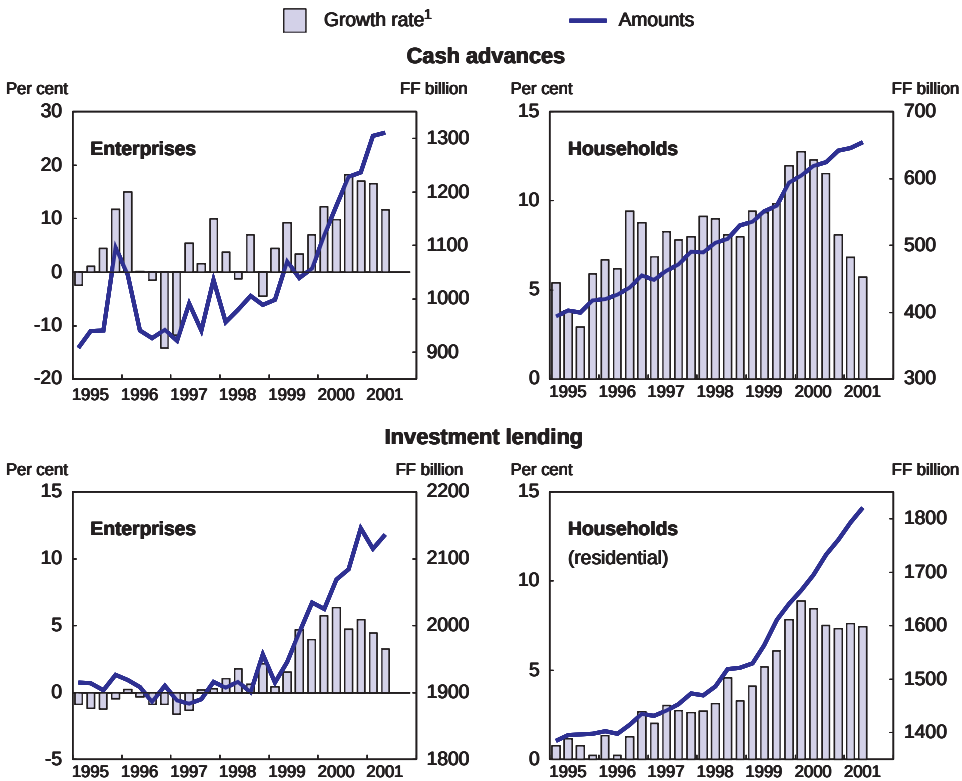
Figure 3. **Credit**



1. Annual rates over four quarters.
Source: Banque de France and OECD.

In spite of this accumulation of capital equipment, the rate of business fixed investment in France (ratio of non-residential investment to GDP in nominal terms) is still one of the lowest among the major OECD countries. Thus the business fixed investment ratio was 16.8 per cent of business sector GDP in 2000, compared with a weighted average of 18.6 per cent for the OECD area. The investment rate declined during the 1990s and the recent pickup has not been sufficient to bring it up to the OECD average. This comparative weakness of business fixed investment in France may be partly explainable by the high level of outward direct investment by French firms relative to inward investment by foreign firms (Box 1).

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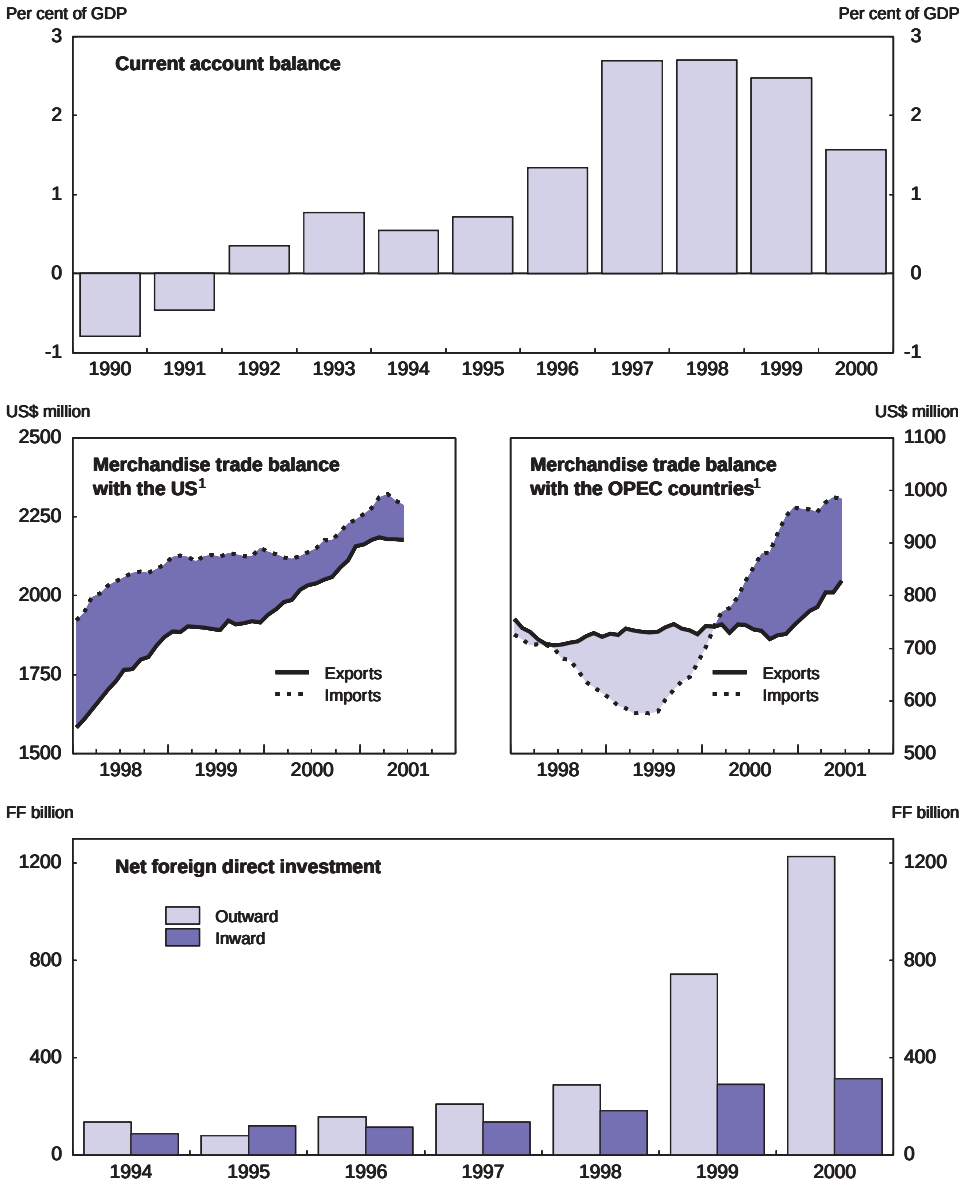
Figure 4. **Foreign trade**



1. Three-month moving averages of monthly rates.
Source: INSEE and OECD.

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... a high level of consumer confidence,...

Households increased their spending in the last three years of the decade, while maintaining a relatively high saving ratio. The growth of household consumption was largely due to the high level of job creation and cuts in personal income tax. Households decided to save part of their gains, especially in 2000, rather than increase their spending still further. The stock market downturn at

Box 1. Why do French businesses invest so much abroad?

French investment abroad showed exceptional growth in 1999 and 2000.[1] Outward direct investment increased by 160 per cent in 1999 and by 65 per cent in 2000, amounting to 113 billion and 187 billion euros respectively in those two years. Inward investment flows to France, although reaching historically high levels, were much smaller and amounted to only 44 billion and 48 billion euros respectively.[2] The net deficit on direct investment is thus very large (Figure 4).

The surge in outward investment is not unique to France. In 1999 total outward investment by the United Kingdom, by Germany and by France was equivalent respectively to 60 per cent, 40 per cent and 30 per cent of outward investment by the United States. In terms of investment/GDP ratios, the Netherlands, Belgium and the United Kingdom are the leading outward investors with rates of over 50 per cent but France has made a big stride towards them, raising its ratio from 25 per cent in 1999 to 35 per cent in 2000. France is now the world's fourth biggest outward investor, both for absolute value and for GDP share.

A first explanation, put forward by the French authorities, is successful globalisation of the economy. With this investment, firms are expanding through integration, notably European, and international "concentration and agglomeration" in order to attain critical mass in world competition. In some cases, the authorities assist integration. The ability of national firms to make this investment can also be taken as an indication of their financial health. Many operations are financed by debt issues on international markets and share swaps, with therefore no drain on national saving even though the amounts concerned are booked as capital outflows.

A second explanation has been offered by certain observers, who consider that the rapid growth of outward investment reflects the low appeal of France for investors. Two official reports have recently drawn the government's attention to the problems of the country's capacity to attract geographically mobile investment (Lavenir 1999, Charzat 2001). In particular, these reports emphasise the problems posed by the high level of corporate and personal income taxation and the complexity of regulations, as in the case of labour legislation.

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Box 1. **Why do French businesses invest so much abroad?** (*cont.*)

Of the two explanations it is difficult to tell which is the more plausible. Both factors – globalisation of French firms and France's low attractiveness to investors – are probably at work simultaneously. A breakdown of the statistics nevertheless lends some credibility to the explanation that some French firms are trying to become world-sized. Investment during the past two years has been concentrated in a small number of large-scale strategic operations.[3] The distribution of investment by sector also shows concentration in activities where market access necessitates a direct presence. Finally, the network industry sector, which accounts for one-fourth of outward investment, is one in which French firms are seeking to become world leaders.[4]

1. Source: *Stock des investissements directs français à l'étranger au* 31 *décembre* 1999, Bulletin of the Bank of France, July 2001. Additional estimates for 2000 were supplied by the French authorities.
2. Consistent with the recommendations of the IMF Balance of Payments Manual, outward direct investment concerns investments by resident enterprises holding at least 10 per cent of the equity capital of a non-resident enterprise. The capital stock is calculated as comprising the owner's equity of the non-resident enterprise invested in, plus earnings reinvested *in situ* by the resident investor, plus loans and deposits from parent company to subsidiary.
3. In 1999, operations of this kind were Renault's purchase of an equity stake in Nissan, the merger of Rhône-Poulenc with Hoechst (the new Aventis group) and the buy-outs of Petrofina by Total, of Nalco Chemicals by Suez and of US Filter by Vivendi. In 2000, the acquisition of Orange by France Telecom and of Seagram by Vivendi were investments of some tens of billions of euros, alone representing 40 per cent of direct capital outflow.
4. At the end of 1999 total outward investment was divided among the following sectors: energy-water 24 per cent (of which electricity-gas-water 17 per cent and mining 7 per cent); credit 14 per cent; chemicals 9 per cent; insurance 7 per cent; agro-food 5 per cent; transport equipment (automotive) 4 per cent; commerce (mass retailing) 4 per cent. Four corporations alone accounted for 92 per cent of investment in the energy-water sector: Suez (Lyonnaise des Eaux), Vivendi (Compagnie Générale des Eaux), TotalFinaElf and EDF International. In 2000 France Telecom joined them by investing heavily in Europe.

the end of the period may have had a negative "wealth effect" on household expenditure although this is still difficult to ascertain from French data. In addition, the oil price rise and the crisis over bovine spongiform encephalopathy (BSE) seem to have curbed consumer spending somewhat, as indicated by the decline in consumption of energy products and the modest increase in food sales. Other categories of expenditure were buoyant, like consumption of ICT products (personal computers, mobile phones and consumer electronics) and health-related expenditure – the restricted scope of the health sector reforms meant that

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households continued their heavy consumption of medical care, and medicines in particular.

... which encouraged job creation

Job creation was especially vigorous (Figure 5). In the three years from 1998 to 2000 the total number of jobs increased by 1.2 million[2] with a spurt at the end of the year. The buoyancy of employment was essentially attributable to strong output growth. But it can also be explained by the increased employment intensity of growth. Here a number of factors exerted a favourable influence: the progressive reduction of employer social insurance contributions since 1993, which increased the demand for low-skill labour; strong growth of activity in the service and construction sectors, both of which are highly labour-intensive; and more recently the shortening of average work hours. The improvement in the labour market also caused a slight fall in jobs with limited tenure (fixed-term contracts, subsidised jobs, apprenticeships). The estimated GDP growth threshold above which the economy begins to create jobs has fallen significantly in the past few years. Recently, GDP growth of about 1½ per cent was sufficient to permit net job creation, but in the 1980s growth of between 2 and 2½ per cent was necessary. This development might represent a permanent increase in the job content of French growth, but it more likely reflects a rise in the equilibrium level of the employment rate, notably as a result of reduced social charges. In this case, the increase in recent years of the labour intensity at growth would be temporary.

With the labour market remaining buoyant there was a marked decline in the number of job seekers, and the unemployment rate (as measured by the labour force survey) was down to 8.8 per cent in March 2001, the lowest level recorded for that month since 1984. The fall in unemployment concerned most categories of job seekers, irrespective of gender, age, level of qualification, social class or region of residence. Thus the situation has improved for youth and unqualified persons, two groups previously highly vulnerable to unemployment and until now unable to benefit from rising employment. Nevertheless, the unemployment rate for youth and persons designated as "low-skilled" (with no academic qualification or just a certificate of primary education) is still well above the national average. Finally, the percentage of long-term jobless is falling.

In spite of this improvement, the scale of underemployment continues to give cause for concern. The unemployment rate is still higher than in other major OECD countries, with the exception of Italy and Spain. The NAIRU, as calculated by the OECD, is slightly down, but it too is still high. Labour force participation rates rose only moderately during the period from January 1999 to March 2001. Thus, while the officially targeted return to "full employment" has begun, it is still far from complete.

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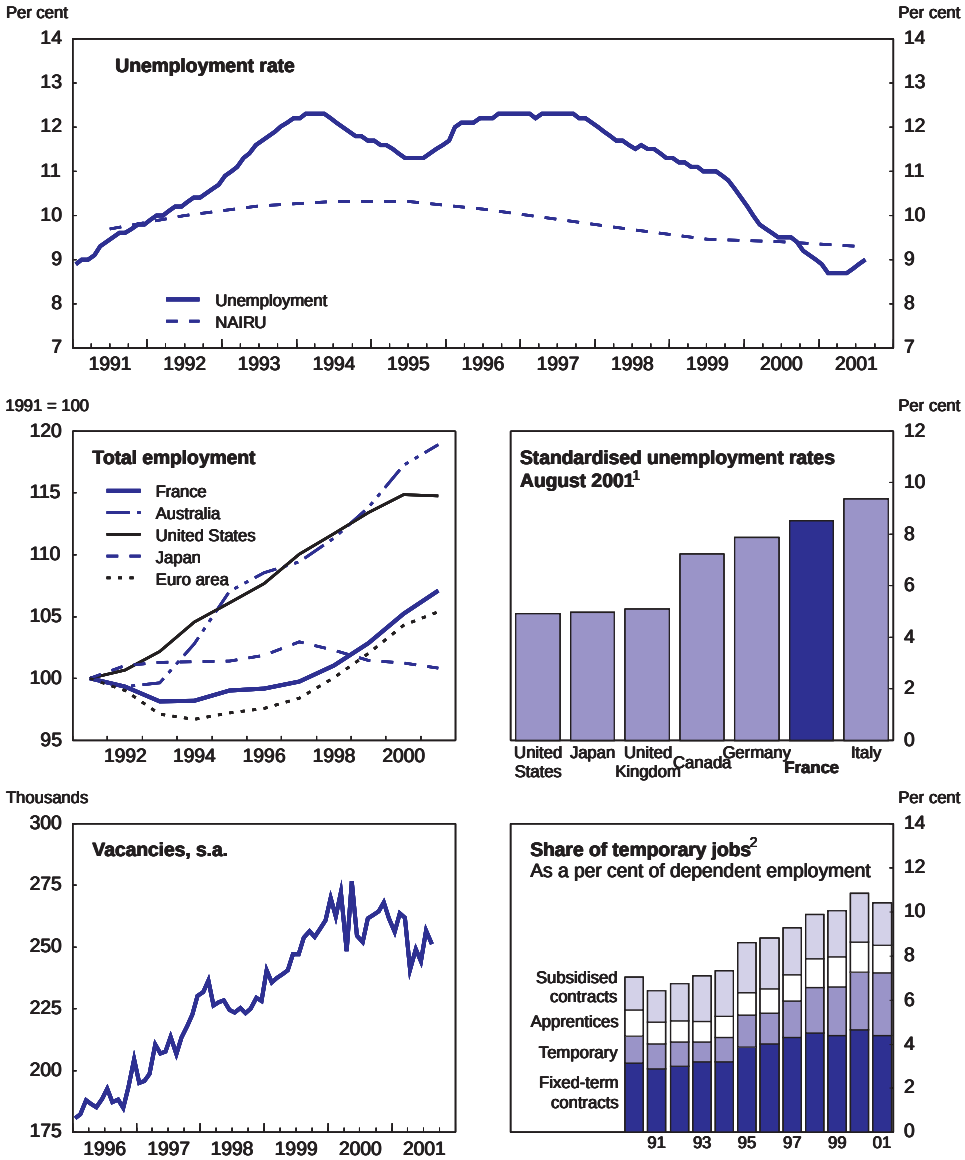
Figure 5. **Employment and unemployment**



1. United Kingdom: June 2001, Italy: July 2001.
2. Employment surveys, March data except for 1999 (January).

Source: INSEE and OECD.

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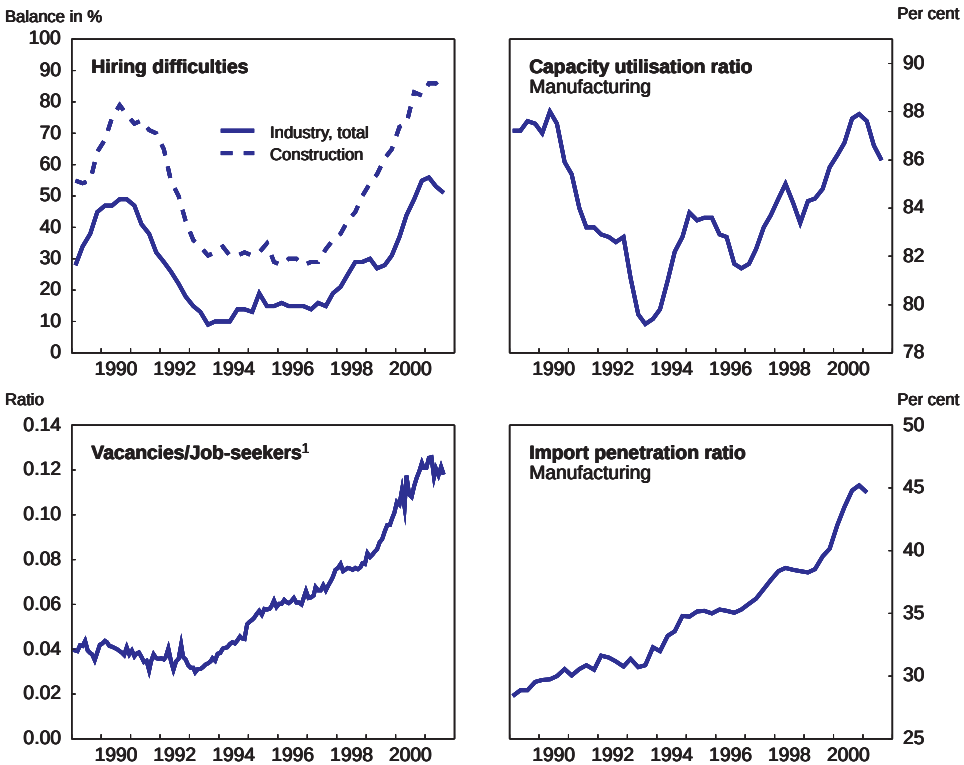
The output gap has closed and strains have appeared

With the acceleration of activity, signs of strain emerged increasingly until end-2000. According to the OECD Secretariat's estimates, the gap between actual and potential output that had opened up in the early 1990s narrowed progressively and closed in 2000. The French economy's productive capacity was being fully used for the first time since the late 1980s and judging from the increasing signs of strain on supply, demand appears to have temporarily outran capacity. Capacity utilisation in industry was near record levels; plant utilisation periods reached an all-time high;[3] an increasing number of firms reported difficulties in meeting orders; inventory levels declined; and supplier-to-producer and distributor-to-customer delivery times lengthened (Figure 6). Despite the investment

Figure 6. **Strains**



1. Category 1, jobless available for work immediately seeking a full-time indefinite-term job.
Source: INSEE and OECD.

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efforts of firms, the productive system showed signs of saturation until end-2000 and because domestic production was unable to keep pace with the very strong growth of aggregate demand, there was increasing recourse to foreign suppliers. The result was a marked rise in the rate of import penetration, particularly for manufactured goods and the instantaneous elasticity of imports with respect to GDP rose very steeply, climbing from 1.2 to 4.6 in the space of one year. A development of this kind is usual in periods of upswing, but not one of this magnitude. One element that might account for this surge in imports was the very strong demand for ICT products, which are largely imported because French production in this sector is limited.

The high rate of job creation provoked signs of strain on the labour market as well which were reflected in recruitment difficulties, particularly in the building and civil engineering sector, certain manufacturing industries and some service activities (Viney and Amar, 2001). Up until the end of 2000 these strains did not affect wages, but some signs of wage acceleration appeared in the first half of 2001.[4]

The emergence of capacity constraints after three years of 3 per cent growth could suggest that GDP increase was underestimated, largely because the "new economy" of information and communication technologies were poorly measured (Annex I and Figure 7). In fact, however, the bulk of these new activities have been factored into the GDP statistics and the emergence of these tensions may reflect certain persistent weaknesses of the French productive system. In such a case, the French economy would have difficulty in sustainably exceeding a "speed limit" of about 2½ per cent a year. A number of factors, however, make the authorities think that potential growth could be raised in the medium term to 3 per cent a year. Such a scenario would necessarily imply further fundamental structural reforms in the labour and goods markets, so as to raise investment and employment rates and boost productivity gains (Box 2).

In 2001 growth slowed distinctly and core inflation picked up

The French economy entered a period of marked cyclical slowdown at the beginning of 2001. Real GDP growth weakened in the first half-year, business expectations were scaled down sharply and foreign trade volumes contracted. Several factors were responsible. First, the world business cycle turned down, directly affecting French exports of manufactures. Second, businesses reacted to the deterioration in their environment by rapidly and steeply reducing their investments and inventories. Finally, production was affected by a number of adverse shocks, such as the cyclical turnaround in the construction sector and uncertainty in the agro-food sector due to the successive meat crises. Coincident with the world economic slowdown, the September 11 terrorist attacks on the United States generated additional uncertainty, shaking the confidence of consumers and investors not only in America but also in France and the rest of Europe.

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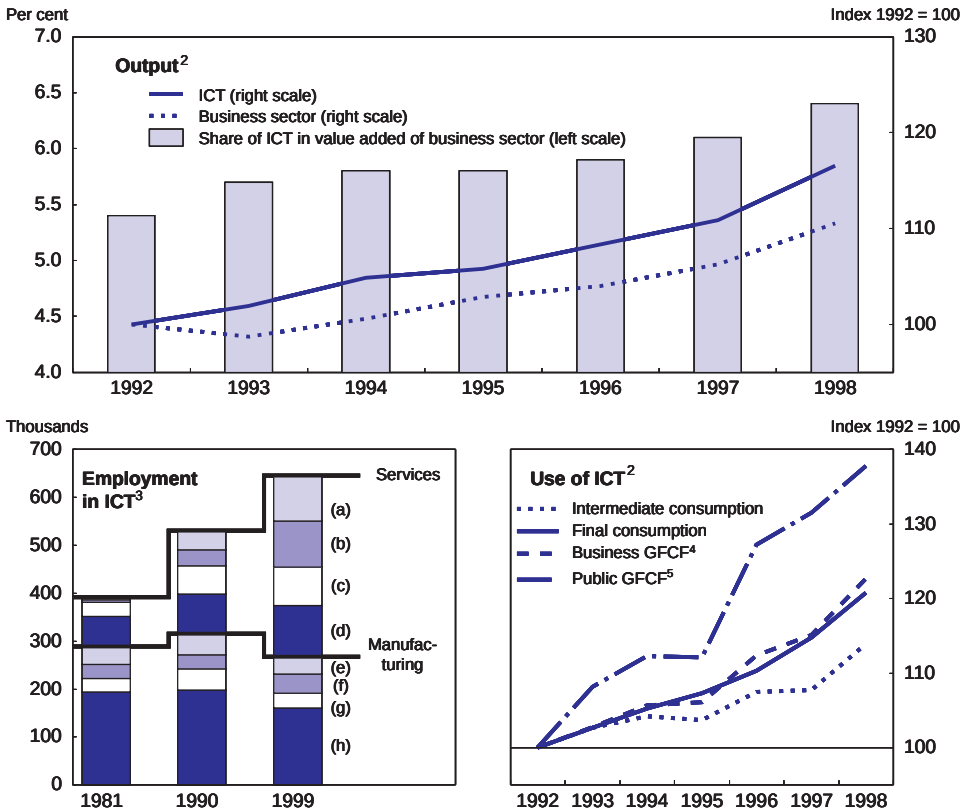
Figure 7. **The new economy**[1]



1. Information and communications technologies (ICT) on the INSEE definition: electronic components, computers, communications equipment, measuring instruments, control and automation systems, wholesale computer trade, computer services and software, telecommunications services.
2. At constant 1995 prices.
3. *a)* Software, *b)* computer services, *c)* wholesale computer trade, *d)* other sub-sectors, *e)* scientific instruments, *f)* telephones, *g)* computer manufacture, *h)* other sub-sectors.
4. Gross fixed capital formation by non-financial companies and unincorporated enterprises.
5. Gross fixed capital formation by general government.

Source: Ministry of Labour, INSEE and OECD.

The world slowdown is weighing upon French business…

Even before September 11 French businesses had rapidly and steeply scaled down their expectations in response to the deterioration in the international environment (Figure 8). The business surveys carried out by INSEE show a decline in expectations, and the Reuters survey of purchasing managers yields a

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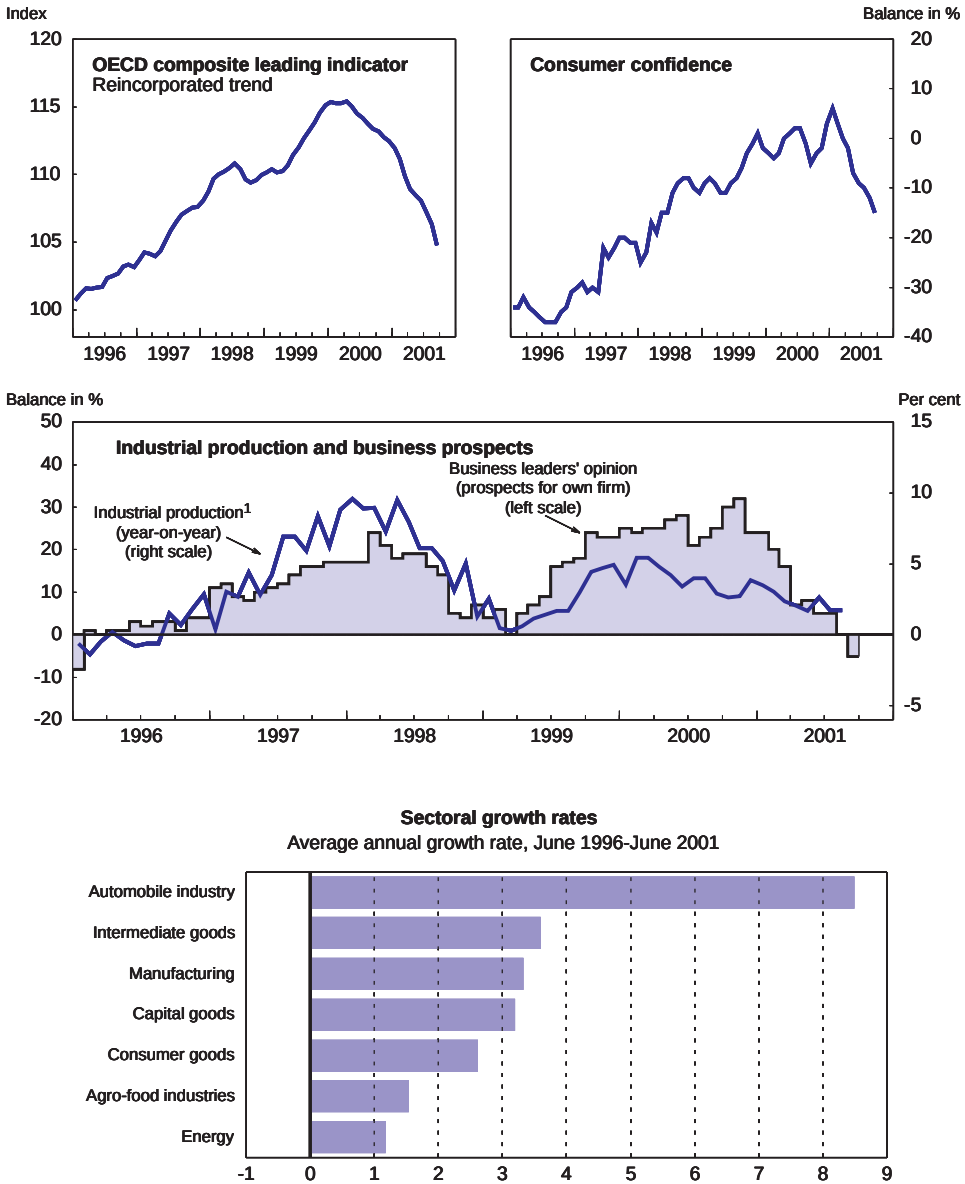
Figure 8. **Cyclical indicators**



1. Seasonally-adjusted industrial production index excluding energy and agro-food industries.
Source: INSEE and OECD.

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Box 2. Can the French economy grow sustainably at 3 per cent a year?

The tensions that emerged in the French economy in 2000, after growing 3 per cent per year for several years, suggest that the output gap accumulated in the early 1990s has been closed. With a saturated productive system, growth can be maintained only if production capacity is enlarged. The pace at which capacity increases depends on numerous factors and it is therefore difficult to estimate precisely the "speed limit" below which growth must stay if the economy is not to become unduly strained. It is nevertheless interesting to produce some estimates, if only to give the authorities an idea of how much scope for action they have in the medium term.

The OECD traditionally interprets potential output as the one that can be produced with the existing capital stock when unemployment is at its structural rate. Hence growth of potential output depends on the rate of increase of the capital stock and labour force, the trend of structural unemployment and the growth of total factor productivity. To produce a realistic quantitative estimate of potential growth it is necessary first to make a number of adjustments: the cyclical component of the different variables must be factored out; the labour has to be measured as the number of hours worked, instead of the level of employment, in order to capture the trend of work time; the "quality" component of the factors of production has in principle to be factored out (which may pose statistical problems) and structural unemployment has to be estimated, since it cannot be observed directly.

Even if the techniques used differ, the calculations of the OECD, the French authorities and economists in the private sector are similar. The OECD's estimates suggest that France's potential growth in the recent period was close to 2½ per cent a year.* This would put France in a median position among OECD countries (Scarpetta *et al.*, 2000) (Figure 9). The IMF (2000) calculates, with the aid of the Hodrick-Prescott and Baxter-King filters, that potential growth is about 2-2½ per cent a year. Chauvière and Didier (1999) estimate it at about 2 per cent a year (on the basis of capital stock data to 1997). Artus (2000) estimates that it is below 2 per cent–which is half the rate he estimates for the United States – essentially because of lesser growth of the capital stock and labour supply. Finally, the French authorities (MINEFI, 2001) estimate that potential growth accelerated in the second half of the 1990s to reach 2½ per cent by the end of the decade.

The French authorities also consider that an increase in the growth "speed limit" to 3 per cent a year is possible** and could be achieve through more buoyant investment and employment. The strength of investment would be due in large part to a rapid development of new technologies. The marked improvement in employment would reflect an appreciable decline in the structural rate of unemployment, the latter being due to cuts in labour taxes and higher exposure of goods markets to competition. The OECD's medium-term projection, on the

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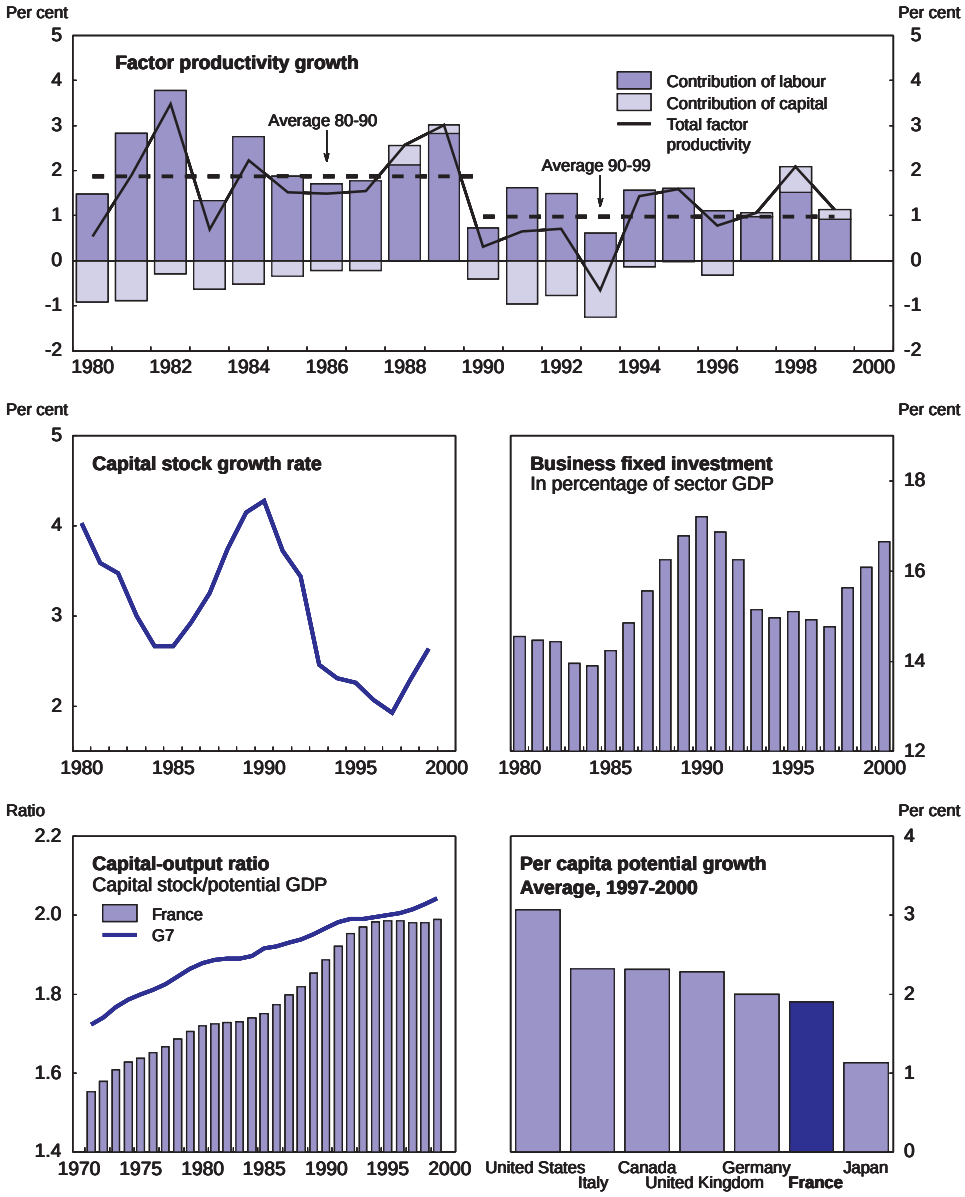
Figure 9. **Supply potential**

Business sector



Source: INSEE and OECD.

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Box 2. **Can the French economy grow sustainably at 3 per cent a year?** (*cont.*)

other hand, is based on potential growth holding at 2½ per cent a year. The assumption here is that the reduction of work hours is liable to lessen the labour factor's contribution to growth during the next few years. A number of other developments might also prevent a decrease in structural unemployment, notably the strong advance of the SMIC minimum wage and the tightening of labour market regulations.

* A breakdown of growth sources is given in Annex II.
** Underlying assumption of the central scenario of the multiyear public finance programme 2002-2004.

similar result. Since autumn 2000 industrial firms have been expecting a further decline in their prospects and accordingly have cut back their investment plans and drawn on their inventories. Manufacturing output therefore slowed sharply in the first and second quarters. Service activity slowed, the construction sector experienced a turnaround in orders, and the agro-food sector was hit by the BSE and foot-and-mouth crises. In consequence, job creation slowed markedly in the second quarter and unemployment has been rising since June.

... whilst core inflation is rising

An inflation spurt occurred in the first half of 2001 (Figure 10), with the year-on-year rise in consumer prices peaking at 2.3 per cent in May. This was due essentially to a conjunction of different shocks. Very rainy weather made fresh fruit and vegetables scarce, whilst the meat crises drove up prices of substitute products. The hike in oil and natural gas prices in 2000 fed through to retail prices of energy and the euro's slide put upward pressure on import prices. Finally, the temporary effect of the VAT cuts in 2000 wore off. Since May consumer price inflation has eased, reaching 1.5 per cent in September. This improvement has been chiefly due to the dissipation of the oil shock and the more recent downtrend in energy prices. The indicators of underlying inflation in fact show a rising rate since the beginning of 2001, with a September figure of 1.8 per cent.

The easing of consumer prices is very welcome and should help to forestall inflationary pressures linked with indexation and upward adjustment of price expectations. Even so, wage performance will be decisive. Admittedly, growth of hourly wages accelerated sharply in 2000 and 2001 (peaking at 5.4 per cent year on year in the second quarter of 2001) with the introduction of the

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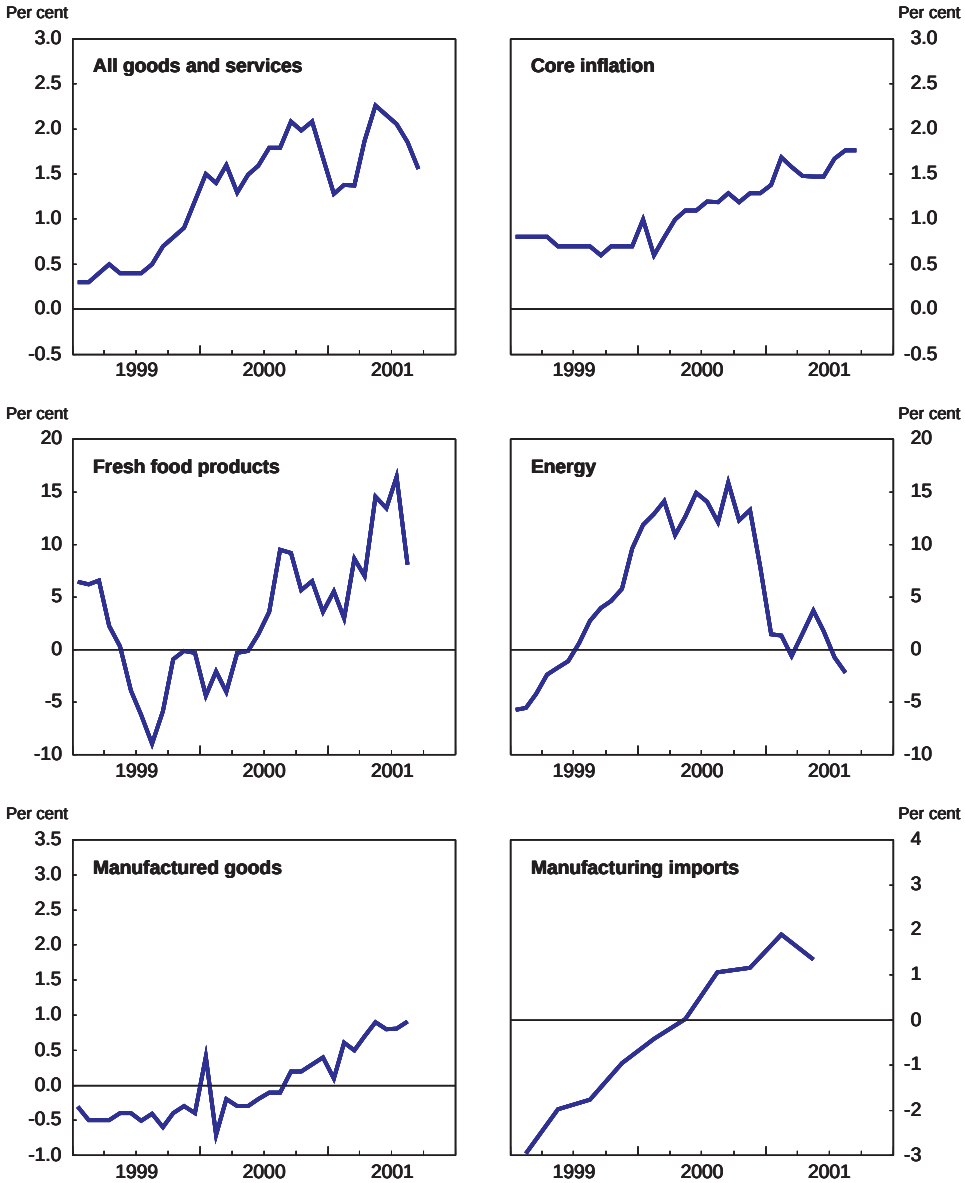
Figure 10. **Inflation**
Year-on-year change



Source: INSEE and OECD.

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35-hour week[5] and the hourly SMIC was increased by 4.05 per cent in July 2001. But per capita monthly-wage growth remained moderate, largely because employees moving over to the 35-hour week agreed to some curbing of their monthly wage increases in return for the shorter hours.[6] Initially, the acceleration of hourly wages did not impact on unit labour costs, owing to hourly productivity gains and the reduction of social charges. However, they began to accelerate as of mid-2000 in response to growth of the labour costs and a cyclical slowdown in productivity increases. All told, French unit labour costs continue to benefit from the moderate trajectory of wages in recent years, even though there has been some acceleration over the past few quarters. In relative terms, unit labour costs in France are still competitive *vis-à-vis* non-European countries (Figure 11).

Activity should pick up in the second half of 2002, but strong risks remain

The international environment of France became less favourable in 2001. Prior to 11 September 2001 the world economy was in an uncertain situation, with some indicators suggesting that the bottom in the slowdown had been reached, while others suggested continued weakness. The American economy slowed in the second half of 2000 and grew only slightly in the first half of this year, and Japan's economy went into recession again. There was a sharp deceleration of world trade growth, which rebounded on the euro area in the first half of 2001. Moreover, other mechanisms of international pass-through of cyclical fluctuations – like parallel swings in stock markets or the reactions of multinational firms – may have amplified the slowdown, although these mechanisms are still not well understood. The published indicators concerning the situation in summer 2001 did not show conclusively that the low point in the slowdown had been reached. In the United States and in Europe businesses still seemed to be experiencing difficulties in adjusting their balance sheets moreover, the correction of the past two years' over investment, particularly in the information and communication technologies, did not appear to be complete. Equally disquieting, the confidence of American households, which had underpinned household consumption, seemed at last to be waning, probably because of adverse developments in the labour market.

The terrorist attacks in the United States on 11 September 2001 contributed to a further deterioration in the economic environment. Those events are having a *direct* negative impact on American business activity, although probably of limited magnitude. The *indirect* impact will doubtless be much greater: a steep decline in household and business confidence and, the postponement of expenditure by consumers and investors until current uncertainties become cleared. True, the monetary and fiscal policies of OECD countries should provide an underpinning for growth, but their effect will probably not be felt in full until the second half of 2002 and in 2003 because of cyclical lags. Altogether, according to the

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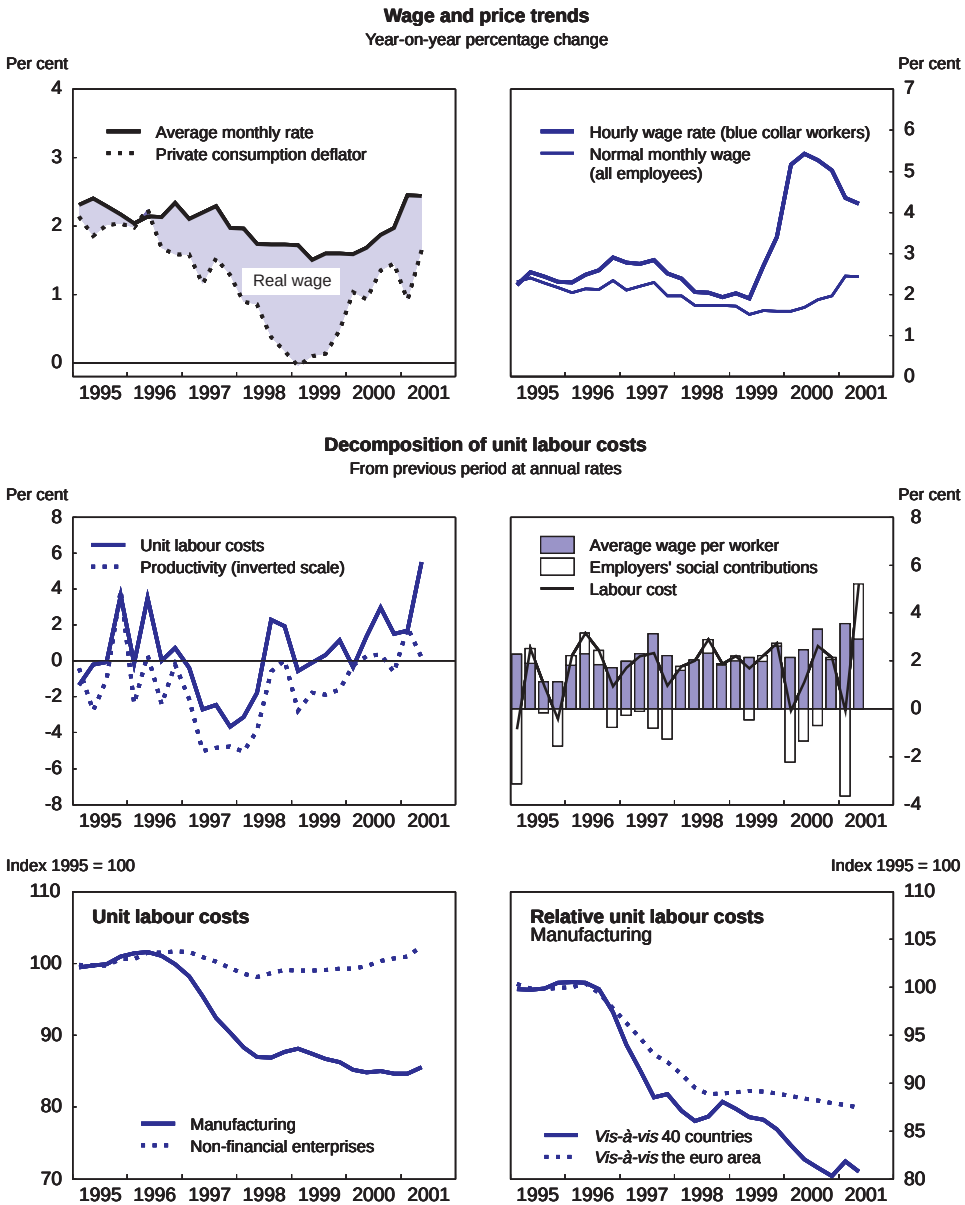
Figure 11. **Wages, labour costs and competitiveness**



Source: INSEE and OECD.

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Table 1. **Summary of short-term outturns and projections**
Percentage changes, 1995 prices

	1998	1999	2000	2001	2002	2003
Private consumption	3.6	2.7	2.7	2.5	1.7	2.4
Government consumption	0.3	2.6	2.3	2.2	1.9	2.0
Gross fixed investment	6.6	7.3	6.2	2.9	0.8	4.1
General government	2.8	2.2	4.2	2.7	0.3	1.3
Household	3.6	8.3	4.6	0.0	0.5	2.7
Other	9.0	8.3	7.4	4.1	1.0	5.2
Stockbuilding[1]	0.6	–0.3	0.3	–0.8	0.1	0.4
Total domestic demand	4.2	3.0	3.6	1.7	1.6	3.0
Exports of goods and services	7.7	4.0	13.4	2.6	1.6	7.6
Imports of goods and services	11.3	4.0	15.2	1.6	1.7	8.0
Net exports[1]	–0.6	0.1	–0.1	0.3	0.0	0.1
GDP	3.3	3.2	3.4	2.0	1.6	3.0
Employment	1.4	1.4	2.3	1.8	0.5	0.7
Unemployment rate[2]	11.8	11.2	9.5	8.9	9.4	9.3
Household saving ratio[3]	15.5	15.1	15.8	16.4	16.8	16.5
Private consumption deflator	0.5	0.4	1.2	1.7	1.4	1.7
GDP deflator	0.7	0.2	0.8	1.7	1.8	1.4
Interest rates (annual averages, per cent)						
Short-term	3.6	3.0	4.4	4.2	3.0	3.8
Long-term	4.7	4.6	5.4	5.0	4.8	5.2
General government financial balance[4, 5]	–2.7	–1.6	–1.4	–1.5	–1.8	–1.4
Current account balance[4]	2.6	2.6	1.6	1.6	1.8	1.6

1. Contribution of change to GDP growth.
2. As a percentage of the labour force.
3. As a percentage of disposable income.
4. As a percentage of GDP.
5. UMTS receipts are treated as current revenue and therefore reduce the deficit for 2001 by 0.1 per cent of GDP.

Source: OECD.

OECD's projections, world trade at constant prices will be flat in 2001 and expand by only about 2 per cent in 2002, before accelerating in 2003. Real GDP in the OECD area is projected to fall slightly in the second half of 2001, level off in the first half of 2002 and pick up around the middle of the year.

As for the French economy, according to the OECD's projections, private consumption will remain the principal source of demand as households benefit from tax reductions and continued – if slower – employment and real wage growth. Nonetheless, the unfavourable international context will weigh heavily on exports and investment during the first half of 2002. Assuming that no additional events

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perturb the international environment, the business climate should improve thereafter and boost output. Consequently, the OECD projects that GDP growth will slow further in the fourth quarter of 2001 and remain weak in the first half of 2002, before reaching in the second half and during 2003 a rate slightly above potential (Table 1). Altogether, in terms of annual averages, real GDP growth

Box 3. Is the changeover to the euro neutral for the economy?

In principle, a "currency reform", namely the replacement of one money unit by another, has no macroeconomic effect. Provided the conversion rate is chosen realistically, the new denomination does not affect the major parameters-prices, assets, liabilities, etc. In the euro area, where the conversion rates were set more than two years ago, economies should not be affected by the changeover to the scriptural euro (cheque books, credit cards, etc.) and the fiat euro (notes and coin) in January 2002. Nevertheless, since this is an operation on an unprecedented scale, there are necessarily some uncertainties. Possible economic repercussions are:

- A favourable impact on production due to the extra activity associated with the changeover. As in the case of the Y2K bug, the cut-off date for the euro money's introduction is necessitating changes in software, new equipment (*e.g.* cash registers), and transport and security operations. Activity in banks, computer companies and security services will probably be stimulated temporarily.
- An inflationary impact produced by price rounding. To obtain euro prices with a minimum of decimal points, producers and traders will be tempted to round upwards. This can even be done ahead of the changeover date: the SNCF (French railways) raised its fares by 1.9 per cent in September 2001, whereas normally it makes the adjustment in January.
- Additional personal spending from holdings of notes in circulation. Not knowing the exact arrangements for exchanging franc notes for euro notes, individuals may be tempted to run down their holdings of banknotes while they are still legal tender. Moreover, the requirement to declare one's identity when converting large amounts at a bank is an incentive to quickly use up money of suspect origin (underground economy, tax evasion, laundering, crime, etc.). As a point of fact, the marked decrease in the amount of fiat money circulating in France since the beginning of 2001, due essentially to the reduced circulation of 500-franc notes, is undoubted evidence of dishoarding. This dishoarding does not necessarily imply additional spending-franc notes can be exchanged for foreign non-euro currencies or substituted for other means of payment–but it is nevertheless likely that it is contributing to increased consumption of durables and semi-durables in the short term (to be offset by reduced consumption later).

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should reach 1.9 per cent in 2001, then slow down to 1.6 per cent in 2002, and rebound to a pace of 3.0 per cent in 2003. The recently announced budget for 2002 indicates that the French authorities intend to apply a neutral fiscal policy next year, stabilising the general government structural balance. With the action of the built-in fiscal stabilisers, however, the general government deficit is likely to be close on 2 per cent of GDP in 2002.

In the current environment, this projection, which builds in a substantial domestic and international reaction to the terrorist attacks in the United States, is subject to unusually large risks. On the upside, if business and consumer confidence (both within and outside France) are less affected than assumed here, then spending levels can be expected to be higher and the slowdown shallower and less durable than projected. Nevertheless, the pre-crisis softening of exports, investment and consumption suggest that even in this "best-case" scenario there will be a slowdown that is likely to extend into 2002. On the downside, there is a real possibility that negative business and consumer reactions could be even more pronounced – particularly if international political developments yield further uncertainty. In this case, the possibility of a deeper and longer period of slow or even negative growth cannot be ruled out. Finally, there is uncertainty about the effects of the changeover to the euro (Box 3).

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II. Fiscal policy issues

As pointed out in the previous OECD *Economic Survey*, French public finances have undergone an intensive process of fiscal consolidation since the mid-1990s. The general government deficit has been reduced from 6 per cent of GDP in 1993 – twice the limit imposed by the Maastricht convergence criteria – to 1.4 per cent in 2000 and the public debt burden has begun to ease (Figure 12). Following this phase of consolidation, French fiscal policy is now based on a multi-year strategy consisting of four pillars: control of public expenditure, reduction of direct taxes and social insurance contributions, free play of automatic stabilisers and a balanced budget target for 2004 or thereabouts. Although this strategy aims at medium-term consolidation, the authorities' projections do not incorporate any reduction in the structural part of the public deficit during the period 2001-2002. Consequently, ambitious measures will be needed in order to lower the structural public deficit in subsequent years.

Fiscal policy in the short term

Public finance strategy for 1999-2001 *is based on control of public expenditure,...*

For some years, pursuant to the Stability and Growth Pact, the authorities have practised a strategy of multiyear consolidation of public finances.[7] Control of government expenditure is the cornerstone of this strategy. Each year, within the framework of its multiyear public finance programme, France adopts a three-year norm for growth of general government expenditure.[8] It comprises separate growth norms for central government expenditure under the budget laws passed each year; for healthcare expenditure used in defining the ONDAM (*Objectif national de dépenses d'assurance maladie*) national health spending targets entered in the social security financing laws passed each year; and essentially indicative growth norms for the other spending categories, notably local government, pensions and unemployment insurance.

For the three-year programme 2002-2004, the authorities have set a target for public expenditure growth of 1.5 per cent a year in real terms,[9] which is close to the average rate during the recent past (Table 2). Most of the pressure will be on

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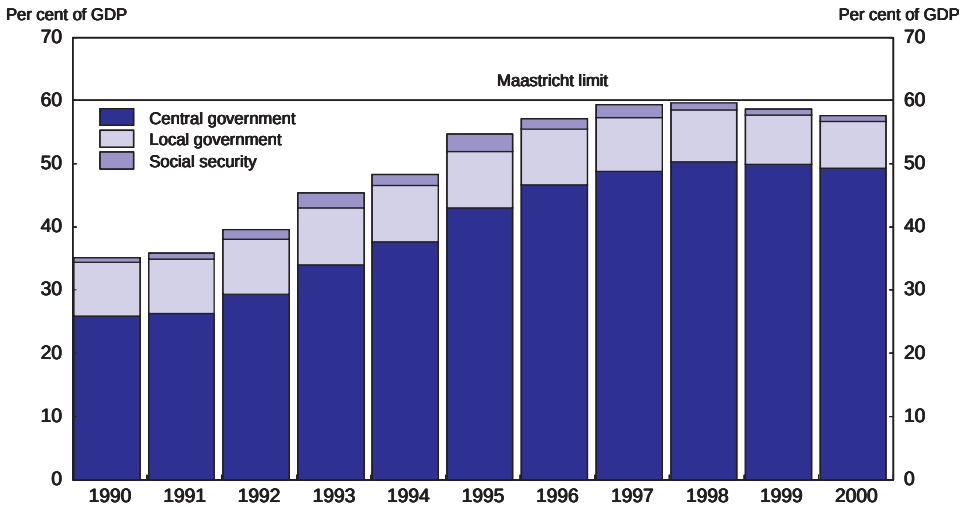
Figure 12. **Trend of public debt by government sector**



Source: INSEE.

Table 2. **General government expenditure**[1]
Annual percentage changes; constant prices

	1994-2000	1999	2000	2002-04 (programme)
General government expenditure, *of which*:	1.6	2.2	1.0	1.5
Central government	0.9	3.4	–0.6	1.4[2]
Local government	2.2	3.6	2.0	1.7
Social security	3.1	2.3	1.4	1.6
Multiyear programme target	n.a.	n.a.	1.2	1.5

1. The annual changes observed may include changes of perimeter.
2. Real growth of 1.4 per cent in central government expenditure on a national accounts basis is consistent with 0.3 per cent growth on a budget accounts basis.

Source: INSEE, national accounts, and Ministry of the Economy, Finance and Industry.

central government expenditure, which must not grow by more than 0.3 per cent a year on average in real terms.[10] Within that limit, about FF 30 billion is to be redeployed each year to finance the government's priorities (education, employment, social security, justice, culture, communication, environment), with commensurately less funding for other categories (notably defence).

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... lighter taxation...

The second strand of the government's fiscal strategy is an easing of the tax burden, which is one of the highest in the OECD area. Like other countries, France has decided to reduce direct taxes,[11] essentially by lowering the rates of personal income tax and corporate income tax. The authorities aim to reverse the upward trend in direct taxation, which reached an all-time high of 45.6 per cent of GDP in 1999. Tax reduction began in 2000 and should continue in 2001 and 2002 down to a level of 43.8 per cent of GDP in 2004. Discretionary tax cuts amounted to 1 percentage point of GDP in 2000 and nearly 0.6 point in 2001. Nevertheless, the economy's overall strength meant that collections were stronger than in the past and tax receipts as a proportion of GDP fell by less.[12]

... free play of built-in stabilisers...

The third component of current fiscal strategy is to allow the automatic stabilisers to operate freely. The authorities consider that tax revenue must be allowed to slow when growth eases and accelerate when it picks up. This cyclical fluctuation allows tax revenue to exert a moderating influence in periods of high growth and to support demand when activity slows. Public finance is thus seen as a regulating element that helps to "smooth" activity.[13]

... and equilibrium of public finance by around 2004

The stated objective of the authorities in their three-year programme is to bring the general government budget into near-equilibrium by around 2004. This goal is to be achieved through control of public spending, exhaustion of tax relief measures and a relative decline in debt-service charges. The authorities intend to move towards budgetary equilibrium gradually. Following a public borrowing requirement of 1.4 per cent of GDP in 2000, the central objective of the programme for 2002-2004 is to bring the general government account to, or close to, overall balance by 2004.[14]

The draft budget for 2002

For the draft 2002 budget the French authorities have adopted a macroeconomic scenario in which the growth of the world economy picks up quickly – a scenario that differs significantly from the one envisaged by the OECD.[15] Within this macroeconomic framework, the 2002 draft budget puts the general government deficit at 1.4 per cent of GDP next year, which is higher than the target set by the government in its 2002-2004 multiyear programme (Table 3). The deterioration essentially reflects the downward revision of growth in 2001 and 2002. To a lesser extent it also reflects the faster-than-expected increase of general government expenditure. Central government spending is set to rise by 2.0 per cent in nominal

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Table 3. **Main components of public finance, 2001-2002**

Per cent

	2001	2002
Central government expenditure[1]		
Multiyear programme 2000-2002[2, 3]	0.3	0.3
Multiyear programme 2001-2003[3]	0.3	0.3
Multiyear programme 2002-2004[4]	0.3	1.4[5]
Finance bill for 2002	0.3	0.5
General government expenditure[1]		
Multiyear programme 2000-2002[2, 3]	0.9	1.0
Multiyear programme 2001-2003[3]	1.1	1.5
Multiyear programme 2002-2004[4]	1.8	1.5
Finance bill for 2002	1.7	2.2
Direct taxes[6]		
Multiyear programme 2000-2002[2, 3]	45.4	45.2
Multiyear programme 2001-2003[3]	44.6	44.4
Multiyear programme 2002-2004[4]	44.7	44.5
Finance bill for 2002	44.9	44.5
General government balance[6, 7]		
Multiyear programme 2000-2002[2, 3]	−1.6	−1.2
Multiyear programme 2001-2003[3]	−1.3	−0.9
Multiyear programme 2002-2004[4]	−1.0	−0.6
Finance bill for 2002	−1.4	−1.4

1. Annual increase at constant prices (deflated by the consumer price index excluding tobacco).
2. Owing to the change of national accounting base, the public expenditure aggregates defined on an ESA95 basis in the 2001-2003 programme are not directly comparable with those of the programme for 2000-2002.
3. Prudent growth assumption.
4. 3 per cent growth scenario.
5. 1.4 per cent volume growth of central government expenditure is consistent with 0.3 per cent growth on a budget accounts basis.
6. As a percentage of GDP.
7. Excluding UMTS receipts.

Source: Ministry of the Economy, Finance and Industry.

terms (0.5 per cent at constant prices) in 2002, a rate slightly higher than in previous years and due in large part to public-service post creations in the sectors of security, justice and education. Public spending in respect of health insurance is also budgeted to increase more rapidly (up 3.8 per cent at current prices) owing to the brisk growth of healthcare consumption and the introduction of the 35-hour work week in public hospitals.

The draft budget for 2002 contains two elements of fragility. First, it is based on an optimistic macroeconomic scenario that carries a strong downside risk. Should this risk materialise, the public deficit would widen appreciably as automatic stabilisers come into play; but there seems to be no danger as yet that the Maastricht Treaty limit of 3 per cent of GDP will be exceeded. Second, there is

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no provision for any budgetary consolidation in 2002. According to the government's projections, the structural part of the public deficit, which was reduced to 1.5 per cent of GDP in 2000, will remain stable this year and next. Thus the 2002 draft budget diverges significantly from France's Stability and Growth Programme for 2002-2004, which provided for a steady reduction of the structural deficit each year to bringing it to balance by 2004. Finally, the authorities have proposed additional measures to support demand in 2002.[16]

It would, of course, be possible to delay the return to a balanced budget by another few years, although this would go against the thinking behind the government's multiyear strategy. The aim of which is to eliminate the structural part of the public deficit and thus provide enough room for the automatic stabilisers to operate without endangering government finance. Indeed, the current situation and the various disruptions experienced in the 1990s serve as a useful reminder that international shocks are recurrent and that cyclical fluctuations are more the rule than the exception. It would, therefore, be desirable for the government to take measures to ensure that its aim is met. An appropriate course would be to undertake reforms that allow public expenditure growth to be reduced to at least the level envisaged in the multiyear programme and preferably below. This would make it possible to reduce gradually the structural part of the public deficit, while allowing automatic stabilisers to operate freely on the revenue side.

A *coherent strategy, but with elements of fragility*

French public-finance strategy, defined and reinforced in the context of the Stability and Growth Pact, provides a coherent multiyear framework. It has the merit of focusing on control of public expenditure, the aim being essentially to improve the quality of spending and notably its cost-effectiveness. The French authorities are also right in trying to reduce taxes, which are still distinctly higher than in most other OECD countries and, in an environment of increasing tax competition, cannot but effect productive capacity. Finally, the authorities are right to set a target of medium-term balanced budget, while allowing automatic stabilisers to act in the short term. Allowing tax revenue to fluctuate as a function of growth can reduce the costs of adjustment during periods of overheating or recession, and diminish the potential effects of hysteresis and permanence (European Commission, 2001).

However, to achieve the objective of a balanced general-government budget by 2004, the authorities will have to take additional measures. The structural balance on the general government account[17] has remained unchanged or has deteriorated slightly between 1999 and 2001, according to the estimates of the French authorities,[18] the European Commission[19] and the OECD[20] (Figure 13). Public spending control has not been sufficient to both offset recent tax cuts and reduce the structural deficit. Admittedly the aforementioned estimates are fragile,

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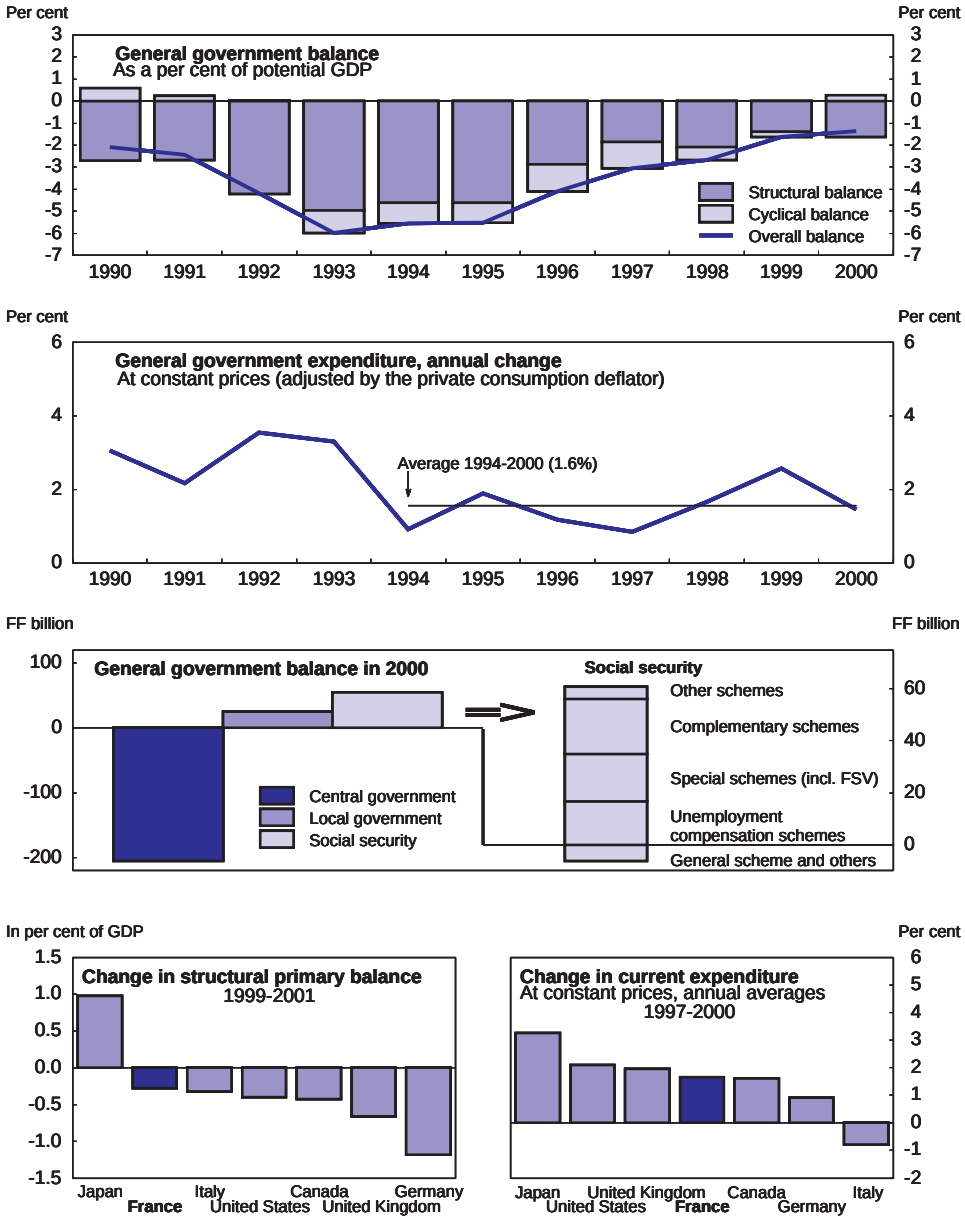
Figure 13. **Fiscal policy**



Source: INSEE and OECD.

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but they suggest that fiscal policy was neutral, on the whole, between 1999 and 2001 and that the downturn in the borrowing requirement as a percentage of GDP simply reflects the action of built-in stabilisers.[21] With all the caution required, it can therefore be concluded that the recent improvement in public finance reflects cyclical factors more than a policy of fiscal consolidation and that an additional 1½ per cent of GDP in spending cuts is required to achieve balance if taxes are not to be raised.

The current growth norm for public spending (1.5 per cent a year at constant prices) corresponds to the average recorded between 1994 and 2000. Given the effort still to be made and the tax cuts the government wants to put through, this norm should be tightened. As the authorities' three-year programme stands, the norm will allow a balanced budget to be reached by 2004 only if economic growth averages 3 per cent a year over the period 2001-2004, which seems beyond the economy's present potential. The spending growth norm should, therefore, be adjusted downward by exercising more restraint in certain categories of expenditure.

Civil service is the category of expenditure showing the steepest increase (in absolute value), whether for central government, local authorities or the social security agencies. Between 1995 and 2000 this spending rose by FF 185 billion in value terms. In the case of central government, staff-related expenditures (civil service and a large part of employment policy expenditure) account for nearly half of total spending (Table 4). As pointed out in the previous *Survey*, the public sector employs about one-fourth of France's total workforce, much more than in most other OECD countries. A gradual downsizing of the civil service will be necessary following the retirement of large numbers of public servants during the next ten years. Since the retirement pensions of civil servants are paid directly from the budget, the wave of departures will cause a steep rise in public expenditure if there is no downsizing. Systematic non-replacement of retiring civil servants could be offset by productivity gains – achieved, for example, through computerisation of administrative functions, the development of e-government, and the simplification of administrative procedures now desired by the authorities. Such steps will necessitate some real thought about the tasks of the State. Moreover, to try and replace all retiring public servants, when the private sector is already experiencing considerable hiring difficulties, would tend to crowd that sector out of the labour market and necessarily reduce the growth of potential. A smaller increase, if not a decrease, in general government staffing expenditure is also justified as an option for the future, given the rejuvenation of the workforce and hence of the wage structure (with a reduction of the career advancement effect).

The State's employment-policy expenditures are another area where savings can be made. This spending category has remained stable overall in recent

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Table 4. **Trend of central government expenditure in 10 categories**

	Initial Finance Acts (FF billion)					% of general budget	Change 1997-2001	
	1997	1998	1999	2000	2001	2001	FF bln.	%
Civil service	636.8	657.0	675.5	693.7	709.8	42.5	73	11.5
Net public debt	232.6	234.8	237.2	234.7	239.8	14.4	7	3.1
Employment policy	143.6	148.8	152.2	151.7	149.3	8.9	6	4.0
Social transfers	110.7	109.8	115.4	125.7	128.9	7.7	18	16.4
Defence (excluding wages)	113.2	104.3	107.1	103.9	104.3	6.2	–9	–7.9
Economic and other interventions	93.2	96.0	98.6	89.5	97.6	5.9	4	4.7
Public agencies and corporations	78.9	81.2	85.0	86.3	86.0	5.2	7	9.2
Civil capital expenditure	71.9	72.2	72.8	73.7	72.9	4.3	1	1.4
Subsidies to special welfare schemes	48.5	48.8	47.2	45.8	42.3	2.5	–6	–12.7
Central government operating expenditure	35.1	33.8	34.2	37.3	37.9	2.3	3	7.8
General budget with constant 1988 structure (1)	**1 564.5**	**1 586.7**	**1 625.2**	**1 642.2**	**1 668.7**	**100**	**104**	**6.7**
Expenditure growth norm, constant prices, Finance Bill		0%	1%	0%	0.3%			
Perimeter changes accumulated since 1998 (2)	–0.5	0.0	45.4	22.6	42.7			
General budget with current structure (3) = (1 + 2)	**1 564.0**	**1 586.7**	**1 670.6**	**1 664.9**	**1 711.4**			
Balance of special Treasury accounts (4)	–0.7	4.6	–3.1	–3.0	–9.2			
Net expenditure of central government (3 + 4)	**1 563.3**	**1 591.3**	**1 667.4**	**1 661.9**	**1 702.2**			

Source: Ministry of the Economy, Finance and Industry.

years – even as, owing to the improvement in the labour market, the number of potential beneficiaries has fallen – because the government considered it important to pursue active programmes targeting certain categories of job seekers (see Chapter III). Furthermore, a new approach to healthcare spending, as recommended in the previous *Survey*, could ultimately help to bring this category under control after the surges observed in 1999 and 2000. Unfortunately, little has been done in this regard (see below). The authorities should also take care that public spending is not increased unduly by new and potentially very costly programmes, like the reduction of work time (see Chapter III).

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Structural fiscal issues

In addition to fine-tuning fiscal policy, French public finance has to contend with several long-term challenges. Tax reform has begun but needs to be taken further in order to encourage the accumulation of human and financial capital. Population ageing will put increased budgetary pressure on retirement pension schemes, but also on spending on healthcare and old age dependency. Finally, a major effort is under way to improve the transparency of public finance.

Taxation: further reform is needed

France has traditionally had a high tax burden relative to other OECD countries, both as concerns average and marginal tax rates (Albert, 2001). In 2000, the general government received resources equivalent to 45.2 per cent of GDP in the form of receipts from taxes and social insurance contributions. With the addition of other public receipts, the general government take from national income amounted to about 50 per cent of GDP, one of the highest levels in the OECD area.[22] Taxes on consumption and labour are based on average rates significantly higher than the European mean, while the average rate of corporate income tax is slightly lower. The top marginal rate of personal income tax is higher than in most other OECD countries because this tax is steeply progressive. Summing up all the various obligatory charges and taxes on personal earned income, the average tax rate of an unmarried employee can be as much as 58 per cent of his earnings. In an environment of pronounced tax competition, the high level of taxes and welfare contributions in France may place French producers at a disadvantage, encourage human capital to move abroad and provoke heavy flows of outward direct investment.

A number of recent reports have helped to reveal the drawbacks of such a tax regime. Bourguignon and Bureau (1999) have shown that prior to the recent reforms designed to reduce inactivity traps, a person receiving basic income support (RMI) was liable to a marginal tax rate of over 100 per cent upon return to employment, once the authorised period for combining income support and earned income had elapsed. Lavenir (1999) considers that the personal income tax system is unattractive to very high earners, because of the combined impact of high social security contributions, a very high top marginal rate of income tax and a stock option regime whose technical features make it largely unworkable. Finally, Charzat (2000) indicates the importance of the tax environment to the development of international groups and suggests that the overall tax burden of such enterprises should be reconsidered.

Conscious of the structural problems associated with the tax system, the authorities have begun a gradual process of modernising it and easing the tax burden. The first relief came in 1993 with the reduction of employers' social security

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contributions for low-wage employees. The movement has accelerated since 1999, at a time when revenues were unexpectedly large and at the same time as many other OECD countries, notably Germany, undertook ambitious tax reforms (Joumard, 2001). The reform in France, under way for some years now, is built around three medium-term goals: *reducing the inequalities* between persons of different income levels, with a focus on tax relief for low-income groups; *supporting employment*, by cutting labour cost and reducing inactivity traps; and *encouraging investment*.[23] A list of the main taxes modified, abolished or created appears in Box 4.

Reducing income inequalities

A major aim of the authorities is to further narrow the income distribution and a number of tax measures have been taken accordingly. Income tax relief has been concentrated in the lower brackets and a tax credit (*prime pour l'emploi*) has been introduced so households below the tax threshold can still benefit from tax relief (see Chapter III). Cuts in indirect taxes, which are largely independent of income level (residence tax, road tax, real estate transfer tax), have also been of proportionally greater benefit to the lowest income groups. On the other hand, indirect taxes, and particularly VAT, are not effective tools for the purpose of income redistribution, since the basic pattern of taxation on households' consumption varies little with income. As a result, it is more costly to achieve a given redistribution target through indirect taxation than by way of direct taxes or the social transfer system.

Encouraging job creation

Various different measures have sought to ease labour taxes. First, the wage share of the *taxe professionnelle* (local tax on businesses), which was equivalent to 2.2 per cent of wage payments in 1995, is being phased out. Its reform significantly reduces the cost of labour and should encourage job creation, particularly among the low-skilled. Second, the cuts in employer social-security contributions for workers earning between 1 and 1.8 times the SMIC also appreciably reduces labour costs and encourages job creation. Third, the authorities have taken steps to "make work pay" by introducing an earned-income tax credit (*prime pour l'emploi*), which supplements the income of low-wage working persons. This measure stimulates labour supply and it may have the indirect effect of moderating wage pressures and thus stimulating labour demand as well. The effect of the VAT cuts, targeted or general, is more uncertain,[24] even if they do help to reduce the tax and social security wedge.

Encouraging investment

Taxation of capital income was increased between 1995 and 1999, largely in the context of achieving the Maastricht convergence criteria: creation of a 10 per

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Box 4. Tax reform

Tax modifications for the period 2000-2003

- *Degressive reduction of personal income tax scales*. The reform aims to lower the marginal rates of income tax between 2000 and 2003. When this adjustment is completed, the bottom rate of tax will have been lowered from 10.5 to 7 per cent and the top rate from 54 to 52.5 per cent. It is also planned to adjust the relief for the poorest households and modify the "family quotient".
- *Reduction of the corporate income tax rate*. For SMEs the rate of this tax will be lowered from 36.67 to 27.5 per cent in 2001, then to 15.45 per cent in 2002 and finally to 15 per cent in 2003. For all companies the 10 per cent increase introduced in 1995 will be phased out, and the rate of tax for large companies will be lowered from 36.7 to 34.33 per cent in 2002, then to 33.33 per cent in 2003. Other company tax measures are: lowering of the threshold for equity holdings of parent companies in their subsidiaries below which dividend payments are not taxed; lowering of the rate of the *avoir fiscal* tax credit that can be used by companies; reduction of declining balance depreciation coefficients.
- *Reduction of* VAT. The standard rate of VAT was cut by 1 point on 1 April 2000; the rate on home improvements was cut from 20.6 per cent to 5.5 per cent.
- *Reduction of oil taxes*. Cut of 20 centimes in the domestic tax on oil products in the fourth quarter of 2000. Suspension of the plan to close the gap between the price of gasoline and the price of diesel oil. Partial refund of oil tax to road haulage contractors and public transport operators.
- "*Employment bonus*". Creation of a tax credit for persons gainfully employed. The bonus maximum applies to persons earning the SMIC minimum wage, with a declining rate applying to persons earning up to 1.4 times the SMIC. The bonus takes account of the number of hours worked and the family pattern of the taxpaying household.
- *Other measures*. Reduction of the *taxe d'habitation* residence tax. Exclusion of wages from the base of the *taxe professionnelle* business tax. Reduction of the real estate transfer tax. Extension of the declining-rate rebate on employer social security contributions to wage payments of up to 1.8 times the SMIC.

Taxes abolished in 2000 and 2001

- Gradual phase-out of the exceptional surtax on corporate income tax.
- Phase-out of the contribution representing the real-estate lease tax.
- Abolition of the annual flat-rate tax for very small businesses.
- Abolition of fifty taxes (duties on ID cards for foreign minors and special cards for foreigners practising certain occupations, registration fees for the *baccalauréat* and *brevet* school examinations, and various fiscal taxes, notably on credit cards).

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Box 4. **Tax reform** (*cont.*)

- Abolition of the regional quotient of the *taxe d'habitation* and readjustment of the relevant tax relief arrangements.
- Abolition of parafiscal levies.
- Abolition of the road tax (*vignette*) on private cars and certain light commercial vehicles owned by indivudals and non-profit bodies.

Taxes created in 2000 and 2001

- Introduction of a social security contribution from profits.
- Introduction of a tax to finance new roads and utility networks.
- Introduction of a tax on the proceeds of sale of sports broadcasting rights.
- Introduction of a State solidarity tax on water agencies.
- Introduction of a special tax on reserves for a rise in oil company prices.
- Introduction of a tax on CDs and DVDs.

Source: Ministry of the Economy, Finance and Industry.

cent surtax on corporate profits in 1995 and another surtax of 15 per cent in 1997, and introduction of a social contribution from profits to finance the measures accommodating the reduction of work hours. This movement was reversed as from 1997 with successive cuts in the surtaxes. The rate of profit tax will have been reduced from 41.7 per cent to 33.3 per cent between 1998 and 2003 (34.4 per cent allowing for the social contribution from profits of the businesses concerned). These cuts have been partly financed by a broadening of the tax base, with the result that the average rate of taxation has not fallen, even though the marginal rate has declined to nearly its 1995 level. An additional effort has been made to help small and medium-sized enterprises, as of 2001 part of their profits are taxed at a 25 per cent rate and this will be lowered to 15 per cent in 2002. In all, the statutory corporate income tax rate is very close to the OECD median, as is the effective rate. However, with the recent measures being taken in some countries, Germany in particular, France may, once again, find itself with a rate above the median.

Overall, the process of tax reform is now underway in France. As recommended in the previous *Survey*, the authorities have designed their fiscal strategy in a multiyear framework, which is more transparent to economic agents than annual programmes and enables them to adjust their expectations better.

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Although the actual amounts of the cuts in direct taxation are comparatively modest as yet – they represented 1 percentage point of GDP in 2000, 0.6 point in 2001 and should amount to 0.5 point in 2002 – they are still a step in the right direction. Pursuit and amplification of the reform will be necessary in the future if France is to become more attractive to business investors and highly skilled personnel, which a number of official reports suggest is not the case today. The general lowering of average tax rates should be continued, taking care to avoid initiatives that target certain occupations – like professional soccer players and performing artists – which only penalise other taxpayers. It is an all-round easing of average tax rates that should be targeted, not category-specific measures. Income declaration and tax collection arrangements also need to be simplified in order to reduce the costs associated with the filing of returns and their verification by the tax authorities. The abolition of the road tax eliminated a heavy cost for users, although its environmental impact may be questioned (Chapter IV). The television licence fee, which is particularly costly to collect, should be abolished. Deduction of income tax at the source would help to make filing procedures simpler and reduce tax collection costs.

Ageing: an inevitable impact

The French population will age considerably over the period 2000-2040, as in many other OECD countries. Three factors will drive this trend: *i*) post-war baby-boomers will reach old age; *ii*) old persons will live longer, thanks to medical progress; and *iii*) the number of young persons will remain flat owing to the trend decline in fertility rates. According to the demographic projections of INSEE (2001), the population aged over 60 will be about 20 million in 2050, twice today's figure. However, the population of working age will remain stable after 2005. In 50 years' time, therefore, France will have eight persons aged over 60 for every ten persons of working age, whereas today the ratio is only four to ten.

The ageing of France's population will occasion steep increases in public spending on pensions and, to a lesser extent, health and care of dependent persons. The demographic shock will begin in 2005, when persons born in 1945 turn sixty, the statutory age of retirement. According to generally accepted projections, pension expenditure will increase from 11.6 per cent of GDP at present to between 15.7 and 16 per cent of GDP in 2040.[25] To finance this increased expenditure, in the absence of reform, a steep rise in contribution rates would be necessary, with a marked increase in labour costs as a result. Recent studies made by Charpin (1999), the OECD (2001*b*) and the Economic Policy Commission of the European Union (2001) converge on this diagnosis. The demographic shock is inescapable and a reform of France's various pension regimes, further to the measures already taken in 1993, is necessary in order to minimise its impact on the economy.

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The French pension system

France's pay-as-you-go pension (PAYG) system was introduced after World War II and its broad principles have remained largely unchanged to this day. The basic principle is solidarity, between generations (working persons pay for retired persons), between occupational categories (pension schemes in surplus subsidise schemes in deficit) and between social categories (the rich pay for the poor). Now covering the entire labour force, the French system is still marked by the diversity of the regimes that constitute it, each with its own operational approach and characteristics: the schemes for employees in the public sector, the schemes for private sector employees and the schemes for self-employed persons (Davanne, 2000). The pensions of public servants are funded from taxes and are designed as an extension of the pay received over the course of a career, a sort of deferred wage. Private-sector pensions are financed by a general scheme and by supplementary schemes requiring wage-based contributions. Finally, the pensions of the self-employed offer a fairly low earnings replacement ratio, the capital of the enterprise being considered the best source of income when retirement age is reached.

The authorities take the view that the French PAYG system has been a success.[26] It has indeed provided a growing number of pensioners with comfortable incomes, narrowed the purchasing power gap between working and retired persons and reduced poverty among the elderly, as evidenced by the 50 per cent decrease in the number of recipients of the minimum old-age pension over the last thirty years. The ease with which this improvement in pensioners' living standards has taken place is essentially attributable to favourable trends in real wages and the labour force over the past few decades (Merrill Lynch, 2000). Thanks to the strong growth of wages, on which the pension contributions are based, the funds available for each pensioner doubled in real terms between 1960 and 1995. This made it possible to raise the purchasing power of pensioners and to lower the statutory retirement age from 65 to 60. However, it is unlikely that such a favourable combination of demographic and economic factors will be repeated in the future. In particular, population ageing will doubtless cause the dependency ratio to rise, putting the system's sustainability over the next forty years in doubt.

Reforms under way

The authorities realise that the pension system needs to be reformed. The Pensions White Paper produced in 1991 at the request of, then Prime Minister, Michel Rocard established a first diagnosis of the medium-term risks. Several changes recommended in this report were introduced by the 1993 Balladur reform of the basic regime in the private sector. Pensions were indexed to prices *de jure* instead of wages (which had been the case *de facto* since 1987) and calculated on the basis of pay over 25 years instead of 10 years as previously. Moreover, the

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minimum period of contribution required for retirement on full pension at age 60 was lengthened from 37½ to 40 years.[27] The private-sector supplementary schemes have taken similar steps on a temporary basis (three-year period, subsequently renewed). While these moves will greatly reduce the borrowing requirements of the private-sector regime, they imply a decrease in the living standards of private-sector pensioners in relative terms, especially if further adjustments are made to the supplementary schemes. It has been estimated that the replacement ratio (ratio of pension net of social security contributions to the last net wage) could fall from about 80 per cent of the net wage at present to less than 60 per cent in 2040 (Davanne, 2000). At the same time the private system's deficit will be much reduced.

In contrast, no steps have been taken to reform the public-sector pension schemes. As a result, about 45 per cent of future borrowing requirements will come from these schemes, even though public employment accounts for only 25 per cent of total employment. Furthermore, the present situation is creating an inequality of treatment between the private and public sectors. Retirees from the civil service continue to draw pensions indexed to salaries and not to prices, and the amount of the public-sector pension is based on pay received over the last six months of a career and not over the last 25 years. Moreover, the period of contribution required for retirement on full pension is still 37½ years as opposed to 40 in the private sector. On the other hand, private-sector pensions are based on total pay, whereas public-sector pensions are computed on basic pay excluding bonuses, which in some cases represent a sizeable share of public servants' earnings. However, civil servants, covered by the public-sector's special schemes, continue to enjoy particularly advantageous treatment. A first attempt to reform the public-sector schemes (the Juppé Reform) in 1995 had to be withdrawn because of fierce opposition from those concerned.

In March 2000 the authorities decided to open discussions on the measures needed to maintain financial equilibrium of the pension schemes, including those of the public sector, until 2020. The thinking in these discussions is founded on five principles: *i*) consensus-based framing and progressive implementation of measures; *ii*) regard for the diversity and identity of the different pension regimes; *iii*) greater equity and increased solidarity among the pension regimes, with the possibility of progressively aligning the contribution period for public servants with that for private sector employees; *iv*) the introduction of more flexibility so as better to respect individual choice; and *v*) the anticipation of demographic developments through the creation of a reserve fund to smooth inter-generational charges. One of the assignments of the new *Conseil d'Orientation des Retraites*, which brings together the social partners, members of parliament, representatives of central government and independent experts, is to analyse the technical reforms needed to ensure sustainability of the pension regimes (lengthening of contribution periods, revision of pension calculation methods, actuarially neutral formulas

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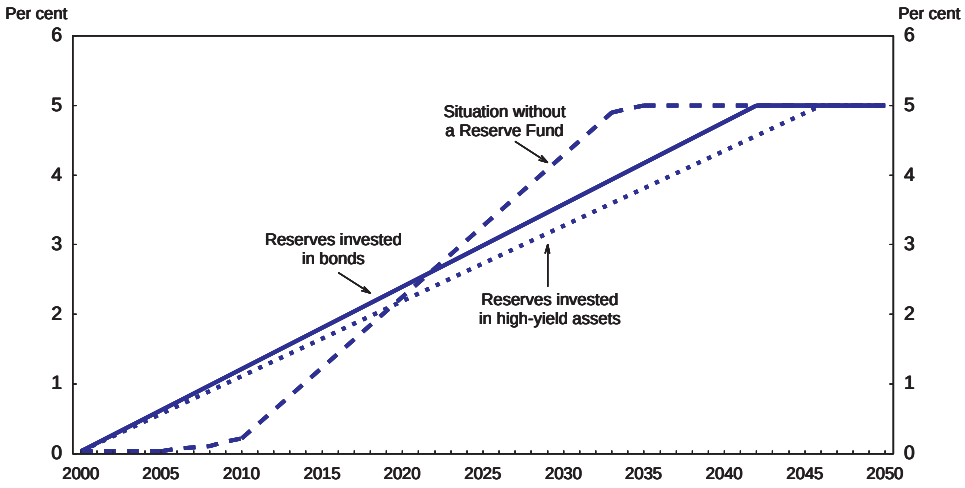
for computing pension entitlements, etc.). Another of its tasks is to bring forth greater social consensus as regards equity of contribution efforts as between schemes and notably between the public and private sectors.

"PAYG with reserves" and personal saving

One substantial measure taken by the authorities was the creation of the *Fonds de Réserve pour les Retraites* (Pensions Reserve Fund) by the social security financing law of 1999, which was followed by an implementing decree in 2001. The basic principle of the Fund is *"répartition provisionnée"* or PAYG with reserves (Davanne, 2000). The idea is not to set up a fully-funded system, but to create a reserve using social security surpluses and other *ad hoc* resources until 2020 so as to reduce the need to pension contribution rates between 2020 and 2040 (Figure 14). The Fund's purpose is to accumulate over the next 20 years reserves from various sources that will be invested. The target is to build up reserves of FF 1 000 billion (constant francs, base 2000) by 2020, then to use those reserves until 2040 in order to smooth the adjustments needed to ensure sustainability of the PAYG compulsory old age insurance schemes covered by the reform. The Fund's resources will come chiefly from surpluses on the various social security funds, grants from public corporations in a surplus position, allocations of tax reve-

Figure 14. **The Pension Reserve Fund**

Retirement pension contribution rates



Source: Direction de la Prévision.

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nue and part of the proceeds from sales of UMTS licences. However, the steep reduction of the price of UMTS licences announced in October 2001 will deprive the Fund of most of those resources. To make up some of this loss, the government has decided to allocate certain privatisation receipts to the Fund, notably those deriving from the sell-off of part of the equity of *Autoroutes du Sud de la France*. As was recommended in the previous *Survey*, the government should continue to allocate privatisation receipts to the Fund, rather than using them to subsidise enterprises still in the public sector. Until the end of 2001 the Fund is functioning without a supervisory authority to oversee its operations and is therefore investing its reserves in essentially monetary assets. Subsequently, as of 1 January 2002, the Fund will be institutionalised as an independent administrative public corporation. The bodies needed to manage the Fund will then be appointed so that it can introduce active management that optimises investment duration and risk, in conditions of transparency and in consultation with the social partners.

The Pensions Reserve Fund has been designed as a means of smoothing unavoidable developments in the PAYG system. In itself it does not resolve the long-term problems, but it does make it possible to avoid sharp swings in contribution rates, replacement ratios, retirement age and borrowing requirements. However, in the medium term the Fund is not macroeconomically neutral. Payment into the Fund of resources independent of the pension schemes (dividends, tax revenue, etc.), resources which would otherwise have been used for another purpose, are, in fact, fiscally equivalent to pension contributions. In addition to its smoothing function, the Fund is thus a transitional source of additional collective financing for pensions over the long term.

Another tangible measure taken recently has been the encouragement of personal saving, particularly by employees. In principle, in a PAYG system, demography should have a neutral influence on the household saving ratio, since each working person is theoretically guaranteed an unchanged standard of living upon retirement. In practice, uncertainties about pension financing, and hence about future replacement ratios, may lead individuals to build up precautionary savings in preparation for their retirement. In fact, households' wealth has greatly increased over the past 20 years or so, representing more than five times gross disposable income in 1997. This growth of personal saving is partly due to demographic factors (CDC, 2001). Life insurance contracts offering a lump sum or an annuity have been much used in this regard and accounted for about 30 per cent of households' financial wealth in 1997.

Other forms of individual or group saving have developed in recent years with the help of tax exemptions. The *Plans d'Épargne d'Entreprise* (PEE) are employer-managed saving schemes that enable employees to build up portfolios of securities. Employees can pay into the scheme the sums obtained from incentives and profit sharing, as well as voluntary contributions that employers may

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supplement. These contributions, which are exempt from tax and social security charges, are invested chiefly in shares and mutual funds. About 26 per cent of all employees have access to these schemes (Jallet and Franceschi, 2001). More recently, the *Plans Partenariaux d'Épargne Salariale Volontaire* (PPESV) and the *Plans d'Épargne Inter-Entreprises* (PEI) were introduced in order to allow more employees to save in this manner, especially those in small and medium-sized enterprises. The new plans will allow large contributions to be exempted from social security charges and income tax and, like the *Plans d'Épargne d'Entreprise*, will entitle employers to a generous reduction of corporate-income tax. Contributions, which may not be withdrawn from the saving scheme for a period of ten years (other than in special circumstances), will be invested in mutual funds or shares. These saving schemes, which are akin to firms' pension plans, will therefore play an important role and encourage employee share ownership.

What should be done?

The ongoing reforms summarised above are useful, but they will need to be supplemented if present trends are to be offset and in order to meet the challenge posed by the financial impact of population ageing. In the absence of more significant reforms, the borrowing requirements of the different pension schemes – and probably of the health insurance funds too – will be such that the level of public debt and debt service will soon become unsustainable. Moreover, expenditure on care of dependent persons may also increase. As in all countries facing a similar situation, the relative level of the pension benefits will have to diminish and contribution periods will likely have to be lengthened.

The are a number of ways in which the extent of adjustments could be limited: *a*) growth of activity could be increased if economic policy and structural reforms are to permit a return to full employment; *b*) a significant upturn in immigration could also support growth while temporarily improving dependency ratios; *c*) pension contribution rates could be raised, provided this solution does not affect labour cost and employees are prepared to forgo some of their take-home pay; *d*) public finance could be readied for the coming shock if a general government primary surplus were achieved sufficient to reduce the public debt steeply as of now; and *e*) changes in individual incentives could be introduced so as to raise the employment rate and improve the ratio of contributors to recipients in the various PAYG pension schemes. Simulations made by the OECD (2001*b*) for a "stylised" country show that relatively modest changes in certain parameters could suffice,[28] if put in hand rapidly. The longer they are delayed, the greater the magnitude of the efforts required.

Raising employment rates and hence the number of contributors is probably the best solution in terms of macroeconomic efficiency, since one of its feedback effects is a temporary increase in growth potential (Visco, 2001). Delaying retirement

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seems possible today because of increased longevity and is necessary given the mounting demographic imbalance between working and non-working persons. However, statutorily raising the age of retirement is not a good solution. International experience shows that such administrative measures often lead to an increase in disability and unemployment benefits among elderly workers. It is therefore essential to determine what are the microeconomic factors that might induce an employee to postpone retirement. Recent studies have revealed a strong correlation between retirement decisions and the incentives provided by social security systems, notably pension benefit scales (Blöndal and Scarpetta, 1997). To stimulate labour supply in older age cohorts, the scales should be adjusted so that the trade-off between income and leisure favours income. Postponing retirement means sacrificing leisure for additional income now (wages are higher than pensions) and in the future (the additional period of work increases pension entitlements). The present French system discourages later retirement, because the additional period of working gives no entitlement to a pension supplement in the public sector, and entitlement to only small supplements in the private sector.

One important path of reform would be to calculate individual pensions in an actuarially neutral manner. Simulations made by INSEE (2001) with individual data suggest that upward adjustment of the pension supplements obtained when the contribution period is longer than the statutory minimum would significantly raise the average age of retirement. This work is still at a preliminary stage, since it is necessary to know the composition of the periods of activity and the preference of individuals for leisure. Even so, this is a promising avenue of research and experimentation. Moves in this direction should be made first of all in the public sector, since its pension schemes do not offer any supplements when individuals accumulate more than the number of contribution years required for retirement on a full pension. Moreover, this contribution period should also be lengthened, because at 37½ years it is 2½ years shorter than in the private sector. According to certain studies made by the new *Conseil d'Orientation des Retraites*, aligning the contribution period for the public sector system and the special schemes with the contribution period for the private sector would cut borrowing requirements by one-fourth in 2020 and one-third in 2040.[29]

Early retirement is one reason for the low employment rates among the older workers. Various state-funded pre-retirement schemes have affected end-of-career trajectories by encouraging early departures. Immediately after the first oil shock, rising unemployment and the financial difficulties of the UNEDIC unemployment insurance fund led the authorities to set up early retirement schemes for the 55 to 59 age group. Private sector schemes include the *Allocation de Remplacement pour l'Emploi* (ARPE) and the *Allocation Chômeurs Âgés* (ACA), which are financed by UNEDIC, and the *Allocation Spéciale du Fonds National pour l'Emploi* (ASFNE). In addition, there are two schemes for public servants: the *Cessation Progressive d'Activité* (CPA) and the *Congé de Fin d'Activité* (CFA). The generosity of these different

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schemes has contributed to a steep fall in participation rates among 55-59 year-olds. For the generations born after 1922, cessation of activity precedes payment of the old age pension by over two years on average (INSEE, 2001). A cutback of early retirement schemes and the introduction of free-choice retirement schemes based on actuarially neutral calculations, which would penalise early retirement and encourage postponed retirement, would therefore be desirable. The authorities have already limited participation in ASFNE to enterprises with redundancy plans and raised their financial contribution to this scheme. A general reduction of early retirement would naturally necessitate a policy aiming to maintain the employability of older workers and notably a different approach to training that would focus on skill renewal over the course of working life.

Health: *how to contain expenditure, while maintaining the quality of care?*

The special chapter of the previous *Survey* dealt with France's healthcare system. It concluded that the system offers many advantages. Medical outcomes are good and the French have average life expectancies a little above the OECD median. Patients may choose freely their doctor, hospital or clinic. Waiting lists are rare and patients are generally treated rapidly, since healthcare supply tends to exceed demand, admittedly with some disparities between regions. Finally, the system is founded on solidarity – between rich and poor, young and old, healthy and sick – which allows all to be treated regardless of income level.

On the other hand, the system is relatively expensive. Nearly 10 per cent of GDP is allocated to health expenditure, this being about 1½ percentage points above the EU median, for medical outcomes that are hardly any higher (Figure 15). Freedom of choice and care quality have a high cost for society. This cost is financed by high taxes on wage income, which are liable to discourage labour supply. The rapid growth of health expenditure is compromising the situation of the employees' general health insurance scheme (CNAMTS), which continued to show a deficit in 2000 despite extremely favourable cyclical conditions for receipts. It is therefore likely that the economic slowdown will lead to a widening of the health insurance deficit in the near future. In the medium and long terms, population ageing will cause a further rise in health expenditure and a stagnation of receipts.[30] If no reform is undertaken, a significant increase in the borrowing requirement of the state health insurance system is therefore likely.

The authorities seek to control health expenditure without undermining the fundamental principles of the French system, namely quality of service, freedom of choice for healthcare consumers and solidarity between different segments of society. On the basis of those principles, the special chapter of the previous *Survey* put forward a number of recommendations for microeconomic incentives to change the behaviour of healthcare providers and consumers. A summary of these recommendations and the record of their implementation to date can be found in Annex III.

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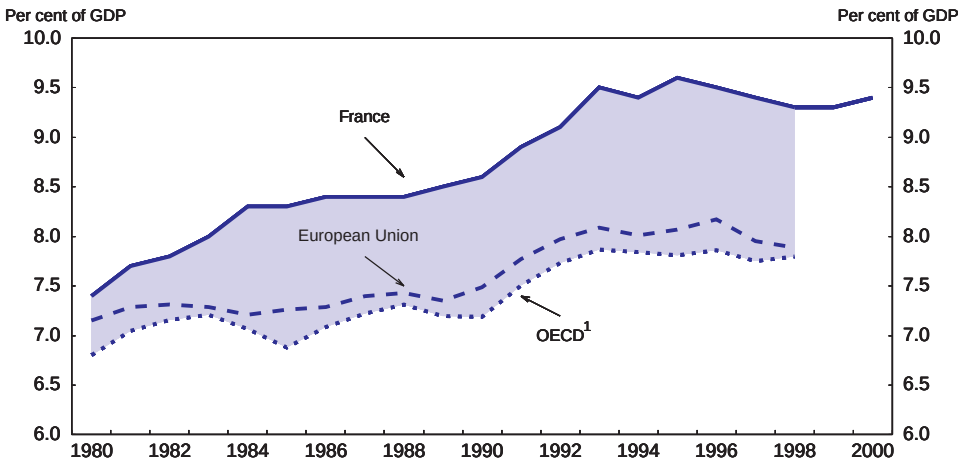
Figure 15. **Total health expenditure**



1. Simple average of OECD countries, excluding Korea before 1985, Mexico and Poland before 1990, Hungary before 1991 and Slovakia before 1996.

Source: OECD Health Data 2001.

On the whole, not much has been done to curb health spending, despite a conspicuous overrun of the expenditure ceilings set by the social security financing law. In the public hospitals sector, a system of payment on the basis of diagnosis-related groups (DRG) has yet to be brought into general use, although this system has been under study and tested for many years. By the same token there are no longer plans to strengthen the role of the regional hospitalisation agencies to the point where they can act as purchasers of healthcare. The labour unrest in hospitals in the early part of 2000 has thwarted efforts to tighten budgets and undone some of the progress made in reducing funding inequalities between regions (De Kervasdoué, 2000). Also, the transition to the 35-hour workweek is likely to result in higher hospital staffing expenditure.

As regards independent practice of medicine, the changes introduced are still modest. The idea of general practitioners screening access to certain over-used categories of specialists has not been implemented. Similarly, a plan to discontinue the system of private patient treatment and private beds in public hospitals, which creates social inequalities in access to care, has not yet been introduced. Nevertheless, independent practice of hospital doctors has been rationalised (payment of fees through the hospital and charge paid by practitioners for use of hospital equipment).

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More significant measures have been taken in the pharmaceuticals sector, where refund rates have been lowered for certain medicines judged to be ineffective. But most such medicines continue to be refundable, although at a reduced rate. Furthermore, the system in operation is still based on price control – pharmaceuticals are one of the few product categories in which prices are still state-controlled – and this is what causes the French, through the play of price elasticities, to be among the heaviest consumers of medicines in the OECD area. Despite the first few measures taken to promote their use, the presence of generic drugs is still minimal, although international experience shows that generics are a proven mechanism for controlling expenditures.

Transparency: the way has been opened up by a reform of the "Organic Law"

As noted in the previous *Survey*, major efforts are being made to improve the transparency of public finance and modernise budgetary procedures. The purpose of this reform is to make the budget process, including the budget review by Parliament, an occasion for defining the government's policies and allocating the resources necessary to implement them. The ultimate intention is that budget discussions should focus on policy aims and not just on the amounts to be allocated. Efforts are being concentrated on the following:

– *Transparency of government expenditure*. The traditional budget statement is concerned with the legal aspect of government expenditure, that is to say with the amount of resources placed at the disposal of ministries. However, this approach cannot be used directly to evaluate the content of government action. Since last year an additional statement has been prepared in order to explain what activities are being financed by each ministry, what the goals are and what results have been obtained to date. For this purpose the 850 existing budget chapters have been grouped into 118 budget aggregates that are calculated each year. Each aggregate relates to a government policy, like a public service provided by the State (education, police, justice, etc.) or an intervention policy (employment, housing, etc.). For each aggregate there is a description of the effectiveness and quality of the public action concerned and the improvements envisaged. Also supplied are indicators of results to date, which permit evaluation and *ex post* monitoring of government action. The aggregate relating to justice, for example, is accompanied by a description of the speed of response to breaches of the law and targets for improvements are put forward. For practical reasons the aggregates still tend to focus on the functioning of each ministry rather than on the policies themselves. But this is only the first stage in a complex process that will have to undergo improvements in the future.

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– *Overhaul of budgetary procedure*. To be really meaningful, the linkages between government expenditure, performance goals and results necessitates a change in budget approval procedures. Parliament has adopted an organic law amending the 1959 legislation on budgetary procedure. The new procedure formalises the new budget statement and approach to expenditure. When it enters into effect in 2006, the 850 budget chapters used to specify appropriations will be abandoned in favour of 100 to 150 aggregates called programmes, in turn included in "missions", some of which may be cross-ministry assignments. To obtain resources, ministries will have to show that these are going to be put to good use. They will be required to report on their observance of targets in an "annual performance report" that will be discussed by Parliament. On the other hand, they will enjoy greater flexibility in resource utilisation. They will be able to redeploy their appropriations without being strait-jacketed by the budget nomenclature. Fungibility of appropriations will be limited, however, in order to avoid slippages in staffing expenditure. The reform will also re-budget extra-budgetary items like parafiscal taxes, State guarantees, special treasury transactions and debt issue.

– *Other budget simplifications*. Other measures, on a smaller scale but important to transparency and simplification, have been taken. The number of parafiscal taxes – taxes from which the revenue is directly allocated to public, semi-public and private agencies – has been further reduced. There were 77 parafiscal taxes in 1981, now there are only 43 and receipts in this category have fallen steeply. By and large, when these taxes have been abolished, the corresponding expenditure has been brought back into the budget. The organic law abolishes the whole system of parafiscal taxes, which are comparatively costly to collect and benefit agencies and institutes not subject to standard budgetary control.[31] On the other hand, little progress has been made in reducing the number of *fonds de concours*, funds for discretionary use by ministries outside the standard procedures applying to non-appropriation receipts, although the sums involved were reduced from FF 69 billion in 1997 to FF 39 billion in 2000.[32] Nevertheless, Parliament's examination of these funds is a move towards transparency. Under the terms of the organic law, the value of *fonds de concours* will have to be entered in the initial finance act, which will also help to increase transparency. A similar effort needs to be made with the special treasury accounts,[33] which numbered 37 in 2000 (of which 12 special appropriation accounts). For example, the appropriation account for the proceeds from sale of corporate securities, equity stock and rights makes it possible, without prior approval by Parliament, to allocate privatisation receipts to public enterprises in the form of capital grants, *i.e.* government subsidies. Thus in 2000 *Réseau Ferré de France* (French Railway Tracks) received

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an endowment of FF 7 billion from the proceeds of privatisation. More generally, it would be helpful if the many and varied financial dealings of the State with public corporations could be aggregated for budgetary purposes, so as to provide a gauge of the nation's financial effort in supporting these establishments. Finally, the development of accrual-basis accounting is continuing with the annual presentation of the *Comptes de l'État*. These accounts chart the financial position of the State as if it were a business. In 2000 a first valuation of the State's liabilities in respect of public service pensions was also published by the Court of Audit.

– *Statistical information on general government*. Substantial progress has been made in circulating statistical information on the budgets of the general government. The situation of the central government budget is described in a monthly note issued by the Ministry of the Economy, Finance and Industry, with a breakdown of the main receipts, expenditures and financial operations. Subannual information on local government and social security agencies is more fragmentary, however. Some information on general government revenue appears in the transaction accounts of the quarterly national accounts, but the trend of expenditure (other than general government consumption) is not described. As stated in the previous *Survey*, the production of quarterly national accounts for general government would increase transparency and be helpful for the guidance of economic policy. In fact, France has decided, along with the other EU countries pursuant to a new Eurostat regulation, to introduce quarterly government accounts by 2005.

– *Information on public service employment*. The need for data on the numbers, pay and skills of public servants has been stressed in different reports by the Court of Audit and economic institutes (for example, see Rexecode, 2000). Not enough is known about staff numbers, working time and absenteeism, making it impossible to gauge the size of the workforce actually available. As one minister said recently, "*today it is impossible for ministries to know what posts they really have at their disposal*".[34] This difficulty is partly attributable to the different practices of government managers. In some cases, posts that are approved remain unfilled. In others, posts are filled but not entered in the budget. A number of initiatives have been taken in an endeavour to remedy these shortcomings. In July 2000 the *Observatoire de l'Emploi Public*, an agency that monitors government employment, was set up, and more information is now being supplied to Parliament.[35] INSEE conducts a survey of government employment, but the results are published with a long time lag.

In addition, the government instituted a major reform of the rules governing public procurement, which have in the past been criticised as being excessively complex. The reform seeks to simplify procedures, clarify rules and

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harmonise the law with that of the European Community. In so doing, the number of articles in the code was reduced by two thirds. Moreover, payment periods and penalties for late payment charges were codified. The reform should increase the transparency by which public contracts are awarded and permit small and medium enterprises to compete for these. Indeed, in the past the complexity of the process represented a significant barrier to such firms.

Fiscal consolidation should be continued

All told, the situation of French public finance has improved in the recent past, but the consolidation effort still falls short of the progress made in most other OECD countries. Even though tax revenue has shown a record increase, the general government account is still in deficit, whereas many countries now have a balanced budget or a budget surplus. It is therefore regrettable that the structural deficit was not reduced between 1999 and 2001. The relatively high level of government debt and the coming challenges posed by population ageing make it essential to restore budgetary equilibrium rapidly, as the authorities plan to do in their multiyear public finance programme. The present strategy is founded, rightly, on control of public expenditure. However, it would probably be wise to tighten the norms for spending growth and ensure that they are complied with by means of appropriate reforms, notably with respect to healthcare and management of the civil service.

On the structural level, important reforms have been initiated or announced. The authorities have made a start on reducing taxation, which is heavy in France, but marginal and average rates are still high for most taxpayers. With the reform of the organic ordinance of 1959 concerning the finance acts, a major effort has been undertaken to increase budget transparency and the effectiveness of public spending choices, but the process will take a long time to complete. On the other hand, reform of the pensions system, especially in the public sector, remains a priority and should be set about quickly.

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III. Structural reforms to raise potential growth

The strains that have appeared after just a few years of growth underscore the weaknesses in the supply potential of the French economy. This weakness is partly the result of the "Malthusian" policies to encourage labour force withdrawals pursued since the oil shocks of the 1970s, which have reduced the contribution of labour to productive capacity (Olivennes, 2000). It also stems from the sectoral policies implemented over the same period to protect the major national monopolies against competition, thereby discouraging productive investment and innovation in the private sector. In the past few years, the direction taken by employment policy has changed. The authorities are now building a new incentive dimension into their measures regarding labour supply, the object being to correct the effects of the past. Nevertheless, contractual relations between employers and employees remain, as they have traditionally been, tightly regulated by the State. As concerns sectoral policies, the major government monopolies in the network industries have begun to be exposed to competition but obstacles to complete liberalisation still remain. This chapter discusses these two questions successively. The main structural reforms that France has introduced are summarised in Table 5, together with the corresponding OECD recommendations.

Towards full employment?

The debate about the return to full employment has gained fresh impetus in the last few years. Several recent studies have made it possible to track certain labour market malfunctions more effectively (*e.g.* Salanié, 2000). Moreover, many empirical studies using individual data and "natural experiments" have provided new and important sources of information, helping to understand microeconomic decisions concerning labour supply and demand in France (*e.g.* Abowd, Kramarz and Margolis, 1999; Crépon and Desplatz, 2001; Laroque and Salanié, 1999 and 2000; Piketty, 1997 and 1998; Gurgand and Margolis, 2001). Thanks to this work, the nature of under-employment in France is now better understood. It is now accepted that unemployment in France today is largely structural in origin. As was observed in the previous S*urvey*, available studies suggest that, even if it has fallen slightly in recent years, the NAIRU is currently between 8½ per cent and 9½ per cent – implying that

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Table 5. **Overview of the progress in structural reform since 2000**

Previous recommendations	Measures since the last *Survey*	OECD comments
I. Labour market		
Reduce wage costs for unskilled labour		
Reassess the level of the SMIC minimum wage. Index it on consumer prices instead of wages.	None.	Revise the method of indexation. (See below, the adjustment of the SMIC to the 35-hour week.)
Reform the unemployment benefit system		
Reduce the risk of unemployment and poverty traps, particularly for those receiving basic income support.	Authorisation to combine temporarily basic income support with income from employment. Property tax, housing benefit and income tax have been reformed. Unemployment benefit is no longer degressive for participants in the PARE (*Plan d'aide au retour à l'emploi*) but support for job-seekers has been strengthened. The employment subsidy provides additional income on top of the SMIC and could ease pressure on the latter.	These measures send out signals that give little incentive to find employment. Reintroduce a gradual decrease in unemployment benefit or introduce strict conditionality in the PARE. The employment subsidy should encourage part-time work, which is a transition from unemployment to employment and could be more targeted for other beneficiaries.
Look at the conditions for the re-employment of young people and adults on the fringes of the labour market.	The youth employment (emplois-jeunes) programme has been extended for a period of 5 years. Launch of the "Fresh start" programme and the PARE/PAP: activation of unemployment compensation expenditure, personalised follow-up of job-seekers, increased access to training.	Give priority to measures that increase the employability of the jobless in the market sector.
Continue to scale back early retirement schemes, which deprive the economy of experienced workers. Make the definition of "arduous" jobs more precise.	The share of public support to early retirement pensions has been scaled down.	Continue the reform of early retirement.
Simplify the administrative regulation of the labour market		
Ease the restrictions on the renewal of fixed-term contracts. Introduce 5-year fixed-term contracts.	The social modernisation bill has reinforced redundancy procedures: a greater role for staff representatives, and redundancy procedures made longer.	Reduce the disparity of the conditions of fixed and indefinite-term contracts.
Enhance active labour market policies		
Evaluate active labour market policies on an ongoing basis.	The employability of beneficiaries of the programmes (the youth employment scheme in particular) is beginning to be better understood.	Pursue efforts to make results transparent.

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Table 5. **Overview of the progress in structural reform since 2000** (*cont.*)

Previous recommendations	Measures since the last *Survey*	OECD comments
Make the shift to the 35-hour week more flexible		
The application of the 35-hour work week to SMEs needs to be made flexible.	Exceptional overtime arrangements are planned for 2002-2003 (when SMEs move to the 35-hour week), with a lower premium and an extension of the overtime quota.	Keep under constant surveillance the circumstances of the move to the 35-hour work week for the 4.5 million employees concerned. Clarify the sources of financing for aid measures and curb their extension.
The impetus given to the hourly SMIC in the context of the 35-hour work week has to be curbed in order not to increase the cost of unskilled labour.	A complex system for adjusting the hourly SMIC to the 35-hour week is liable to impact on the SMIC in 2005. This question is being discussed with the social partners.	Amend the law to prevent an impact on the SMIC in 2005.
II. Product market competition		
Enhance competition in network industries.	The air transport sector was opened to competition in 1997 and telecommunications in 1998. The electricity sector was partially and minimally opened up as of August 2000. "Unofficial" and minimal opening of the natural gas sector occurred in August 2000. Rail transport and postal services remain closed to competition.	Extend the scope for competition in the electricity sector to all consumers; adopt the European directive on natural gas; open up rail transport and postal services to competition.
Apply regulations that protect consumers and ensure supply security in a pro-competitive manner.	Public service targets in telecommunications and electricity have been defined in detail and costed so as to ensure funding that is compatible with competition.	Apply this approach to opening up all network industries to competition.
Continue restructuring network industries.	Air France and France Télécom have progressed towards business enterprise status by associating their employees with the change and making them shareholders in the companies.	Mobilising employees' support of reform by having them share in the benefits of market opening should be an objective in all network industries.
Increase the legal and material resources available to the competition authorities.	The Competition Council has handed down numerous opinions in many areas, but cannot on its own initiative inquire into anti-competitive practices.	Give the Competition Council the means to take up anti-competitive practices on its own initiative. Extend these prerogatives to areas where competition is restricted because of "public service delegations".

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Table 5. **Overview of the progress in structural reform since 2000** (*cont.*)

Previous recommendations	Measures since the last *Survey*	OECD comments
III. Taxation		
Reduce taxation for people with middle and high incomes.	The marginal rate of income tax has been lowered.	Marginal tax rates are still excessive (as high as 70 per cent) for certain categories of income. The authorities should therefore continue and accelerate the pace at which they are lowered.
Ease the burden of tax and social insurance contributions.	The rate of corporation tax has been reduced, but remains above the average for the other European countries.	Bring corporate income tax down towards the European average.
Modernise tax collection.	More than sixty, costly to collect, taxes have been abolished since 1997, in particular numerous stamp duties and the road tax.	The cost of collecting certain quasi-taxes is excessive in relation to the proceeds (*e.g.* the television licence fee). Do away with them and abolish their allocation.
IV. Pensions		
Restore the sustainability of long-term pension schemes by introducing forms of collective capitalisation or employee savings schemes.	A reserve fund has been set up an instrument for smoothing the chocs in the pay-as-you-go scheme. It is to have FF 1000 billion in capital by 2020.	Increase collective savings to fund future pensions expenditure. Add to these savings all surpluses derived from public finances.
Promote actuarially neutral methods of pension calculation.	No concrete measure. Working beyond the statutory retirement age and retiring before are penalised.	The statutory age is still a constraint and distorts incentives. The authorities should therefore make pensions actuarially neutral; allow wage -earners to choose for themselves when they retire and what their pension income will be.
Ensure greater equity between public and private pension schemes.	The contribution period remains shorter in the public than in the private sector.	Extend the contribution period in the public sector.
V. Health reform		
Detailed recommendations concerning the reform of the health system were made in the previous Survey, and are difficult to summarise in this table. Annex III outlines the measures taken by the French Government in response to those recommendations and the OECD's comments.		

Source: OECD.

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cyclical unemployment has now, to a large extent, been eliminated. In these circumstances, a further decline in unemployment would require a combination of favourable structural policies and appropriate macroeconomic policies.

A return to full employment – often defined as an unemployment rate of 5 per cent and participation rates higher than present levels (Pisani Ferry, 2000) – means reducing structural unemployment and requires microeconomic solutions. To bring structural unemployment down from 9 to 5 per cent in a little under ten years would mean reducing that rate by approximately 0.4 per cent per year; this is not unattainable but it is in the upper reaches of what other OECD countries have achieved. Working groups put together in various areas to debate employment policies have resulted in the publication of a number of well-documented reports, such as that by the *Conseil d'analyse économique* entitled "Full Employment" (Pisani-Ferry, 2000) or that by the *Commissariat général du plan* on the reduction of working time (Rouilleaut, 2001). The negotiations between social partners concerning the back to work programme (*Aide au retour à l'emploi*, ARE), unemployment compensation, supplementary pensions and "social modernisation"[36] have also prompted a wide-ranging public debate. The issues involved in returning to full employment may be divided into four categories: how to reduce disincentives to labour supply? How to adjust working time and make it more flexible? What gains can be hoped for from a reduction in the cost of unskilled labour? How can the labour market be made to function better? These questions are addressed below.

Lease discouraging labour supply

Since the oil shocks of the 1970s, France has pursued various policies aimed at halting the rise in unemployment (DARES, 1997). On the employment front, welfare spending and the number of public-sector jobs have been increased, with a parallel rise in tax and social insurance contributions. In addition, public policies promoted selective labour force withdrawals, lowered retirement ages and reduced working time (Olivennes, 2000). Recent analyses suggest that, at best, these policies have done no more than mask underemployment and in many cases have made it worse (Pisani-Ferry, 2000). The result is that participation rates are low, particularly at the two extremes of the age pyramid. In the 15-24 age bracket, only one man in three is working and only one woman in four, while the average age of those completing initial training was 20.6 in 2000 (Minni, Poulet-Coulibando 2001). Similarly, in the 55-64 age bracket, only two men in five are working and just one women in three. Policies, that delay arrival on the labour market, such as that for education, or that encourage early withdrawal have rarefied labour supply. As a result, even though hiring continues, after just a few years of cyclical upswing, employers are faced with recruitment difficulties. The existence of such hiring difficulties alongside underemployment – an unusual position on the Beveridge curve – is uncommon among industrialised countries.

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Basic income support and financial incentives to work

Analysis shows that many government schemes discourage labour supply: basic income support, taxation, early retirement schemes and the pension calculation system. The various basic-income support schemes available to the non-active population – the aim of which is to combat poverty – have acted as a particular deterrent (Piketty, 1997 and 1998; Laroque and Salanié, 1999 and 2000). In fact, France has no less than 8 different benefit systems covering 3.4 million people (Caussat, 2001), the most important of these being the social minimum income (RMI) which is received by over 1 million people (Table 6). Being in receipt of these benefits usually gives access to related entitlements such as tax relief and exemptions, as well as supplementary benefits such as rent allowances and specific forms of aid disbursed by the local authorities. These different systems act like taxes on labour: compared with the combined revenues from these welfare benefits, the financial incentive to taking a job paying the minimum wage, especially on a part-time basis, is often very weak. In some cases, the decision to do so can even result in a loss of income.

Table 6. **Main characteristics of basic income support**

Allowance	Year created	Managing agencies	Number of recipients (thousand)	Maximum allowance (FF per month)	Resource ceiling (FF per month)
Basic old age pension	1941, 1956	Basic pension system	942	3 471	3 555
Basic disability pension	1930	Sickness insurance scheme	99	3 471	3 555
Allowance for handicapped adults	1975	Agencies paying family allowances	631	3 471	4 825
Single parent allowance	1976	Agencies paying family allowances	163	3 680[1]	3 680[1]
Insurance allowance for widows/widowers	1980	National old age insurance fund	17	3 107[2]	3 884[2]
Young first job-seekers' allowance	1979	ASSEDIC	15	1 815	5 449
Special solidarity allowance	1984	ASSEDIC	516	2 577	6 014
Social minimum income (RMI)	1989	Agencies paying family allowances	1 049	2 138[1]	2 138[1]

Note: The number of recipients is for the whole of France (metropolitan France + DOM) as of 31 December 1996, except in the case of the RMI, for which the end date is 30 June 1997. The maximum amounts and resource ceilings for the different benefits correspond to the scales in force at 1 January 1998 for a single person, except in the case of the single parent allowance, for which the scale used is for a single parent with one child.

1. After deduction of the "flat-rate housing allowance".
2. Scale for the first year of widowhood.

Source: CSERC (1997), Ministry for Employment and Solidarity.

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Realising the danger, the authorities changed the direction of employment policy so as to encourage labour supply. However, remedying those measures that discouraged labour force participation is not easy. The solutions lie in rationalising the welfare and tax systems, making benefits conditional and supplementing income from work. Initially the authorities undertook to rationalise the welfare and tax systems. Recipients of basic income support are now authorised to combine benefits and income from employment temporarily, within the context of so-called profit-sharing mechanisms, so as to smooth the transition from non-employment to employment. Thus, the RMI can be combined in full with an income from work during the first six months after returning to work and then partially during the following six months. In addition, other technical measures have been taken; these include the reform of property tax, of personal income tax and of the rent allowance. The management of these mechanisms is, however, complex and they do not provide sufficient incentives to provoke a substantial return to work (Pisani-Ferry, 2000).

Employment tax credit

The authorities have put into place a new more decisive course of action with the creation, in 2001, of the employment bonus (*prime pour l'emploi* – PPE), a tax credit aimed at providing an additional income for people in low-paid jobs – along the lines of the Earned-Income Tax Credit in the United States and the Working-Families Tax Credit in the United Kingdom. The credit seeks to "encourage employment among those receiving the lowest levels of pay".[37] All types of employment are concerned, whether wage jobs or self-employment. Since the object is to make low-income work better paid, the amount of the credit is progressive until the equivalent of the SMIC is reached, and then degressive up to 1.4 times the SMIC. The programme seeks to encourage full-time work rather than part-time work, so the amount of credit is highest for the full-time SMIC. The PPE also takes account of an individual's marital status and is more generous for couples with children and even more so for single-parent families. The scheme is to be phased in gradually because of budgetary constraints and should be fully operational in 2003. For an unmarried recipient of the SMIC in a full-time job, the amount of the credit is FF 1 500 in 2001 and will be FF 3 000 in 2002 and FF 4 500 in 2003.[38] It is estimated that some 8 million households will receive an employment credit in 2001, at a total budgetary cost of FF 8 billion, *i.e.* approximately FF 1000 per household on average.

The creation of the employment bonus constitutes a big step towards reducing the disincentive effects of basic income support. Its actual implementation – with a first disbursement having been made in September 2001 – will give a better idea of its potential impact which, for the time being, remains uncertain and will have to be gauged over the medium term. At all events, the reform of the

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welfare benefits and tax system has to continue. France has a number of basic income support allowances that are appreciably larger than in other OECD countries, while its tax system is particularly complex and the interaction between these different measures is liable to distort labour supply behaviour. A particular example being the parental education allowance, which is thought to have encouraged 100 000 women to withdraw from the labour market (Pisani-Ferry, 2000). So rationalising the system remains important.

Reduce working time or make it more flexible?

Working hours, which in France are traditionally short,…

Work time in France is among of the shortest in the OECD, behind Germany, Norway and the Netherlands.[39] In 2000, employees in France worked, on average, less than 1 500 hours per year, taking into account part time work, public holidays and paid leave but not taking account of absenteeism (Figure 16). The proportion of part-time workers in the population is within the average but the hours they work are considerably longer. In contrast, the effective work time for full-time work is lower, mainly due to the number of public holidays, which exceeds the European average. Reducing work time has been a traditional social aspiration. France's social history has been punctuated by major reforms in working time, such as the introduction of the 40-hour week and two weeks' paid leave in 1936, the move to the 39-hour work week, the introduction of the fifth week of paid leave in 1982 and, most recently, the legislation bringing in the 35-hour work week in 1998. It is true that the statistics on actual working time are averages encompassing different types of reality: on the one hand, there are the doctors, executives and self-employed persons who may work more than fifty hours a week, while on the other hand public enterprises such as EDF have already brought in the 32-hour work week (in return for a slight cut in pay), *i.e.* the four-day work week, and some big companies frequently give their employees extra days leave (on average 3 weeks), which in extreme cases can bring their total paid leave to as many as 13 weeks per year.

… have fallen still further with the transition to the 35-hour work week…

Following the move to the 39-hour work week in 1982, the effects of which are a matter of controversy (Crépon and Kramarz, 1999, Cette 1992), statutory working time remained stable for nearly fifteen years. The law of 13 June 1998 (Aubry I), supplemented and amended by the law of 19 January 2000 (Aubry II), obliged the social partners to resume negotiations aimed at bringing in a further reduction in working time. The Aubry laws set statutory working time[40] at 35 hours as of 1 January 2000 for firms with more than 20 employees and as of 1 January 2002 for the others. A system of aids was introduced to help firms cope

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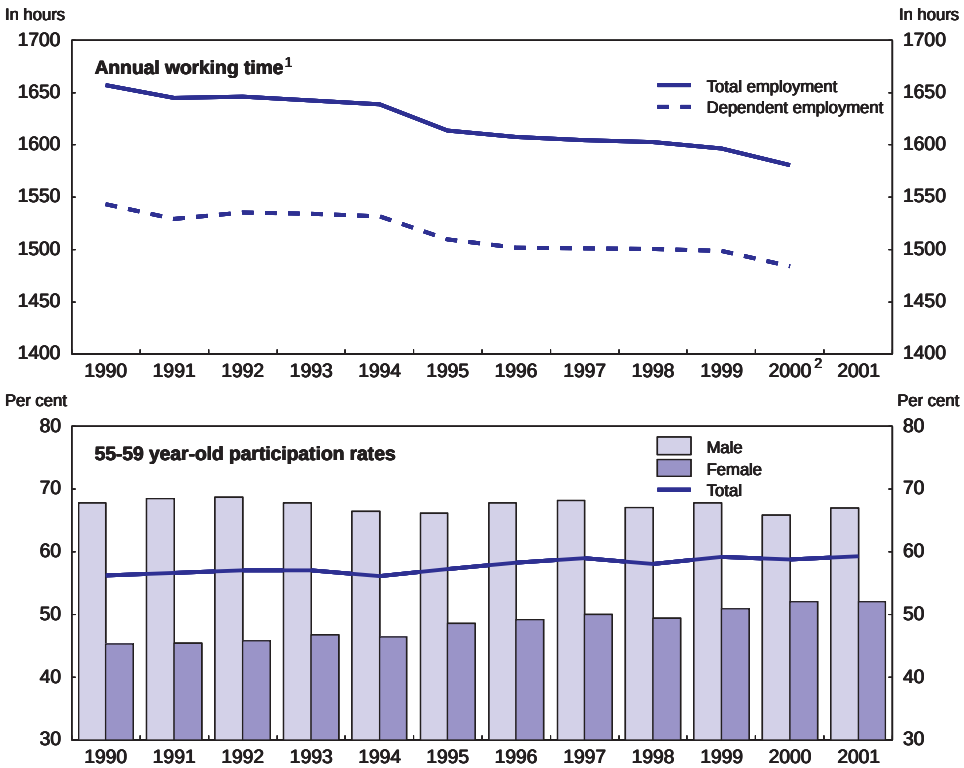
Figure 16. **Working time and participation rates**



1. Total economy, see definition in *OECD Employment Outlook* 2001, Table F.
2. Estimates supplied by DARES.

Source: INSEE, DARES and OECD.

financially with the transition: on the one hand, temporary incentives for firms that reduce working hours before the reduction in the statutory working week; in the other, long-term aids consisted of a per employee lump-sum and a larger amount for low-wage employees. There are specific provisions for overtime and for managers. The reduction in effective working hours is attenuated by the possibility of modifying the way in which hours worked are counted, introduced by the second Aubry law. A detailed description of the legislation is contained in Annex I of the *Survey* of France published in July 2000.

The new arrangements have been brought in rapidly. In April 2001, firms implementing the reduction in working time accounted for nearly 6.9 million employees, of which 6.1 million were full time. About one employee in two (among

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full-time members of the workforce in the non-agricultural competitive sector) work 35 hours per week, whereas in 1996 the proportion was negligible. The extent to which the 35-hour work week has been adopted varies with the size of firms. Whereas the majority of employees in firms with more than 20 employees are covered by the arrangements, this is true of only a minority in small firms. In addition, all government-controlled corporations have moved to the 35-hour work week, if not fewer hours. A frequently used method of reducing working time is to award extra days of leave. According to the survey on the reduction in work time and life styles carried out by the DARES, the majority of employees say that the reduction in working hours has improved their everyday lives but opinion is less favourable when it comes to working conditions, employees saying that the tempo of their work has been stepped up.

… the theoretical framework of which is uncertain

The theoretical literature does not offer any clear-cut conclusion on the impact on employment of the reduction in working time – especially the long-term impact. Some authors conclude that it will have a negative impact due to the additional production and reorganisation costs this type of reform generates; others predict a positive impact (see Cette, 2000*a* for an overview of the literature). The absence of a clear-cut conclusion is probably due to the complexity of the micro- and macroeconomic effects of this type of policy. As regards the specific case of the introduction of the 35-hour work week in France, empirical studies carried out by official bodies have tried to determine the conditions that have to be met for there to be a positive impact on employment (see Rouilleault, 2001, for a an overview of these studies). These official studies conclude that the reduction in working time is beneficial if it has a neutral impact on unit wage costs. Although the reform usually leaves the monthly wage unchanged (and the hourly wage is thus higher), three factors would make it possible to avoid a steep increase in unit wage costs: *a*) hourly productivity gains accruing from work reorganisation, *b*) temporarily slower wage growth, and *c*) States aids that accompany the reduction.

On the basis of favourable assumptions, simulations conducted by the Ministry of Employment and Solidarity in 1998 suggested that the creation of 700 000 jobs in the market sector was a likely scenario.[41] Today this figure seems high. The authorities consider that a number of parameters have reduced the impact on employment. In particular, the actual reduction in work time has been smaller than anticipated in many firms, since they have modified the way in which they calculate working time to arrive at a 35-hour work week with a smaller reduction. It is now generally accepted that the creation of 700 000 jobs in the market sector today seems out of reach.

Economic analysis has in the meantime become more refined and consensual. Most analysts recognize that reducing work time has no effect on structural unemployment and is therefore a policy designed to reduce Keynesian

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unemployment (Kessler and Trainar, 2000; Fitoussi and Heyer, 2000; Cette, 2000*b*). The assertion that reducing work time is neutral as regards structural unemployment reflects the assumption that there is no impact on unit wage costs. There are, admittedly, some further factors that can increase structural unemployment (such as the reduced utilisation of plant and equipment) but these are considered to be offset by others (such, for example, as more flexible working hours). If the reduction in working time only reduces Keynesian unemployment, its impact will be restricted to the short term and, in the medium term, unemployment will return to its equilibrium level. Finally, if one accepts that unemployment in France is characterised by hysteresis[42] (a hypothesis that is rejected by most studies), then a fall in Keynesian unemployment could result in reduced structural unemployment. Nevertheless, the process is symmetric and the existence of hysteresis would mean that the fall in structural unemployment would be reversed during the following cyclical downturn.

An uncertain effect on employment…

In view of the theoretical and empirical uncertainties, gauging the impact on employment of the reduction in working time is a difficult exercise. The authorities calculate the impact on employment by monitoring a sample of firms that have introduced the 35-hour week and another that have not (Fiole, Passeron and Roger, 1999 and Passeron, 2000). After making adjustments for the effects of sector, size and past trends, the difference between the two samples gives an estimate of the number of jobs created as a result of the reduction in working time. On this basis, the authorities calculate that 240 000 net jobs had been created thanks to the new legislation at end-2000 (Rouilleault, 2001), *i.e.* approximately 1.5 per cent of the workforce of the non agricultural market sectors. The effects appear to have been slight in big industrial companies and banks, where reduced working hours have above all curbed the labour shedding engendered by productivity gains. Conversely, job creation has been strong in government-controlled corporations. By extrapolating these trends, the authorities believe that the process will ultimately generate 500 000 net jobs (3 per cent of the workforce in the non-agricultural market sectors), assuming a reduction in working time of 3 hours per week concerning some 80 per cent of full-time employees. These figures tally with the macroeconomic simulations described in the Rouilleaut Report (2001) and those carried out by the *Observatoire français de conjoncture économique* (Heyer and Timbeau, 2001). However, they are higher than the results of the simulations carried out by the OECD – admittedly on the basis of very different assumptions (see Annex IV).

… but a high cost for public finances…

The positive impact on employment reflects a significant cost for the public purse. In order to encourage firms to adopt the scheme and lessen the impact

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on labour costs, the authorities have undertaken to provide exceptional assistance and long-term aid under the heading of the "offensive" part of the first Aubry Law. Furthermore, the second Law undertook to give firms adopting the 35-hour work week an increase in the rebate on employers' social insurance contributions that had already been provided for as part of the so-called "Juppé rebate". For these firms, this degressive rebate applies to employees paid up to 1.8 times the SMIC, whereas previously it was limited to 1.3 times the SMIC. These different types of aid take the form of reduced employers' Social Security contributions.

Under the provisions of the Veil Law of 25 July 1994, the State is obliged to compensate the Social Security system fully for the loss of revenue caused by the measures reducing social contributions. To do this, the authorities have created a new off-budget fund called the F*onds de financement de la réforme des cotisations patronales de la sécurité sociale* (FOREC) which, on the revenue side, receives various taxes and transfers earmarked by the State and, on the expenditure side, pays compensation to the Social Security. FOREC's expenditure in respect of the different reductions is put at between FF 72 and 76 billion in 2000 (depending on the method of accounting), *i.e.* some 0.8 per cent of GDP. This expenditure will increase in coming years, the available projections suggesting a total of approximately FF 125 billion in 2004. After deducting the so-called Juppé rebates which are not part of the 35-hour scheme, the amount of government aid granted per job created is thought to have been nearly FF 130 000 per job[43] (FF 31 billion for 240 000 jobs) in 2000. The high cost has, admittedly, to be looked at alongside the effects of possible financial returns deriving from the scheme, the jobs created result in additional tax and social insurance receipts and reduced expenditure in respect of unemployment benefit. Some authors therefore put the *ex post* cost at one-third of the *ex ante* cost (Heyer and Timbeau, 2001). Nevertheless, the 35-hour work week is proving, like the reduction in employer social security contributions, to be an extremely costly employment policy. Its associated net expenditure are a burden on the central government budget, increase the budget deficit and prevent taxes from being reduced more rapidly.

… which is liable to be increased by the reduction in working time in the civil service

The cost to public purse will be increased by the creation of government jobs when the civil service moves to the 35-hour work week. The authorities have undertaken to apply the reduction in working time to the 5½ million public service employees as of 1 January 2002 (State, hospitals and local government). They would like to effect the transition with no change in the number of employees, a goal thought achievable because some government departments are already close to the new statutory working week and because latent sources of productivity may be mobilised. However, initial negotiations would suggest that additional jobs will be created, particularly in public hospitals and in local government. The

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agreement reached at the Paris City Hall provides for the creation of 1 000 additional jobs (a 2½ per cent increase in staffing), while public hospitals could create more than 45 000 new jobs. Actual implementation of these commitments has been delegated to Ministries, the aim being to reorganise work and bring more flexibility to existing arrangements. One innovation is that working time is to be measured not on a weekly basis, but over the year, the target being to reach 1 600 hours of work per year.

The move to the 35-hour week poses specific problems for small firms…

As in the civil service, statutory work time in small firms with fewer than 20 employees will be reduced to 35 hours as of 1 January 2002. Some 4.5 million employees of small firms are concerned, *i.e.* rather less than half the number in large firms. The shift to the 35-hour work week is going to be more difficult for small firms than for big ones. First, there is the problem of indivisibility: whereas a big company can increase employment by a few percentage points in order to offset the reduction in working time, for a small firm to recruit one more employee can mean a considerable change in size. Second, small firms are not necessarily large enough to take advantage of work reorganisation, such as annualisation or distribution arrangements. Third, they do not have the internal administrative resources needed to cope with the regulatory complexity of the scheme or to handle the transition in the most efficient way possible, meaning that their fixed hiring and conversion costs will probably be high. Last, they will be moving to the 35-hour work week just when growth is slowing appreciably and will not therefore benefit from the favourable economic climate that the big companies enjoyed in 2000. All told, it is likely that, for a number of firms, the 35-hour week will result in a fall in the length of the firms' work week, and, therefore a loss of turnover.

Gradualist elements in the law are supposed to ease some of these difficulties.[44] Furthermore, as part of support-cum-advisory arrangements designed to help businessmen shift to the 35-hour work week, the State is paying for all or part of consultants' mission expenses. The complexity of the negotiations will be eased for some small firms thanks to collective agreements signed at the industry level, which establish the practical details of implementation. An agreement is being envisaged, for example, between the social partners in the hotel, cafe and restaurant sector whereby the reduction in working time will be phased in up until 31 December 2006 (instead of 1 January 2002), making it possible to reduce the statutory working week in this sector from 43 currently to 35 hours. Nevertheless, the risk that the 35-hour work week will reduce the weekly activity of small firms and result in a loss of turnover for them cannot be entirely ruled out. The authorities, therefore, have decided to permit more flexible arrangements for these firms. They have agreed to raise, temporarily, the ceiling on overtime to 180 hours in 2002 and to 160 hours in 2003, after which the normal quota of 130 hours will apply, the rules regarding the way overtime is counted remain the same.

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… and is liable to result in a negative wage shock for employment

The government has pledged to ensure that employees earning the SMIC are paid as much for 35 hours as for 39 hours. The second Aubry Law contains a provision to this effect called the "monthly remuneration guarantee".[45] Since not all employees are going to start working reduced hours at the same time, the amount of the "remuneration guarantee" will depend on the date of the company agreement to reduce working time.[46] In the second half of 2001, for example, there were already four different monthly guarantees.[47] In order to avoid people doing the same work receiving different wages for any length of time, the government has pledged to take measures to deal with the problem before December 2002. In the absence of such measures, the hourly SMIC would have to be raised in 2005 by 11.4 per cent.[48] This adjustment will be on top of the gradual adjustments to the hourly wage between 2000 and 2005 (up to 50 per cent of 11.4 per cent),[49] *i.e.* 17.7 per cent all told in terms of purchasing power, not including any extra increment from the government.

The authorities and the social partners are aware of the dangerous implications of this "wage shock" both for the employment of those on low wages and for firms' financial viability (Cette, 2000*b*; Pisani-Ferry, 2000; Rouilleault, 2001). Various remedies to the situation have been considered, but none of the solutions seems completely satisfactory, bearing in mind the pledges made that employees' wages would be maintained when the 35-hour week comes in. Moreover, the said solutions would mean amending the second Aubry Law and would therefore require Parliamentary approval. This question is currently under discussion with the social partners. Pending the arrival of 2005, the absence of an additional increment from the government would somewhat reduce the scale of the problem.

Reducing labour costs

The employment impact of minimum wage hikes has been a longstanding topic of academic controversy and the results of macroeconomic studies have been tentative and ambiguous. Up to the start of the 1990s, it was generally considered in France that the SMIC was not an obstacle to employment. Empirical research has recently been enriched through an examination of "natural experiments" using sectoral or individual data. These efforts have concluded that increases in the SMIC have had a negative impact on employment and in particular on job opportunities for low-skilled and young workers. Bazen and Skourias (1997) conclude, on the basis of sectoral data, that SMIC hikes of 10 per cent in June 1981 reduced youth employment. Abowd, Kramarz and Margolis (1989) estimate that the chances that a worker paid at the SMIC will still have a job after an increase are reduced by 1.5 per cent for every 1 percentage point of that increase. It is now generally accepted that the cost of labour at the SMIC rate thus has a strong and swift impact on the demand for labour and hence on workers' employment prospects (Pisani-Ferry, 2000).

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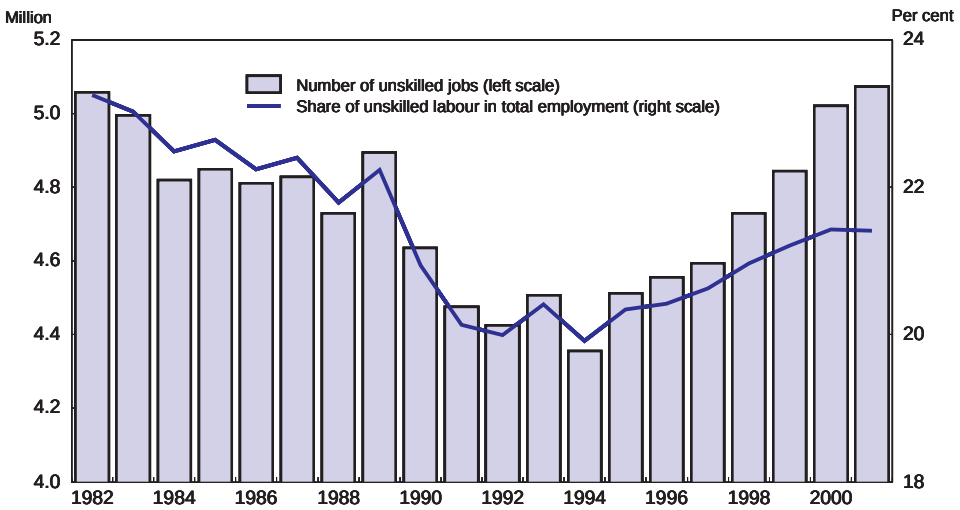
To date, however, any reform in the method of fixing the SMIC has not been deemed desirable. To prevent the SMIC from penalising low-skilled workers, policy has focused on lightening the burden of employers' social contributions that are calculated on net wages and that in 1997 amounted to about 70 per cent of net earnings for those paid at the minimum wage. These measures have made it possible to substantially reduce employers' labour costs, while preserving the purchasing power of the SMIC, as perceived by employees. Reductions in social contributions began to be implemented in 1992, in an effort to encourage part-time work, and relief has since been generalised to all low-wage earners, starting at 1.1 times the basic SMIC, and going as high as 1.8 times the SMIC for workers moving to the 35 hour work week (Gubian, 2000). As a result of all these measures, social charges at the level of SMIC earners are now only 19 per cent for employers and 4.5 per cent for workers (see Annex I of the 2000 *Survey*).

The decision to reduce the social contribution burden on low-skilled jobs was accompanied by a stabilisation in the share of such jobs in total employment: after declining for several years, that share has risen appreciably since 1993 (Figure 17). Besides the reduction in social contributions, several other factors seem to have contributed to this trend. The surge in part-time employment, encouraged by specific measures, concerned low-skilled workers above all. On the other hand, demand for low-skilled labour-intensive services has grown rapidly, in

Figure 17. **Unskilled jobs**



Source: INSEE.

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such areas as retail sales, industrial cleaning, mothers' helpers and day-care centres and the fast food industry, which has offset the decline in industrial activities (Chardon, 2001). Empirical studies on individual data have isolated the influence of lower social contributions from these other influences. Thus, Crépon and Desplatz (2001) estimate that social contribution cuts have had a determining influence and that the measures taken in 1995 and 1996 had an impact of between 400 000 and 500 000 jobs.

Overall, it has now been clearly demonstrated that the cost of labour at the SMIC level has a strong and swift impact on the demand for low-skilled workers. Policies to alleviate social contributions have helped to revive demand for this type of worker, admittedly at a high budgetary cost. To preserve the positive impact of the reduction in social contributions, increases in the SMIC must remain moderate in relation to average wages. In any case, the future of the SMIC will inevitably have to be rethought since, as noted earlier, without such reform, the SMIC increases imposed by the Aubry II law would, in effect, cancel out the positive impact of contribution relief over most of the last 10 years.

Measures relating to low-skilled labour must not obscure the broader question of wage flexibility. In effect, differentiations in wage trends are strikingly weak, given the diversity of the unemployment risks facing different occupational groups. The risk of unemployment is sharply differentiated according to age, sex, level of education and skills. These differing risks of unemployment should normally translate into different trends in wages for various categories, but this has not been the case (Figure 18). The lack of wage flexibility between categories is one reason for the high equilibrium level of unemployment, since it makes a portion of the population less employable (Pisani-Ferry, 2000). But the differing risks of unemployment may also reflect a queuing phenomenon, whereby the most educated employees fill unskilled jobs, crowding out the least educated. The reasons why the wage distribution has changed so little are not well understood, and deserve further study.

Enhancing the functioning of the labour market

The simultaneous existence of massive underemployment and severe recruitment difficulties underlines the dysfunctional aspects of the French labour market. In May 2001 there were 2.3 million job seekers on the books of the National Employment Agency and at the same time nearly 160 000 unfilled job offers. Recent studies suggest that the French labour market has a dual nature: the risk of losing one's job is low but the probability of escaping from unemployment is also low. It would seem as if the fact of finding a job means a permanent increase in a worker's employability, while a period spent in unemployment reduces employability – just as permanently. It is not surprising, then, that groups of persons particularly vulnerable to the risk of unemployment encounter difficulties in finding a job (particularly

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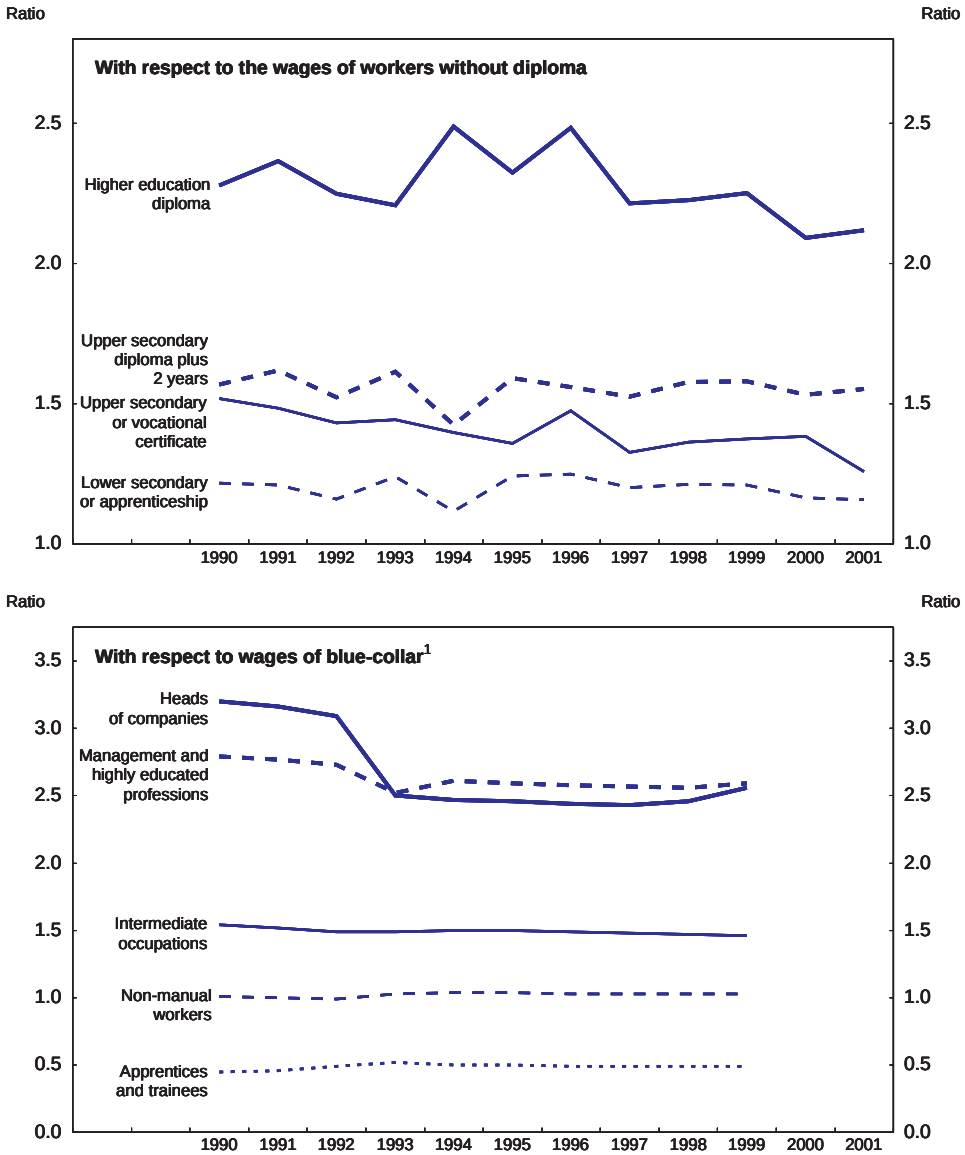
Figure 18. **Wage ratios**



1. In 1993 important methodological changes were introduced in the way INSEE uses its annual social data reports (DADS). As a result, 1993 wage levels are comparable neither with those before 1993 nor those after 1993.

Source: INSEE.

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unskilled and inexperienced youth) coexist in the labour market with employers who have trouble in recruiting skilled workers. These difficulties do not just indicate an inadequate labour supply; they may also reflect the difficulties firms have adapting to the new pace of hiring and a mismatch between the skills available on the market and those sought by firms. Under these conditions, public policy faces two challenges: on one hand, to build bridges from unemployment to employment, by enhancing the employability of job seekers, and on the other hand, improving the match between labour supply and demand, through systems that assist job finding.

Enhancing the employability of job seekers

The authorities have set themselves the objective of giving the unemployed and in particular the long-term unemployed, a chance to return to stable employment. This priority has translated into a significant increase in subsidised employment (*emplois aidés*), the aim of which is, of course, to reduce unemployment, but also to provide a springboard to stable employment. The Central Government budget earmarks something more than FF 50 billion a year to financing such jobs, under various programmes (apart from social contributions relief). While "passive" spending (unemployment benefit, early retirement) has fallen sharply since 1997, spending on jobs has remained stable for several years. Instruments have evolved towards greater support for people in difficulty and the young. Employment programmes for people in difficulty have been refocused, notably the number of people starting "employment-solidarity contracts" that finance limited-term jobs in the non-commercial sector, has fallen but this has been partly offset by people starting "consolidated-employment contracts" (DARES, 2001). The "Fresh Start" programme run by the ANPE, which caters to those in the greatest difficulty (*inter alia* RMI recipients, young people who have been unemployed for six months and adults who have been unemployed for a year), has yielded satisfactory results in getting the least employable back into employment (Micheau, Poujoly and Pommiter, 2001). In addition, the TRACE (Trajet d'accès à l'emploi) scheme, which has been in force since October 1998, provides personalised support for the least employable young people to help them find a job and become integrated in society. By the end of 2000, almost 95 000 young people had joined the programme and more than 50 per cent of those exiting it, did so towards a job. The government is planning to double the number of places on the programme (*i.e.* 120 000 in 2002) and to introduce a monthly "job access grant" of 300 euros (up to a maximum of 900 euros over six months) during periods when young people are neither employed nor on a training scheme.

At the same time, allocations to youth employment contracts have risen sharply. This device, launched in 1997, now employs 275 000 young people. These

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jobs, financed by the Central Government at the rate of 80 per cent of the minimum wage, are for the most part in the education sector (teachers' assistants), the national police (security guards) and the Ministry of Justice as well as in associations, public undertakings and municipal government. The programme appears to improve the employability of participants. Indeed, three quarters of those who leave it prematurely are earning a salary six months after leaving the programme and a third of them are working in the market sector. Nevertheless, the great majority of young people find that, as the end of their five-year youth-employment contract nears, they have been unable to find work outside of the program. The government therefore decided in 2001 to renew it for another five years and to increase spending in order to enhance the occupational prospects of beneficiaries. Overall, the gross cost of the programme (including employers' residual share of the cost and ancillary expenditures, in particular on training) was FF 70 billion over the period 1997-2001, and will amount to FF 105 billion for the period 2002-2006.

Back-to-work assistance

Although primarily based on insurance principals, the unemployment compensation system also provides incentives to return to work. Eligibility criteria are relatively strict, less than half of job seekers receive an allowance. Allowances are degressive and therefore encourage recipients to return to work after an initial period. Finally, penalties can be imposed for refusing a job offer. In practice, however, the incentives are less strong. In particular, penalties are rarely imposed (OECD, 2000*b*), which means that job offers can be turned down with impunity. Business and labour have thus launched discussions to recast the unemployment insurance system and in January 2001 they signed an accord on assisted return to work and unemployment allowances. At the same time, agreements signed by the UNEDIC, ANPE and the central government made it possible to set up an enhanced job assistance service (*Plan d'Action Personalisé* (PAP), run by the ANPE. These agreements introduce two important changes. First, the PAP extends the "Fresh Start" programme to all job-seekers, whether in receipt of unemployment benefit or not. Second, job seekers applying for an unemployment allowance must sign a back-to-work assistance plan (PARE), which establishes access to the PAP. The degressivity of unemployment allowances is abolished. The new "back-to-work assistance" (ARE) is not degressive and the duration of payments is contingent solely on the period during which unemployment insurance contributions were paid. In addition to these changes, the ANPE's human resources have been expanded and the UNEDIC covers part of the personnel and training costs. Yet the accord does not impose strict conditionality on eligibility for the ARE. If the current application of existing eligibility rules persist, whereby the unemployed can refuse job offers without a legitimate reason, the new system might provide even less incentive than the earlier one. Not only does it end the degressivity of unemployment allowances, but it does nothing to condition these allowances to any real effort to find a new job.

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Regulating redundancies

France has a complex set of regulations governing redundancies and layoffs. It is true that administrative authorisation of redundancies has been abandoned and the government cannot in principle prevent layoffs as long as they are conducted in accordance with the regulations. However, for companies with more than 50 workers and for layoffs involving more than 9 employees over 30 days in any firm, the employer must prepare a corporate plan for avoiding the layoffs or reducing their number. The employer must also attempt to reclassify workers whose redundancies cannot be avoided. In order to avoid redundancies, firms must adjust work time, turn to voluntary part-time work, offer early retirement or resort to partial unemployment. Various procedures must be examined in an effort to reclassify and reallocate workers, such as grants for geographic mobility or new business creation. The head of the firm must consult the employee representatives about the contents of the corporate plan and must report it to the administration, which will judge it for consistency with the regulations and, if there are shortcomings, insist that the procedure begin again.

The announcement of several redundancy plans at the beginning of 2001 led the authorities to recommend, under the Social Modernisation Bill, that current provisions be reinforced. Redundancy plans were renamed "job preservation plans" and several additional conditions were added to redundancy procedures: consultations with personnel representatives are to be broadened, layoffs must be conditioned upon reducing hours worked and redundancies are prohibited in companies that resort to overtime. New players are introduced into the corporate plan procedure, including the labour inspector, who may intervene at various stages of the procedure, the departmental prefect, who will be involved in the production-site redevelopment phase and the courts, to which the parties concerned may turn. The risks inherent in tightening employment protection legislation in this way are well known. Indeed, it is far from proven that stricter redundancy regulations reduce unemployment even as they are likely to reduce the fluidity of the labour market and result in reduced hiring. Moreover, they could encourage employers to resort to part-time work or to fixed-term contracts in order to avoid the administrative constraints involved in redundancies. They could also lead to wage pressures from workers in jobs (OECD, 1999). An initiative of this kind, then, is likely to depress employment over the long term.

How can network industries be opened to competition?

One important area for structural reform within OECD countries relates to opening network industries to competition. According to recent studies,[50] such moves are leading to significant productivity gains in all OECD countries, radiating positive externalities throughout the economy. The results include lower prices

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and better services for consumers, higher earnings for workers and greater returns for investors.[51]

As in other countries, France has begun to subject its network industries to competition. These reforms are beginning to produce results that are just as positive as elsewhere. They start, however, from a tradition of organisation for these services that poses obstacles in the move to market mechanisms.[52] Once reforms are underway, however, network industries tend to become dynamic players on the international market. An interim assessment is offered here, surveying progress with reforms in the principal sectors and their impact, as well as the responses that in some cases have contributed to the concerns that are holding up further liberalisation.

Promising first steps

The degree of exposure to competition differs from one network industry to the next, ranging from almost complete liberalisation (as in telecommunications) to an almost total absence of competition (postal services).

Telecommunications

The 1996 telecommunications law embodied comprehensive European guidelines. The objective was to achieve full liberalisation of telecommunications services in Europe as of January 1, 1998. After three years of implementation in France, this policy has achieved a largely positive record.

Long distance telecommunications have become highly competitive. New entrants[53] had won a combined market share of more than 30 per cent by the end 2000 by offering rates 5 to 15 per cent lower than those of the incumbent operator,[54] depending on destination. In the wake of this intense competition, France is now one of the countries in which long distance communications would seem to be least expensive (Figure 19), even if international comparisons of these rates must be approached with caution.

In mobile telephone service, where three operators compete with each other,[55] call volumes rose from 10 billion minutes in 1998 to 35 billion in 2000. Although France still lags behind some other countries in this area, the penetration rate of mobile service among the population was 54 per cent in June 2001. In terms of Internet access services, where 13 suppliers are competing, the volume of communications soared by 150 per cent in 1999 and 160 per cent in 2000.

Local calls are the last segment still dominated by the old public monopoly. In this area, the lack of competition seems to cause "black spots" in the French market's performance. The authorities have announced that this segment will be

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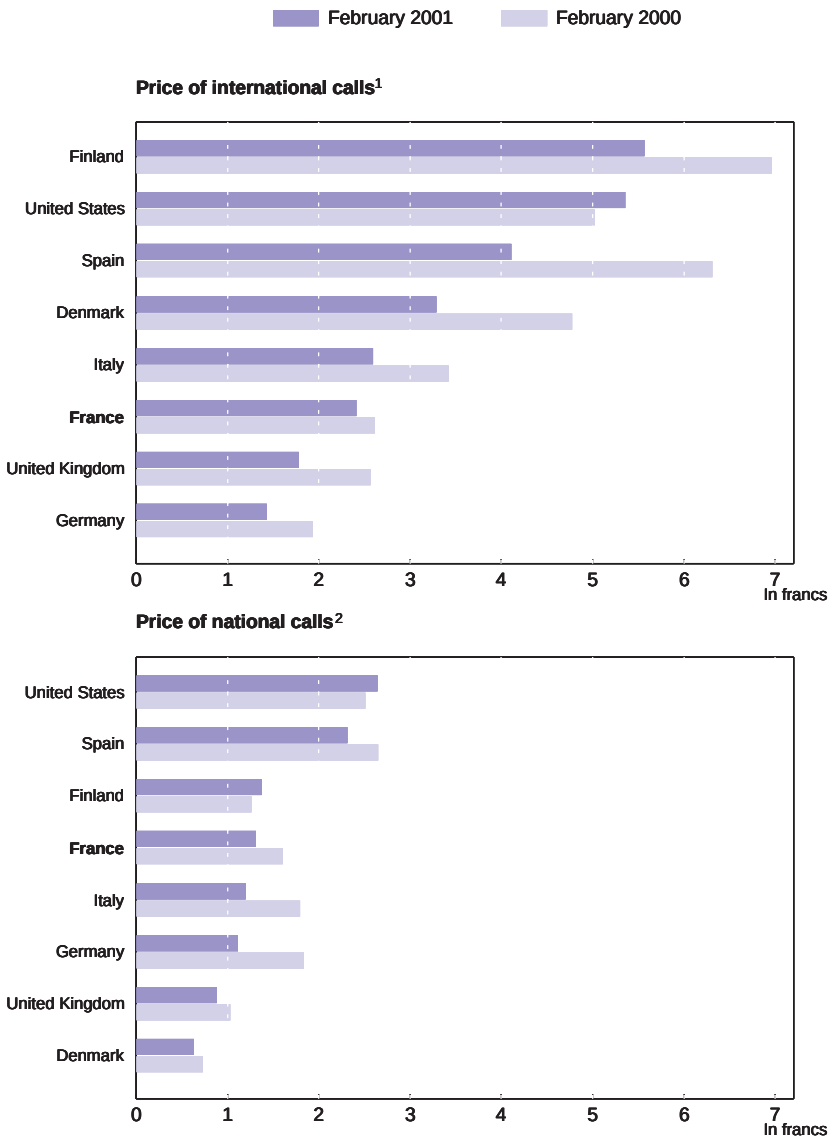
Figure 19. **Rates for long-distance calls**



1. Standard business rates for calls under three minutes to the United States (to London for the United States). Rates take account of reductions but do not include taxes. When several operators provide the service, an unweighted average of the rates was used.
2. Standard business rates for a three-minute call to a subscriber 320 kilometres away.

Source: National Utility Service Consulting Group, 2001.

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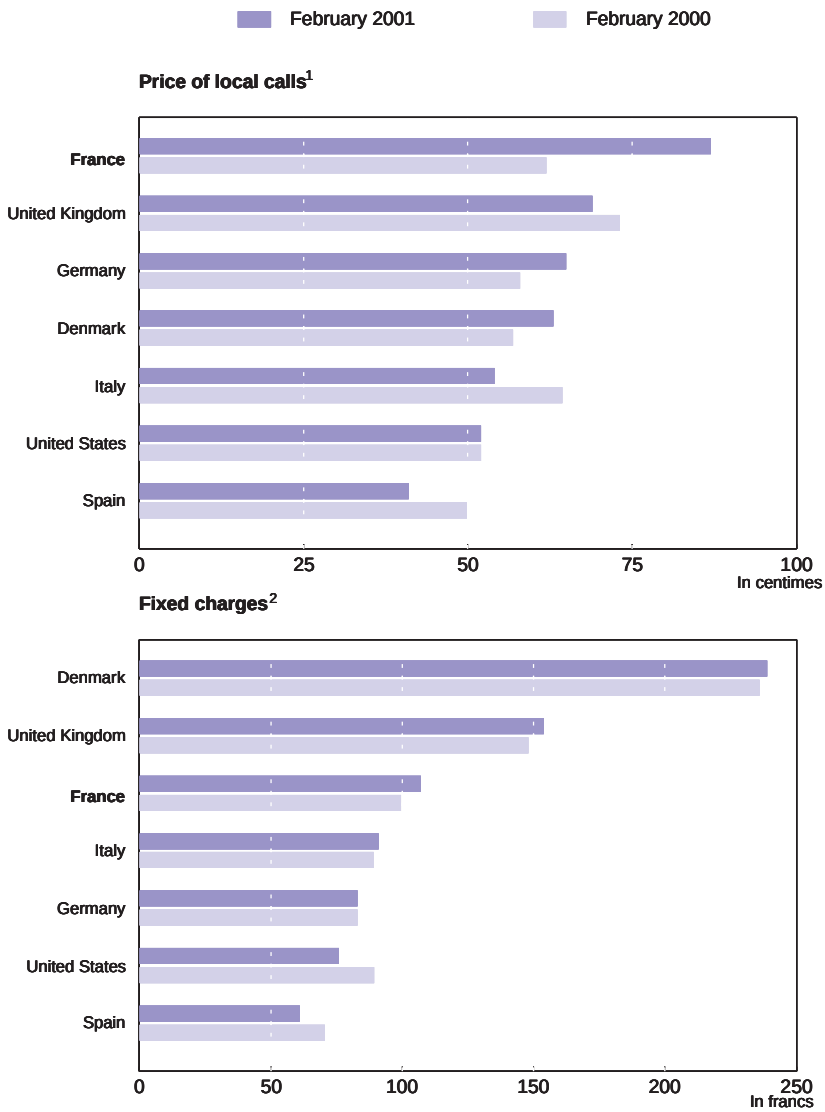
Figure 20. **Rates for local calls and fixed charges**



1. Standard business rates for 3-minute calls. Rates take account of reductions but do not include taxes. When several operators provide the service, an unweighted average of rates was used.
2. Standard rates for a business line. When several operators provide the service, an unweighted average of rates was used.

Source: National Utility Service Consulting Group, 2001.

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opened up more to competition from 1 January 2002, with better facilities for operators carrying calls within *départements*. It should be noted that the price of such calls declined by substantially less in 2000 than for long distance calls, and subscription rates and charges for calls of intermediate duration actually rose. Although this change reflects "tariff balancing" that is natural in a competitive environment, France is today one of the most expensive countries for these services[56] (Figure 20). There are also indications that France Telecom charges relatively high prices for the universal services that it is obliged to offer and for which it invoices the additional costs to a special fund (see below).

Air transport

Air transport is a second area where European initiatives at deregulation have been concentrated. This policy led to the creation of a "single European aviation market" in April 1997. European airlines have since then had access to all internal routes, domestic and international, within the Union, and they are free to set their prices, subject only to *ex post* monitoring by the competition authorities. The French government transposed this directive without reservation.[57] Two years later, a bilateral aviation agreement between France and the United States provided for additional liberalisation of transatlantic traffic.[58]

The impact of these measures has been felt primarily on international traffic, which accounts for 70 per cent of the total air transport market in France. The major European and international airlines have increased their flights to and from France, forcing Air France to restrain its own pricing. While fare cuts are still less spectacular than those following deregulation outside Europe,[59] the results of greater competition are more visible in France than in many other countries. Business and economy class fares and special fares of the Apex type offered through Paris airports in 1999 compared very favourably with those observed in other major countries (Figure 21).

A significant aspect of liberalisation has been the appearance on the air transport market of a new category of "no-frills" flyers, offering cut-rate prices in exchange for simplified service: these have proven popular among passengers with a high price elasticity.[60] Today they are operating on a limited number of French routes. These are primarily companies based across the Channel, and serving secondary continental airports where congestion is low. Their services are limited to two per cent of air traffic in France and, therefore, despite their substantially reduced fares, their overall impact on the air transport market is marginal (Figure 22). The underdevelopment of such low-cost services is a weakness of the French market. The placement on the market of the takeoff and landing slots released by the AOM-Air Liberté group following its restructuring could spark an expansion of such low-cost services.

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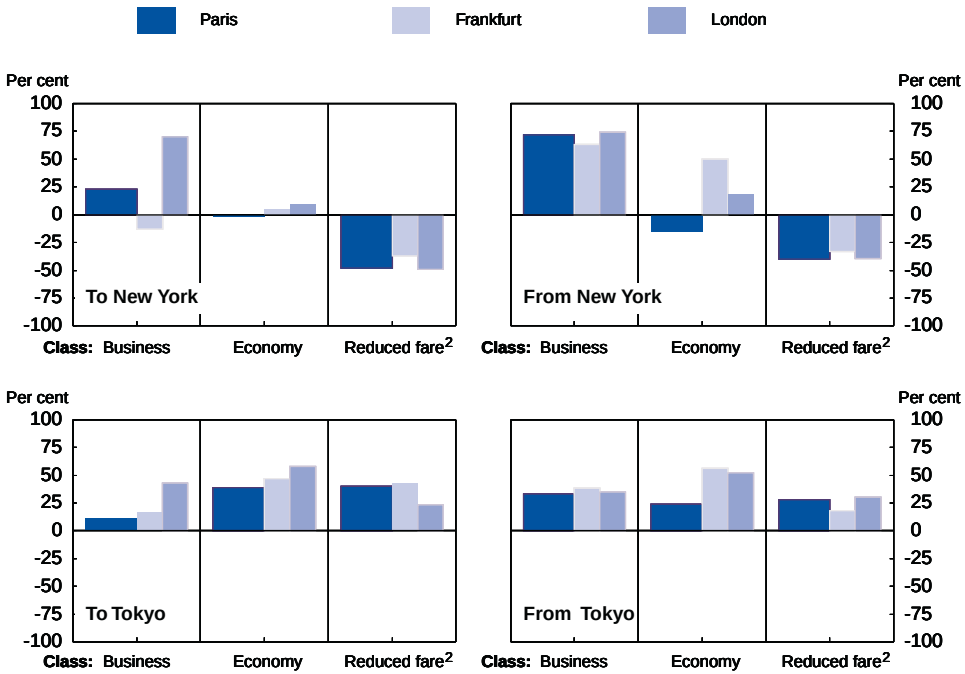
Figure 21. **Air fares from and to Paris**

Deviations from international average[1]



1. Differences in per cent between the fare of the largest carrier, from or to Paris, Frankfort, London, and an estimated average fare (one hundred international routes adjusted for economies of scale over the distance). For the methodology, see R. Gönenç and G. Nicoletti, OECD, Economics Department, Working Paper No. 254, "Regulation, Market Structure and Performance in Air Passenger Transportation".
2. Apex-type.

Source: OECD.

Electricity

Deregulation of the electricity sector is more recent. There is a European directive setting initial and intermediate objectives for 2000 and 2003, respectively. That directive contains quantitative objectives and organisational requirements intended to make monopolistic portions of the networks (transmission and distribution grids) more readily accessible to new entrants. France transposed this directive by means of an Electricity Law, which opened 30 per cent of the market to competition in 2000, and 35 per cent by 2003.[61] It also created the Electricity Transmission Network (RTE), a network management body that enjoys financial and managerial independence within EDF and whose strategic priorities and financing would be

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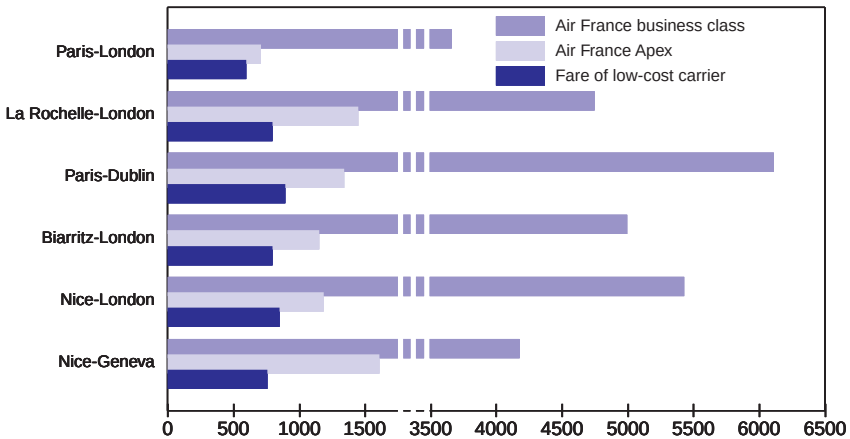
Figure 22. **Fares of low-cost airlines**[1]
August 2001, in francs



1. Return fares. The low-cost airlines are: Buzz for Paris-London and La Rochelle-London, Ryannair for Paris-Dublin and Biarritz-London, Easy Jet for Nice-London and Nice-Geneva.
Source: OECD.

clarified by total separation from EDF. RTE is required to provide producers with free and equal access to its transmission network. The electricity prices offered to eligible customers are freely negotiated. For non-eligible consumers, prices will continue to be set by the government after consultation with the Electricity Regulation Commission (CRE, 2001). This deregulation, while only partial, represents a veritable revolution in this traditionally monopolistic sector.

Because transmission costs are low, particularly between European domestic networks, this opening has led swiftly to sharp competition in the market for eligible consumers, who may now turn to a variety of alternative national[62] and, above all, foreign producers. At the end of April 2001, 52 customers (70 sites) or nine per cent of eligible clients had cancelled their contract with EDF. This situation put pressure on prices, resulting, in substantial price cuts[63] for bulk clients. For non-eligible clients, rates have declined less, although they are below the European average thanks to EDF's low nuclear power production costs (Figure 23). The pricing discrepancies between eligible and non-eligible clients, which the CRE and the government are monitoring closely but which they have had to tolerate in light of EDF's recent financial weakness,[64] imply a cost in terms of household welfare and in terms of distortions to competition between companies of different sizes (Figure 24).

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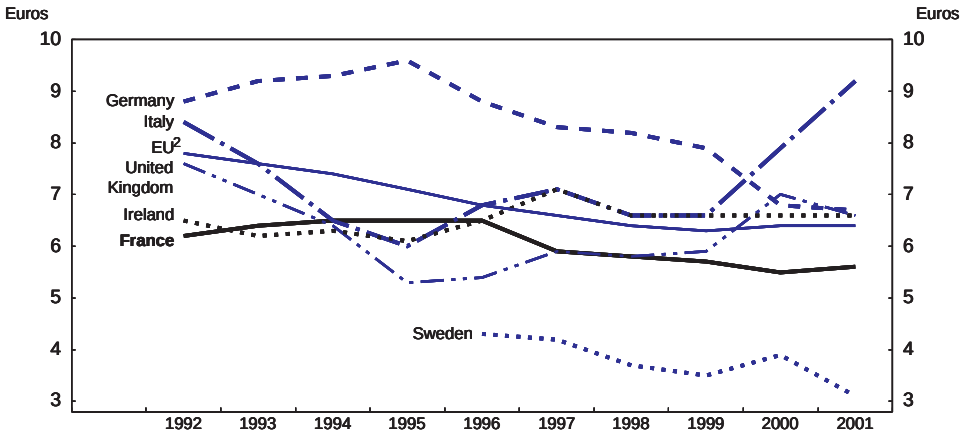
Figure 23. **Electricity tariffs for industrial users**[1]

In euros per 100 kWh

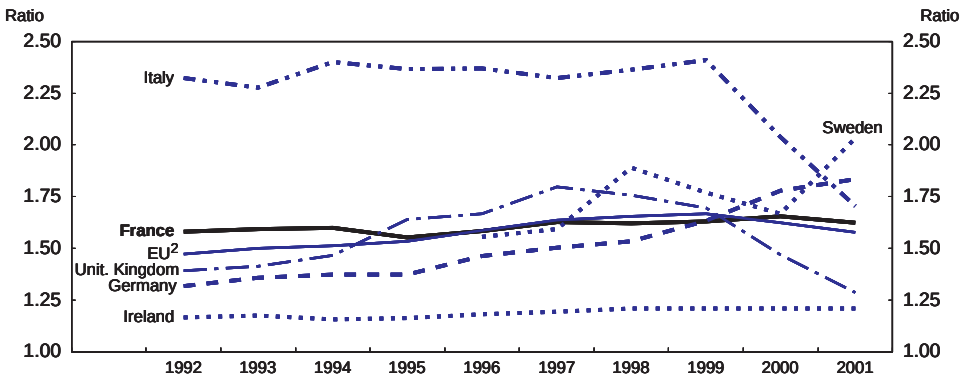


1. Tariff as of 1 July for an average industrial user with an annual consumption of 2 000 MWh.
2. Average weighted by national consumption.

Source: Eurostat, *Energy Statistics*.

Figure 24. **Relative electricity prices by type of user**

Household/industrial user prices[1]



1. Price as of 1 July. Ratio of the average price for an average household with an annual consumption of 3 500 kWh, of which 1 300 kWh at the night rate, and an average floor area of 90 m^2 and the price for an average industrial user with an annual consumption of 2 000 MWh.
2. Average weighted by national consumption.

Source: Eurostat, *Energy Statistics*.

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Rail transport

In the rail transport sector, European liberalisation directives work in two directions: *i*) separation, at least for accounting purposes, of physical infrastructure activities and actual operation of trains, and *ii*) gradual opening of the freight sector to competition, beginning with international freight service as of 2003, and expanding to embrace all goods transport.[65] The French government transposed these directives by means of a decree signed in December 1998.

The SNCF has been split into two entities. The new French Rail Network (RFF), which is completely independent, has inherited the entire track network and a major portion of the debt of SNCF. SNCF, with its financial burden now relieved, is responsible for passenger and goods traffic and has retained the railway stations. RFF is for the moment essentially a paper entity and has only a skeleton staff: SNCF is responsible for management and maintenance of the track network, and is paid for these services. Strangely, the remuneration that RFF pays to SNCF to maintain the network is greater than the fees that SNCF pays to RFF for running its trains over the tracks. Fees are set at a level that does not cover maintenance and network renovation costs and does not provide for amortisation of existing facilities,[66] which is aggravating the financial situation of RFF. The latter would thus be unviable were it not for the significant public subsidies it receives. On the other hand, application of the directives has not yet been enough to call forth any competitor to SNCF in the freight sector.

Postal services

European initiatives at liberalising postal services have for the moment encountered only mixed success.[67] France has transposed the existing postal directive by incorporating specific clauses into its Regional Planning Law rather than by framing a new law.[68] The market has been opened *a minima* and there is some competition in non-reserved services (parcel and express services). In the absence of further liberalisation and competitive pressure, the price of a stamp for ordinary mail seems higher than in more competitive countries, though still within the European average.[69] Increased volumes carried with the same number of employees generated productivity gains in the 1990s but the workforce rose by a further 6 500 in 2000 after an agreement reducing work time was signed. As *La Poste* is today the largest employer in France, with 310 000 workers and a large number of retirees,[70] any plan to transform it poses both social and budgetary challenges. The opening of the postal market to competition should strengthen incentives to restructure and accelerate productivity gains.[71] The existence of a large "reserved domain" (packages under 350 grams), the maximum authorised by the Directive, seems to play a pivotal role in preserving *La Poste's* financial balance.

Natural gas

A 1998 European directive establishes conditions for gradual opening of the gas market to competition.[72] This directive requires that separate accounts be

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kept for supply; distribution, storage and transport; and other non-gas related activities. It also states that all suppliers must have access to the pipeline networks under equal conditions. A deadline of August 2000 was set for transposing these provisions into national legislation.

France has not yet transposed the European directive. *Gaz de France* (GDF) remains therefore, in principle, in a monopolistic situation. However, the government has adopted a "transitional regime" for the opening of the gas network to competition. Eligible French consumers are allowed to choose their supplier freely anywhere in Europe. In addition, GDF and its two sister companies[73] are required to post gas transport rates for third parties and GDF has offered a formula for automatic restitution of imports contracted in its name (in order that the import monopoly, which remains legally its own, would not be an obstacle to imports by third parties). This formula has failed to satisfy France's European partners, however.[74]

The French authorities maintain that this "unofficial" deregulation is just as effective as formal application of the European directives.[75] New suppliers of gas from the North Sea have penetrated the market and in 2001 won provisioning contracts for 14 per cent of eligible consumers. The impact on provisioning prices, the ultimate test of effective competition, has not been documented. GDF's financial performance for the year 2000 would not seem, in any case, to have suffered from competitive pressures.[76] The price of gas for residential consumption, a significant segment of the market, remains higher in France than in those countries where the degree of market liberalisation exceeds what is called for under the directive.[77] It does however remain lower than in most other European countries, in third place behind the United Kingdom and the Netherlands, which are producers.

Overcoming concerns within a competitive framework

The first French deregulation reforms have begun to produce positive externalities, in terms of lower prices, more diversified supply, improved quality and impact on economic activity. Efforts to proceed further with the reforms are meeting with a lack of enthusiasm, however. There are four factors that appear to be sapping their momentum: *i*) a fear that their "public service mission" will be compromised, *ii*) a concern that employees will lose their social benefits; *iii*) worries about whether the regulated industries can promote innovation and industrial development as they have done in the past; and *iv*) the perceived risk of "market failure" in a competitive environment. Initial experience in France as in other countries shows that these concerns can be answered in a competitive environment. It will be important to do so, considering that international expansion by traditional French operators would seem increasingly at odds with a domestic market that is relatively closed.[78]

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Maintaining the mission of public service

Traditionally, French network industries have been required to perform certain public services, *i.e.* they are to offer their services throughout the national territory at identical and generally affordable prices for the entire population. Assuring universal accessibility was the principal justification for the special legal status accorded to network industries, *i.e.* their monopolistic privileges and their ownership by the State.[79] The onset of liberalisation has evoked fears that this public service mission would be called into question. Yet the initial moves to open telecommunications and air transport show that specific measures can be taken to keep these services accessible to the large majority of the population.

In *telecommunications*, public service obligations were spelled out in detail before the sector was opened to competition. Several types of public services have been defined: the universal telecommunications service (telephone service, public telephone booths, information service, universal phonebook, special "social" rates, free routing of emergency calls, all at prices that are standard and affordable throughout the country); compulsory services (leased lines, integrated services digital network, and packet-mode data transmission), which must be available nation-wide but without any subsidies and which are financed jointly by all operators. Public interest missions (such as higher education and public research and development in the area of telecommunications) in contrast, are mainly financed by the State. The net financial burden that these universal service missions generate has been estimated and a special fund was created to finance the additional cost, to which all market operators contribute in proportion to their the volume of their traffic. France Telecom, which is responsible for providing the universal service, is compensated for the net costs incurred in providing it. The next stage will be to foster more open competition in fulfilling public service obligations, through more systematic tenders that will take account of the competitive advantages that participation in the services provides. The Telecommunications Regulatory Authority (ART) has gradually reduced its estimate of the net cost of fulfilling universal service obligations, which for 2001 amounted to the relatively modest sum of FF 2.7 billion.[80] ART monitors production costs and quality for these services in order to prevent overbilling.

In *air transport*, the consolidation of activity within a single group (Air France), which henceforth will pursue purely commercial objectives, is creating concerns about the continuity of services and their future price, particularly for destinations where demand is less sustained and less regular (Senate, 2001*b*). There are no formal public service obligations in air transport. However, an air service intervention fund (FIDA) is available to cofinance air service for cities or regions that might be excluded from commercial service.[81] Twenty-two routes were deemed eligible for support initially but few regional authorities have requested assistance from the fund to date.[82] There may be a need to provide further clarification of the minimum

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service objectives in air transport as well as the conditions of eligibility for the intervention fund.

The *Electricity* Law of 2000 spells out in detail the public service objectives in the sector. The first objective is to guarantee a standard price and quality of service throughout the country (including Corsica and the overseas departments and territories) for all non-eligible clients. The law provides for two equalisation funds for financing the additional costs of electricity production and distribution in disadvantaged regions.[83] The second objective concerns electricity as a basic necessity, which no household should be forced to do without, even in situations of financial distress. The 2000 law provides funding for this activity from the Electricity Equalisation Fund (FPE). The third objective of public service is to guarantee energy security over the short and long term[84] and to promote environmentally-friendly electricity sources. The extra costs generated by these objectives are split between electricity suppliers via the Electricity Generation Public Service Fund (ESPE) and between distributors via the FPE.

Projects for liberalising *railway transportation* are hampered by concerns over the future of public services in this area. Opening this service to competition and adopting "profit-oriented objectives" are seen by rural and outlying urban districts as likely to result in less frequent service, of lower quality and at higher prices. These concerns persist despite the fact that the SNCF has long since abandoned attempts to cross-subsidise fares and guarantee uniform frequency of service.[85] In an interesting new initiative, seven regions are now negotiating directly with the SNCF for the type of service they want and how to handle associated costs.[86] This financial empowerment of the regions has led to a clearer formulation of needs, greater transparency with respect to costs and a number of adjustments to service. Regional traffic grew by 12 per cent from 1996 to 1999 in the experimental regions, compared with 6.5 per cent in other regions and bus-rail competition in public transport is beginning to make itself felt in the context of regional transport schemes.[87] A decision to extend this empowerment to all the regions was recently taken under the Solidarity and Urban Regeneration Law of December 2000.

In *postal services*, public opinion and municipal authorities alike fear any backtracking from the post office's traditional presence in rural areas and the concept of "universal postal service" that today shapes most of the activities of *La Poste*. According to the Post and Telecommunication Code, this universal service includes "the offer of national and cross-border mailings of up to 20 kg, registered mail and insured mail. Pickup and distribution service is provided on all working days".[88] The post office also offers certain other services, the purpose of which appears to be more social than commercial, at net cost, although it is not legally required to do so. Chief among these are financial services to households and individuals who, because of their geographic location, the modest size of their accounts[89] or their legal history,[90] are in practice excluded from normal commercial banking services.[91] One study estimated that 3 million individuals use these

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services for lack of an alternative (Gautrat and Gounouf, 1995). If this service is deemed to meet a social objective that the post office is best equipped to provide; for accounting and financial transparency it would be advisable to define it formally and measure the net cost.

Concerns about the public service mission of rail transport and postal services generally leave little room for cost considerations and any reduction of service is viewed as a net loss to the community, whatever economies may thereby be achieved. Public debate tends not to take into account production costs, which are also a burden on the community. Discussions over future policy would benefit from a more accurate identification of the common services that are desirable to make available throughout the country, together with the appropriate financing sources.

Associating employees with the process of change

In the post-war period, public enterprises in the network industries (notably EDF, GDF and SNCF) served as a model for labour relations (Boitreau *et al.*, 1998). They provided their employees with work, pay and welfare conditions superior to those prevailing in the labour market. The special status of the personnel of public network enterprises stems from legislation of 1950 that laid down specific rules of employment for Air France, EDF-GDF, the SNCF and the RATP (Paris Transport Authority). Other aspects, not covered by the initial statutes, were incorporated subsequently. Current labour standards in these enterprises present a number of special features:

- A pay scale patterned on that of the civil service. Nominal wages are subject to across-the-board increases. Bonuses are paid for work in difficult conditions.
- Pension and health insurance schemes are special and generous. Conditions as regards pension computation, contribution periods and retirement age are more favourable than in the private sector.[92]
- An employment guarantee, not specified in either the EDF-GDF or the Air France statutes, is now recognised as a key labour standard in these enterprises.
- The interests of employees are represented in numerous bodies.[93] The Public Sector Act of 1983 allowed staff representatives to be elected to boards of directors. The right to strike, which is restricted in principle, is used more than in other sectors.[94]
- Their enterprises' works councils are the best-financed in France and offer considerable in-kind benefits.[95]

The future of this special-status employment in a competitive environment rightly prompts questions. Very marked differences between the labour standards of public enterprises and those of new market entrants do not seem

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sustainable. Increased productivity in the older highly capital-intensive incumbents that have already written off a great part of their investment may offset some of these differences, but only temporarily. The firmly entrenched tradition of special status explains why insiders are resisting privatisation.[96] Concerns of employees are matched by those of retirees, the generosity of whose special pension schemes would be exposed to the financial pressures of competition.[97]

Experience to date shows that these dilemmas are more easily addressed from the bottom up, by associating employees with the new prospects of business development afforded by a change in ownership status. So doing would help to mobilise employees around new principles of business operation and new corporate projects. The way the transition was tackled by Air France and France Telecom provides some insights. Their opening to competition was not without difficulties but was easier than originally foreseen. In the case of *Air France*, the first step was to obtain a consensus of the employees on the inevitability of opening up to competition and the impossibility of remaining competitive with the structures inherited from the past.[98] The second stage was a formal undertaking by the management to go ahead with the reform while avoiding: *i*) any layoffs and *ii*) any mandatory pay cuts. Next, a restructuring plan with a focus on productivity gains (30 per cent on average for the personnel as a whole) was submitted directly to the employees and received their approval. Finally, in order to contain wage costs, the management proposed a 15 per cent reduction of the wage bill in return for a distribution of 10 per cent of the company's share capital, a proposal that was generally well received. This was carried out in 1998 and 1999, pilots receiving 6 per cent of the capital in return for pay cuts and the other staff categories 4 per cent in the form of voluntary subscriptions at advantageous prices (20 per cent discount). Today Air France is the European airline with the largest employee shareholding. The operation enabled the management to put in hand an ambitious development plan that made the company more profitable and helped it to win market shares in 1999 and 2000.

The process at *France Telecom* was similar in a number of respects. Opening to competition was accepted by the employees as part of an aggressive corporate strategy, in return for an undertaking by the State to preserve their entitlements. Since the company had been formed from a government department (*La Direction Générale des Télécommunications*), employees were assured of retaining their status as civil servants – but newly hired employees will not have that status. Their civil service pension entitlements were also guaranteed under an arrangement whereby the State assumed the implicit pension debt in exchange for the company's payment of a lump sum (equivalent to 0.5 per cent of GDP in 1997). The employees, 92 per cent of whom took part in the operation, obtained 3 per cent of the share capital with a price discount of 20 per cent. As in the case of Air France, an aggressive development strategy carried this liberalisation project through and made France Telecom a world player in the space of three years. It

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has encountered some stock market turbulence in 2001, however, owing to the strain of its high indebtedness and limited profitability.

Air France and France Telecom are both operating in relatively dynamic markets. This may have made it easier for them to rally their employees around restructuring and development projects. Even so, when these companies began to expose themselves to market competition in the second half of the 1990s it was not at all clear that their efforts would be successfully, socially or commercially. Their entrepreneurial initiatives, led by motivated managers, were supported by the State with innovative schemes concerning employee entitlements. Continuation and completion of privatisation in network industries would do even more good (Box 5 and Table 7).

Maintaining innovation in the network industries

There are questions concerning the future of the technical and economic externalities traditionally provided by public corporations. These companies have played a pioneering role in certain innovations, contributing to the design, financing and testing of major technical systems which, over time, have become areas of French industrial excellence.[99] Innovation models based on co-operation between network firms, research laboratories and equipment manufacturers include nuclear technology supported by EDF, big aeronautical projects piloted by Air France, the digital communication techniques developed for France Télécom, high-speed rail transport system designed by the SNFC and the water treatment techniques developed by the water companies.[100] An emerging fear is that this technical capacity may suffer after opening up to competition.

One of the important aspects of government monopolies' technical ability was the security of supply that they gave France. Typifying this is the electro-nuclear programme which, in the wake of the first oil crisis, was able in a relatively short space of time to guarantee a reliable source of energy, independent of supply risks and at a cost and quality level that was recognised internationally.[101] Even when they imported their technologies,[102] the government monopolies had a time horizon for their economic calculations, a volume of demand and a capacity to co-ordinate interdependent activities that enabled them to internalise economies of scope in a way that might not have been possible in other countries. In some cases, this enabled them to launch big technical projects that proved commercially profitable. The high-speed train is one example that proved profitable on certain lines[103] (though not on others) by making the most of the public corporations' planning and co-ordination capacity. On the other hand, the public sector has also had its "white elephants" (expensive and rigid public projects which were difficult to stop).[104]

The network industries' innovative capacity can be preserved in a competitive framework. The big laboratories like the CEA (nuclear), the CNET

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Box 5. Privatisation: the current situation

Two major privatisations have been carried out since the previous *Survey*, together with a few other operations on a smaller scale:

- A new capital operation was put through in *France Telecom*. Intended to finance the acquisition of Orange at a total price of 39.5 billion euros, the operation included a share issue of 21.4 billion euros and a capital increase of 18.1 billion euros reserved for Mannessmann in exchange for its shares in Orange. The State's holding in France Telecom was thus reduced to 55.5 per cent, without generating any privatisation receipts.
- For the creation of the EADS aerospace company through the merger of *Aérospatiale*, DASA (Daimler) and CASA, the State reduced its holding through an initial public offering in June 2000. On the basis of assets contributed by several partners, the issue generated FF 7.3 billion in receipts for the State. The latter now owns 15 per cent of EADS (through the SOGEPA and SOGEADE holding companies).
- *Thomson* SA, a 100 per cent state-owned company, reduced its stake in *Thomson Multimédia* and *Thomson*-CSF. The stake in *Thomson Multimédia* was diluted at the time of the company's introduction on the stock exchange by a a share issue at the end of 1999, and then by the sell-off in 2000 of part of the Thomson Multimédia shares held by Thomson SA; the receipts of FF 4 billion from this sell-off were entirely used to reduce the company's liabilities. Part of the share capital of *Thomson*-CSF (renamed *Thalès*) was exchanged with Alcatel for the shares which the latter held in Framatome and which the State has recovered for the reorganisation of the nuclear energy industry. The privatisation generated no revenue.
- Other share sales took place in SEITA (Altadis), for FF 888 million, the Caisse Nationale de Prévoyance (where part of the State's share stake was transferred to the Caisse d'Épargne savings banks) and the SOFREMI company (where a small public shareholding of FF 2 million was sold to Thomson CSF).

Receipts from these operations are being used to make capital increases and share purchases in the public sector. Even though the small number and the special character of the privatisations carried out have restricted liquid receipts, inflows of FF 11.4 billion were recorded in 2000 (as against FF 17 billion initially foreseen). These receipts have been used to make the following capital increases:

- A capital increase of FF 7 billion in Réseau Ferré de France (French Railways). Total grants from the State over the past four years (1997-2000) amount to FF 42.5 billion.
- An additional capital increase of FF 2.1 billion in Charbonnages de France (French Coal Board). A total of FF 11.8 billion has been injected into CDF over the past few years.
- EPFR (Établissement Public de Financement et de Restructuration, the debt defeasance body of the Crédit Lyonnais) has received FF 1.5 billion. Since 1996 capital grants to EPFR have totalled FF 45.65 billion.

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Box 5. **Privatisation: the current situation** (*cont.*)

– Venture capital funds managed by the Caisse des Dépôts et Consignations and seed funds directly financed by the State have received endowments totalling about FF 190 million.

Following these sell-offs and reinvestments, the public sector at end-May 2001 covered the enterprises and activities listed in Table 7.

(digital communications), INRIA (applied mathematics) and the SNECMA (aeronautics) have remarkable material and human resources that they can turn to commercial advantage on an open market for applied research at both the national and international levels. Development contracts awarded by the government, international organisations and domestic and international companies can take the place of the exclusive orders placed by government monopolies, with whom they were virtually integrated. The move towards more open financing of large-scale applied research is a global trend in which France could participate by making the most of its specific resources.

Countering "market failures" with efficient regulations…

A fourth concern is that opening up to competition may result in "market failures", in particular an accentuation of the monopoly positions of network operators. Good international practice in countering such risks is still evolving. In France, the initial reactions. of operators exposed to competition in some cases confirmed these predictions. France Télécom and EDF (Air France's polices are less closely monitored) have on a number of occasions been called to order by the regulatory and competition authorities when they showed tendancies to raise prices in monopolistic segments. These risks were perceived as being more moderate in the historical model. In open markets, this type of threat is controlled by regulations governing competition that are based on collaboration between the competition authorities (the *Conseil de la concurrence* in France) and independent sectoral bodies specially created for this purpose. Regulatory bodies set up in the telecommunications (ART) and electricity (CRE) sectors, soon to be followed in the gas industry, have made a promising start. Competition authorities are also active in sectors other than the network industries (Box 6). Sectoral regulatory agencies do not operate in the aviation, rail transport and postal sectors,[105] and there competition monitoring remains the sole responsibility of the traditional competition authorities.

Regulatory agencies are needed for those activities in which the technology does not allow several operators to co-exist – *i.e.* natural monopolies.[106] However, in an evolving technological environment, the boundaries of a natural

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Table 7. **Main public or partially public enterprises and main public corporations**

	Principal activity	Stage holding[2] at 31/07/01 %	Market value of the company at 31/07/01 (euros million)
Listed enterprises with a direct or indirect State holding			
France Télécom	Telecommunications	55.5	60 748
Air France	Transport	55.9	4 306
EADS[1]	Aerospace	16,7	18 322
Crédit lyonnais[2]	Banking	11.7	14 990
Thomson Multimedia[3]	Electronics	35.9	9 730
Thalès (ex-Thomson CSF)[4]	Electronics	33.4	7 494
Renault	Motor vehicles	44.2	12 434
Dassault-Systèmes[4]	Automation	15.7	5 518
Bull	Computers	16.3	315
Caisse Nationale de Prévoyance	Insurance	1.2	4 996
Other public enterprises and corporations			
Framatome	Nuclear energy	19.7	
GIAT Industries	Plant and machinery	100	
SNECMA	Aerospace equipment	97	
Commissariat à l'Énergie Atomique	R&D	100	
EDF	Electricity	100	
GDF	Gas	100	
La Poste	Postal service	100	
SNCF	Transports	100	
RATP	Transports	100	
RFF	Rail way infrastructure	100	
Aéroports de Paris	Auxiliary transport service	100	
Caisse des Dépôts et Consignations[5]	Finance	100	
SAPR	Autoroute Paris-Lyon	98.9	
SANEF	Autoroutes du Nord et de l'Est	99.5	
ASF	Autoroute du Sud	98.9	

1. Held by SOGEPA and SOGEADE.
2. Of which 1.7% held by employees.
3. Held by Thomson SA.
4. Held directly since the merger of Dassault-Participations and Dassault Systèmes on 31 May 2001.
5. The holdings of the Caisse des Dépôts et Consignations are not included in the table. These holdings represent indirect public investment managed according to the financial objectives of a private investor.

Source: Ministry of the Economy, Finance and Industry.

monopoly can shift rapidly. For example, advances have made it possible to open up local telephone communications to competition ("unbundling of local loops"). In such a rapidly changing environment, regulators must be able to decide where the boundary lies between the two types of activity. The Electricity Regulation Commission (CRE) has made a similar effort to identify activities that can be opened to competition and decided in 2000 that "balancing management" and

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Box 6. The retail distribution sector also has competition problems

Competition-related problems do not exist solely in the network industries. A government Order of 1986 set up the *Conseil de la concurrence*, a body which is attached to the Ministry of Economic Affairs, Finance and Industry, but is independent in its decision-making.* The *Conseil* is responsible for the surveillance of competition conditions and is empowered to hand down decisions of a jurisdictional nature. In 2000, the *Conseil* imposed a record penalty of FF 1 billion on the banking establishments responsible for anti-competitive agreements in determining the contractual provisions relating to mortgage loans. By virtue of the Order of 1986, the *Conseil* was also given two other functions: to give the Minister for Economic Affairs and Finance opinions regarding mergers (a dozen opinions per year) and opinions on matters relating to competition (20 to 30 per year).

Competition conditions in the retail distribution sector have recently attracted the attention of the authorities. They focussed on the exercise of competition following the acceleration of acquisitions, mergers and integration in the sector. Surveillance of the sector was stepped up when *Carrefour* (1 229 shops and sales totalling FF 134 billion) launched a public offer for the shares of *Promodès* (1 979 shops, FF 112 billion of sales) in August 1999. The Minister for Economic Affairs, Finance and Industry asked the *Conseil* to investigate and it produced an analysis of the 99 local and regional markets that were affected by this integration and handed down an opinion on the proposed operation. The *Conseil* concluded that "the operation gave *Carrefour* a predominant position which was liable to be detrimental to competition" in 27 regions. On the basis of this analysis, the Minister for Economic Affairs requested *Carrefour* to sell 34 shops within one year, and authorised the merger on that condition. *Carrefour* pledged to undertake this restructuring.

A second case of competition in the retail sector concerned fuel distribution on motorways. Following the merger between Elf and Total, it was initially decided to sell some of the motorway service stations to a third party (*Carrefour*). However, both motorists and the authorities noticed a systematic alignment of prices. In 2000, two additional measures were taken with the object of making the market more competitive: *i*) greater transparency and contestability in the granting of service station concessions; and *ii*) using the resources of motorway companies to advertise on motorways the prices of petrol in service stations near to exits. Sixty different sites were chosen to begin posting prices on an experimental basis in August 2001.

* The *Conseil de la concurrence* is an independent administrative agency specialised in analysing and regulating market competition in order to safeguard public economic order. Its responsibilities and activity are described on the following Internet site: *www.finances.gouv.fr/conseilconcurrence/*.

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the supply of "smoothing currents" are activities that can be opened to competition. Up to now, this "active market-opening" role by a sectoral regulator has been performed neither in the air transport or rail transport system, nor in the postal system.

Price surveillance is another important regulatory activity. Prices in monopolistic segments must be approved to prevent any abuse of market power. This concerns both monopolistic markets (local telecommunications, local electricity distribution) for final users and "network cores", to which it is essential that other firms have free access in order ensure competition. Next, the price of "public service obligations" in the form of direct compensation to the firms that fulfil them, must be monitored to prevent any abuse. In France, regulatory authorities monitor closely both types of prices in the telecommunications and electricity sectors. The ART and CRE have developed advanced models for calculating costs, which they use to make their own price proposals and to evaluate the proposals by enterprises and the government.[107]

Lastly, regulatory authorities are responsible for taking steps to ensure that competition develops properly in the future.[108] They must oversee investment trends in network infrastructure and ensure that capacity grows satisfactorily so as to safeguard supply to consumers and access to new entrants.[109] The award of infrastructure licences provides an opportunity to verify the level of investment and, if necessary, to demand that it be increased, including by the provision of economic incentives. The authorities also need to form a long-term understanding of structural change in these industries, including the redrawing of the boundaries between them and other activities, and to address the competition issues that arise in consequence.[110] When the authorities have no legal mandate in an area that they consider to be crucial for the future, they have the possibility of making recommendations to policymakers and firms.

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IV. Towards ecologically sustainable growth

Introduction

Protecting the environment has become increasingly important in France over the last 30 years. Virtually ignored until 1970, the environmental dimension has gradually been integrated into government policy choices and public concerns. This interest in the living environment and in the management of natural resources has now become an important aspect of daily life. The public is insisting on better air quality, purer water, a more stable climate and better-protected natural areas.

As in other OECD countries, the objectives and instruments of French environmental policy have evolved considerably in recent years. In the context of transposing European directives, the initial stress was on the regulatory approach and, in particular, on respecting technical standards for eliminating or limiting emissions. This regulatory approach has brought about a considerable reduction in industrial pollution and untreated urban sewage discharges but probably not at least-cost. Compared with economic instruments that rely on the transmission of price signals, regulatory tools are rarely cost-effective (Box 7).

Recent years have therefore seen a growing emphasis on economic instruments, and in particular on environmental taxation and on the emergence of user prices that are more closely aligned with the costs of the externalities borne by society. This new approach has not been without its problems, however. It is not easy to move from the notion that the environment is a "free good" to an approach that takes account of the costs imposed on society. Farmers, truckers, industrialists and automobile drivers are for the most part strongly opposed to this approach. Energy-intensive firms particularly fear that they will lose international competitiveness, since there is no international co-ordination on the use of economic instruments. This chapter discusses these various problems, focusing on a few selected areas of environmental policy. After a brief review of the institutional context, the chapter goes on to discuss: *a*) the role of taxation, in particular for reducing greenhouse gas emissions, *b*) urban air quality policy, and *c*) issues relating to water pollution, especially of agricultural origin. The third section offers some conclusions and recommended actions.

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Box 7. The advantages of economic instruments

Emissions or discharges of pollutants constitute negative externalities, in the sense that the resulting costs are borne by the community and not by the agent who caused them. It is often possible to assign a price to environmental damage, representing the value to society of avoiding such damage. An economic instruments, such as a tax on emissions or a cap-and-trade scheme, transmits this cost to enterprises and obliges them to treat that cost in the same way as internal costs, such as those associated with energy or labour. Moreover, if prices of goods and services reflect environmental costs, households will take them into account in their spending behaviour. By "internalising" these externalities in this way, economic instruments can help reduce environmental damage and to do so at least cost.

A fundamental characteristic of economic instruments, in fact, is that by providing the same marginal incentive to all players they can equalise marginal abatement costs (*i.e.* the costs of emissions reduction) and thereby minimise the total cost of achieving a given environmental objective. By leaving it to agents to decide how they will reduce their polluting emissions and by how much, economic instruments in effect allow agents with the lowest abatement costs to contribute the most to pollution reduction efforts, something that regulatory-type instruments do not achieve. Moreover, while regulatory instruments must be re-examined and revised as technologies or economic activity evolve, economic instruments provide a permanent incentive to reduce polluting emissions.

The institutional setting

Responsibility for environmental policy in France has historically been shared among a number of players. At the central level, there are several ministries, besides the one specifically in charge of the environment that have responsibilities in the environmental field. Issues relating to marine pollution thus fall largely to the Ministry of Transport, the French Navy and the Ministry of the Interior. Questions relating to nuclear installations are in part handled by the Ministry of Industry, while the Ministry of Public Works, Transport and Housing is partly responsible for those relating to urban pollution. Several players, including an interministerial mission, are responsible for combating the greenhouse effect. The management of environmental resources is broadly decentralised among local governments and other decentralised agencies that have funding but do not always have regulatory powers. At the local level, the overlapping of responsibilities can lead to tensions, as in the relationships between the departemental prefects (representatives of the Central Government) and local elected mayors, both of whom have prerogatives over automobile traffic and air quality surveillance. A

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consensus on the choice of policy instruments to be used has not yet emerged, as illustrated by the recent debate on environmental taxation which pitted the finance ministry against the environment ministry.

The ministry of environment was created in 1971, but until the early 1990s it was subordinate to other ministries.[111] Early in the last decade the ministry was given regional directorates for environment and public entities, in particular the Agency for Environment and Energy Management (ADEME).[112] The Ministry of Regional Planning and Environment was further strengthened in light of the emergence of major environmental initiatives, such as the Water Law and the Waste Law of 1992 and the Air Law of 1996, which marked a reinforcement of environmental policies. Today the environment ministry deals with a wide variety of additional issues including the management of water resources, public wastes and discharges, climate change and regional planning. There is no environmental police to enforce regulations. As a result the environment ministry relies on the inspection services of other ministries (agriculture, public works) and has joint services with the industry ministry. This dispersal of responsibilities explains some of the shortcomings in the enforcement of regulation that were noted in the OECD's 1997 Environmental Performance Review.

The emergence of these new horizontal issues has served to enhance the regulatory approach – often based on Community directives – by focusing it on efforts to change behaviour through the use of economic instruments. A cultural shift of this kind is not straightforward, particularly in the case of agencies responsible for inspecting or supervising economic agents with the help of regulatory powers. This shift was symbolised by the creation, in 2000, of an environment ministry directorate responsible for economic studies and environmental evaluation, whose task is to ensure that the environmental dimension is taken into account in public policies, including those relating to energy, taxation, health and transportation. The result should be a greater use of economic analysis in these areas, which has been increasing over the last years but is still relatively undeveloped in France. The directorate is also supposed to supply the authorities with expertise in economic regulation of the environment. However, at the political level, the debate over the use of economic instruments for protecting the environment has just begun, and this is reflected in the difficulties encountered in implementing environmental taxation.

The use of tax policy to change the behaviour of economic agents is problematic because it collides with the tradition of earmarking the proceeds of environmental taxes to agencies that then pass them back to the payers. Water agencies, the role of which is described below, are thus designed to ensure a "fair return". Until recently, environmental tax proceeds were allocated to ADEME, which like many other agencies in France, operated outside conventional budgetary procedures. The allocation of environmental taxes to agencies that return them to

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the payers in the form of cleanup or "depollution" subsidies is not consistent with the "polluter pays" principle as defined by the OECD. The French authorities accordingly decided to put an end to the earmarking of these taxes, to bring the corresponding expenditures of ADEME into the budget of the environment ministry and to issue transparent criteria for allocating subsidies on the basis of a project's environmental and economic merits. This decision led to a sharp drop in the number of grants.

Ecotaxes and the greenhouse effect

France has gradually introduced a fairly broad range of environmental taxes and charges, such as for example water charges, waste taxes, taxes on atmospheric emissions or on noise. Generally speaking, these taxes aimed at, or having an effect on, the environment are out of line with environmental costs, which in many cases are still underestimated. They are thus not fully effective as incentives and the achievement of environmental objectives is left to regulatory instruments. A determination to offer equal incentives to all sectors in order to minimise overall abatement costs seems, on the other hand, to have been behind the proposed carbon/energy tax on intermediate consumption by businesses ("TGAP-énergie"). This tax, which was to have financed the cuts in social welfare charges for businesses, was expected to contribute significantly to the greenhouse gas emissions reduction plan for meeting the objectives of the Kyoto Protocol. The Constitutional Council rejected this plan in December 2000. The government is studying altenatives, notably the negotiation of voluntary commitments to reduce greenhouse gas emissions.

Existing taxes are not well aligned with environmental externalities

Energy and transportation

The domestic tax on petroleum products (TIPP) represents the EU harmonised excise duty and is the most important energy tax in France. In 2000, it accounted for about 11 per cent of Central Government revenues or 1.8 per cent of GDP. Transportation fuel taxes represent the bulk of these revenues (more than 90 per cent), although heating oil and gas are also taxed. Generally, these taxes, introduced for fiscal policy reasons, bear no direct relationship to the environmental externalities that fuels produce, in particular the carbon dioxide emissions that contribute to the greenhouse effect (and which are directly linked to the quantity of fuel used (Figure 25). Coal, which is the most polluting of fuels, is not taxed at all and thus enjoys an indirect subsidy.[113]

The level of gasoline and vehicle taxes is high in France as compared with North America and other EU countries. If the transportation system is to remain efficient, it is essential to cover marginal costs for every kind of transport (and not

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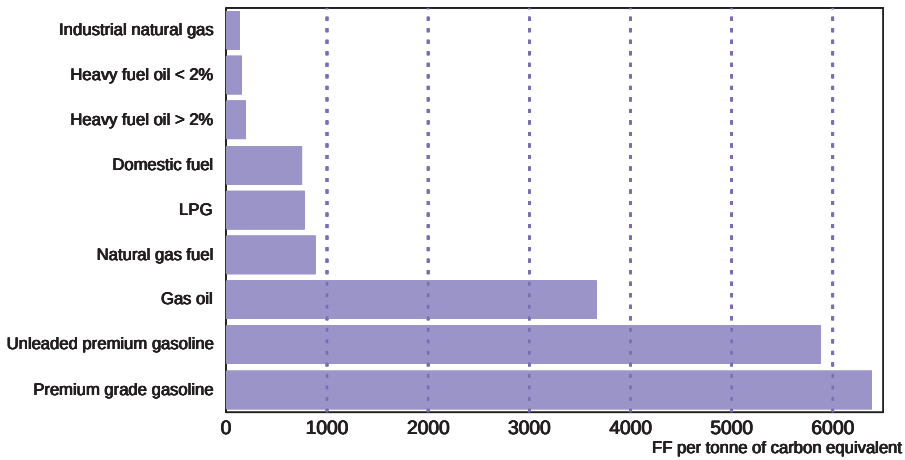
Figure 25. **Taxes on the carbon content of fuels**
January 2000



Source: Ministry for Territorial Development and the Environment, Impact Study of the bill submitted by the government to Parliament. Extension of the general tax on polluting activities to intermediate energy consumption.

merely average costs). Although very simple to administer, fuel taxes are not the best instrument for taking into account the external costs, in particular the environmental costs, associated with vehicle use, because (with the exception of CO_2 emissions) these costs vary sharply as a function of the place of use (city or countryside) and the technical characteristics of the vehicle (Figure 26). In particular, the various regulations concerning emissions standards have greatly reduced the level of pollution generated by a litre of gasoline in later-model vehicles.[114] Other social costs (infrastructure wear, congestion, noise, accidents) not included in the chart are thus becoming more important. The transport tax base should therefore evolve towards one that relates more closely to infrastructure use, location and environmental characteristics of vehicles. Technological progress should make this easier to achieve (see section on air pollution).[115]

There is, however, one area where fuel taxes provide incentives with negative fallout for the environment: the gap between diesel and gasoline taxation. This differential is relatively wide in France (Figure 27) and has encouraged a sharp shift towards diesel engines in the automotive fleet (Figure 28). This is a result, in part, of the attractive but misleading argument that diesel is more economical and that its use should be promoted[116] and it also stems in part from the pressure brought by transport companies. In terms of externalities, however, a litre of diesel pollutes more than a litre of gasoline (see Figure 26). In 1990, the

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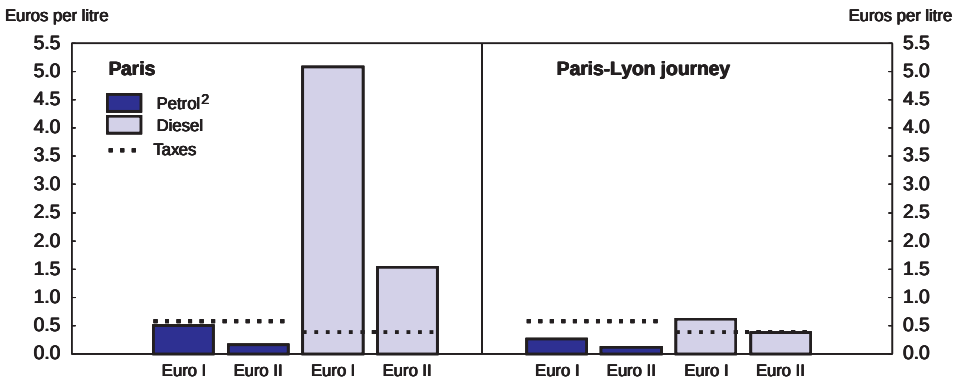
Figure 26. **Cost of air pollution by private cars**[1]



1. Cost for health, agriculture and buildings, of emissions of nitrogen oxides, particulates and carbon monoxide. An estimate of the costs of greenhouse gas emissions – 66 euros per tonne of carbon – is also included.
2. Euro I: 1993 European standard, Euro II: 1997 European standard.

Source: Spadaro and Rabi (2001).

government committed itself to reducing this differential to the average European level within seven years. After two years of reduction, the process was suspended in the autumn of 2000, following the sharp jump in oil prices, and despite their subsequent easing it has not been resumed.[117]

A number of tax exemptions or reliefs are provided for particular sectors, which are difficult to justify in terms of marginal social costs. Trucking companies enjoy a partial rebate of TIPP, which was significantly increased in autumn 2000 to a level representing 14 per cent of excise taxes for that year and has been budgeted at 10 per cent of all excise taxes for the year 2001.[118] At the same time, the rebate was extended to public passenger transport companies. This came on top of the special reduction in the TIPP (the "fiscal bonus") and the tax stabilisation mechanism, which allows consumers to recoup surplus VAT revenues following an oil price increase, measures that were taken to mitigate the impact of the rising price of oil.[119] Moreover, taxi drivers are exempt up to a limit of 5 000 litres per year for each vehicle. Public transit is also exempt from TIPP on LPG and natural gas, which is a relatively inefficient way of subsidising use of these fuels, since they do not have a zero marginal social cost (they produce CO_2, at least); incidentally, if they had no social cost, this exemption should be extended to all agents. The fuels used for air and sea transportation are also exempt, in accordance with the provisions of international agreements. Fuel consumption by farmers, moreover, is taxed at the home-heating rate, which is only one seventh of the excise tax on diesel.

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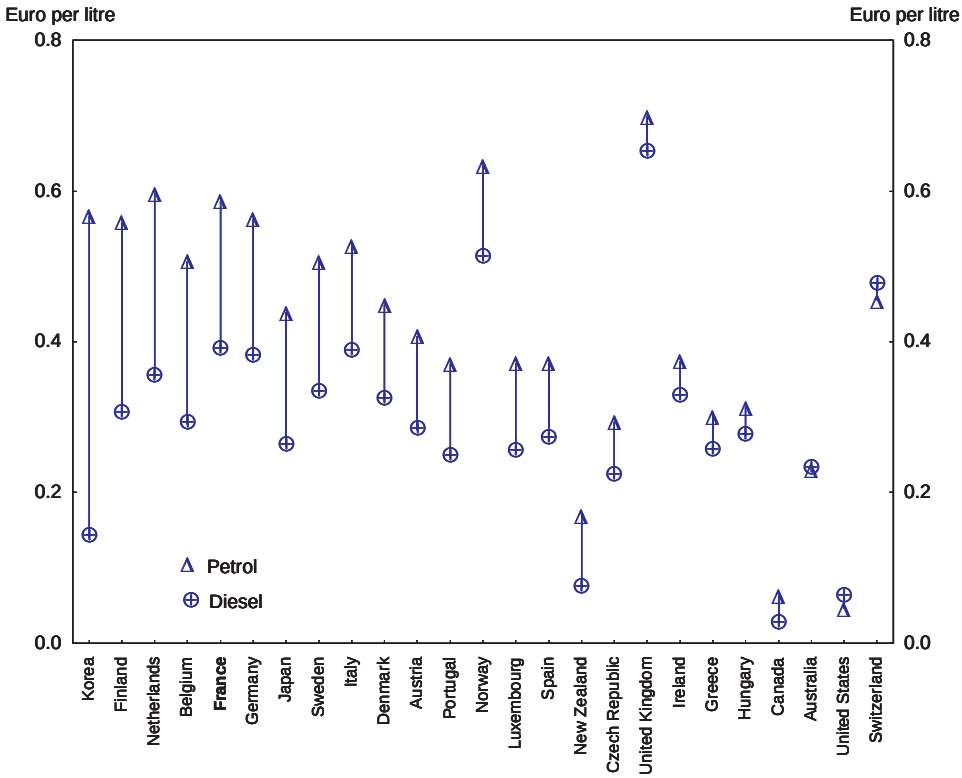
Figure 27. **Diesel and gasoline taxes**
1 January 2000



Source: OECD/EU database on environmentally-related taxes.

Another measure taken in the fall of 2000 was to abolish the annual tax on automobiles, that was amended in 1999 to differentiate between the emissions characteristics of vehicles; this is a further illustration of the lack of policy continuity in this area.

Directly targeted environmental taxes

Various taxes and charges[120] on polluting emissions or products (atmospheric emissions, wastes, base oils, aircraft noise) have been introduced since the early 1980s. The tax on stationary-source atmospheric emissions, for example, has been in place since 1985.[121] Most of these taxes, allocated to the ADEME

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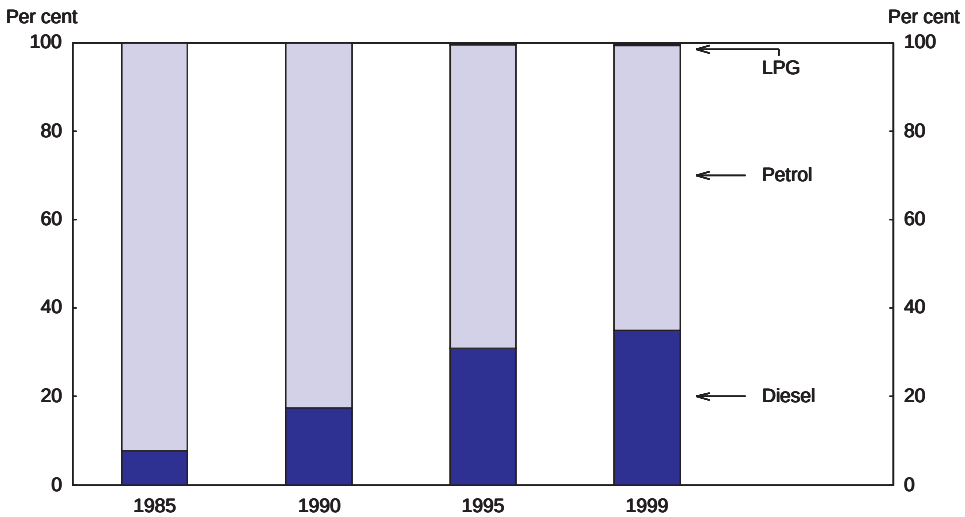
Figure 28. **Breakdown of the stock of motor cars by type of fuel used**



Source: Comité des Constructeurs Français d'Automobiles.

budget, were earmarked for pollution cleanup expenses and not intended to provide an incentive proportional to the cost of pollution-related damage. The charge on atmospheric emissions was used, for example, to finance emission reduction equipment, pollution monitoring networks and research and development costs. To achieve environmental objectives, regulatory instruments are used, such as atmospheric discharge permits or the setting of targets for recycling or energy-recovery incineration rates.

In 1999, these taxes (which amounted in total to about FF 2 billion) were grouped under the general heading of TGAP (general tax on polluting activities): they were assigned initially to the general budget, and since January 2000 to the FOREC (social contributions reform fund) that finances tax relief for employers on their welfare charges (following the example of other European countries), specifically in the context of the shorter working week. Their level and structure have not been modified,[122] however, and in many cases they fall short of what would be required to take adequate account of pollution damage, which often is still not evaluated from an economic standpoint.[123] The new pesticides tax, on the other hand, is highly differentiated to reflect the harmful effects of different categories of products. Generally speaking, the system of taxes and charges relating to domestic waste management also relies heavily on a financing-based rather than an incentive-based approach (Box 8).

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Box 8. Waste taxes: not much incentive

It is not all that easy to apply a system of "economic" management to household wastes. For one thing, waste-associated costs vary widely depending on the product and how it is treated. For another, it is difficult to transmit a price signal to agents that will induce them to modify their behaviour and so reduce the associated economic and environmental costs. Making producers responsible for handling the wastes arising from their products is one way of resolving the incentive problem. This has been used in France since 1993 for packaging waste reduction: producers or importers are expected to make provision for, or at least contribute to, the removal of packaging wastes, either through their own initiative or by contributing to firms such as "Eco-emballages". This company, in turn, undertakes to help municipalities establish selective collection and sorting systems and to valorise their wastes, either through recycling or through incineration with energy-recovery, at a rate of 75 per cent as established by legislation.

Because it offers a choice between two types of waste treatment, this objective is more cost-effective than a recycling target, since while it does not permit optimisation of treatment across all available options (including storage), it leaves Eco-emballages with the possibility of minimising costs. In practice, however, Eco-emballages has translated this goal into recycling targets for each material, without attempting to rationalise the approach through cost trade-offs. Moreover, contributions from firms are based on the weight of their waste output and on the different materials involved but the contribution rates have been set so as to finance the achievement of these recycling targets by type of material, rather than to reflect (as far as possible) the cost of collecting and processing the packaging and the associated environmental costs.

The greenhouse effect and green tax

Compared to other countries of similar size and level of development, France produces relatively few emissions of carbon dioxide, the principal greenhouse gas (Figure 29). This reflects in large part the country's heavy reliance on nuclear energy, which since 1970 has been seen as a means of meeting concerns over the security of supply following the two oil crises. Nuclear power now accounts for 80 per cent of electricity production. In the future, the nuclear share in electricity output is likely to decline in favour of gas, in particular, and the current level would seem to exceed what is judged to be economically efficient.[124] A recent forward-looking economic study of the industry, including environmental externalities such as CO_2 emissions and nuclear wastes (the evaluation of which remains highly uncertain) suggests, however, that the potential for substitution will remain greatly dependent on the trend of fossil fuel prices and that it would be worthwhile in any case to extend the life of existing nuclear power stations.[125]

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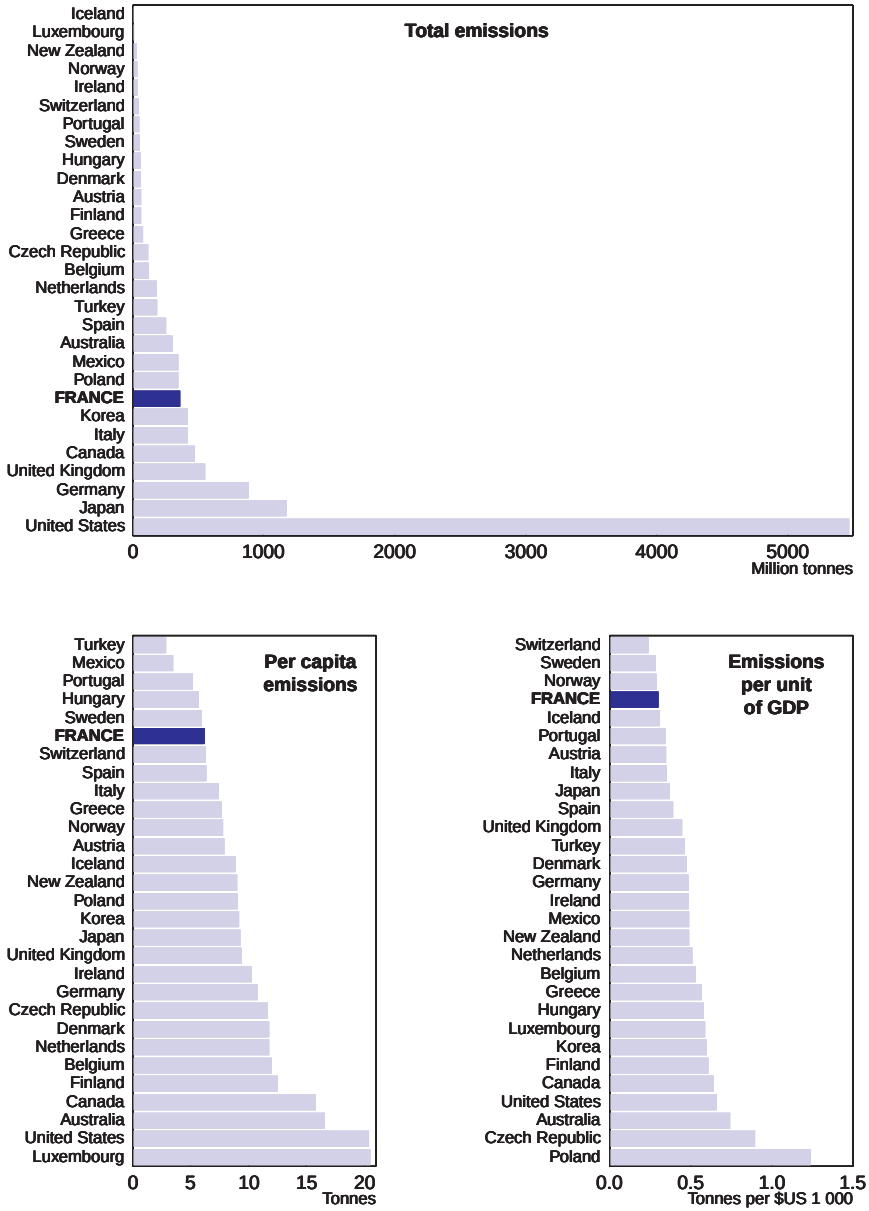
Figure 29. **Carbon dioxide emission intensities**[1]



1. In 1997.
Source: OECD.

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The target that France has negotiated within the European Union as a result of the Kyoto Protocol is to maintain greenhouse gas (GHG) emissions for the period 2008-2012 at their 1990 level. Emissions estimates for 2000 show that France has kept to the voluntary commitment made under the United Nations Framework Convention on Climate Change (UNFCCC) to hold GHG emissions in the year 2000 to their 1990 level.[126] Looking ahead to 2010, however, GHG emissions are expected to grow by 11 per cent, primarily due to transportation, energy production and, to a lesser extent, construction and tertiary activities. The level of carbon taxation required to stabilise energy-related carbon emissions at the national level, according to simulations, is FF 1 300 per tonne, which would imply a reduction in GDP of 0.2 percentage point to the year 2010. Resorting to the flexibility mechanisms allowed under the Protocol would significantly lighten this burden. Prior to the Bonn negotiations, it was estimated that, with an international carbon price of between 500 and FF 600,[127] domestic measures would have constituted slightly more than half of the reduction effort. The favourable treatment agreed at Bonn for carbon sinks should reduce the overall cost of the agreement, especially for countries with large forested and farming areas and bring about a sharp reduction in the international price of carbon, thereby also reducing the domestic effort to cut emissions.

To contain the growth of GHG emissions, a national programme to combat climate change (PNLCCC) was adopted in January 2000.[128] The Programme aims at stabilising French emissions in 2010 at their 1990 level, without using international trading mechanisms. Established under the aegis of the Interministerial Mission on the Greenhouse Effect (MIES), it identifies a large number of sector-specific measures (for industry, transportation, construction, agriculture, waste management, energy, etc.). Some of these reinforce and broaden existing measures: they are aimed primarily at strengthening energy conservation and reducing transport emissions, through regulations, subsidies and voluntary agreements,[129] and would be low-cost or "no regrets" (*i.e.* cost-free) or they would have significant ancillary environmental benefits.[130] The bulk of supplementary emissions reductions required to achieve the Kyoto objective would have to be obtained through economic instruments, which are thus a significant feature of the programme. Structural measures relating to transportation infrastructure and the promotion of renewable energy are to supplement the emissions reduction programme. The entire set of measures was based on a cost for each tonne of carbon avoided around 2010 comprised between FF 500 for tax measures and FF 1 000 for the others.

"TGAP-E*nergie*", *a missed opportunity*?

The introduction of a tax limited to intermediate energy consumption by businesses was motivated in part by the fact that businesses currently enjoy energy tax advantages over households[131] (Figure 25) and that they thus have

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room to reduce emissions at lower cost. The plan was to tax fuel inputs as a function of their carbon content, based on an initial rate of FF 260 per tonne of carbon, and at the same time to tax intermediate electricity consumption using the same rate applied to a hypothetical carbon content, equivalent to one-half that of the best gas-based production technologies. It was planned thereafter to increase the taxation rate gradually to about FF 500 per tonne of carbon by 2010.[132] This tax would have been added to the existing components of the TGAP and its revenues (expected to be about FF 4 billion) would have been earmarked for the fund financing social security expenditures. The Constitutional Council overruled this project in December 2000, however, on the grounds that the planned exemptions and abatements would create inequality among taxpayers and that the electricity tax was not sufficiently tied to the objective of reducing GHG emissions.

Compared with the carbon/energy taxes in effect in some other European countries, this plan had the advantage of initially offering the same marginal incentive for all businesses, while reducing the financial burden for energy-intensive firms with a tax allowance (rising steadily with a firm's energy intensity up to 95 per cent). The scheme was designed to limit the cost of the tax to a maximum of 0.3 per cent of value added, without making this a rule for calculating the tax.[133] In principle, this system amounted to a free distribution of emission credits on the basis of past emissions ("grandfathering"), used in most existing trading schemes, particularly the market for sulphur dioxide emissions in the United States. While it did not apply the polluter-pays principle as such, it did retain the efficiency aspects of an economic instrument: by transmitting the same price signal to all agents, equalising marginal abatement costs and minimising total abatement costs. This is not the case with the mechanisms in place in other European countries such as Denmark, Finland, Norway, the Netherlands, Sweden and the United Kingdom, which, by reducing the tax rate or simply exempting energy-intensive industries, transfer the burden to other agents who will most often have higher abatement costs.

A tax on electricity consumption is not a well-targeted instrument for reducing greenhouse gas emissions. It would have an impact on the demand for electricity, but not on the structure of supply (whether between gas, oil and coal, or between fossil fuels and other energy sources such as nuclear or renewable). From the viewpoint of global warming, it would be more effective to tax inputs directly as a function of their carbon content, in the same manner as intermediate energy consumption; this would also meet the objections of the Constitutional Council. Two considerations may explain why a tax on electricity consumption was considered preferable: reluctance to harm the competitive position of French production in an increasingly integrated European electricity market and a desire not to favour nuclear power over the longer term. The competitiveness argument is rather weak in the case of the French electrical industry, first because 80 per

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cent of supply is nuclear, on which the tax would have no effect, and second because neighbouring European countries (the only possible competitors) also have set greenhouse gas emission reduction targets. The concern not to favour nuclear power can be explained by the fact that the externalities to which it gives rise, in particular nuclear wastes, would not be properly taken into account. If this is the case, it would have been better to tax those wastes; the specific treatment of nuclear power in France makes it difficult, however, to implement such taxes, or even to propose them, as can be seen from the abandonment of the proposed charge for radioactive wastes discussed earlier.[134] The great uncertainty concerning the valuation of these externalities would also be an obstacle. In any case, the nuclear industry is in a situation of excess supply in the base market on which it sells and the risk at this time of favouring growth in the country's nuclear plant is very low.[135] If on the other hand the taxation of inputs implied creation of a rent for the nuclear industry, which would benefit from higher electricity prices without any increase in its own costs, this windfall could be taxed.

On the other hand, the proposal contained provisions that reduced its effectiveness. In particular, it held open the possibility for energy-intensive businesses to negotiate voluntary agreements to reduce emissions after 2002, thereby allowing the amount of tax due to be reduced, with an incentive equal to twice the normal tax rate.[136] Given the administrative costs of implementing such agreements (see below), this provision would have made the mechanisms significantly more cumbersome and complicated. If the objective was to preserve an incentive to reduce emissions beyond the allowance threshold, this could have been accomplished through a subsidy equal to the tax rate. If the intent was to alleviate the financial burden on enterprises still further, it would seem more appropriate to increase the tax allowance. Moreover, the many exemptions (government agencies, municipalities, agriculture, forestry, fisheries, railways, river ports, airports, etc.) were unjustifiable given the objective of minimising abatement costs; it would also have been desirable to subject household consumption to the TGAP, even if it meant overhauling the existing tax structure, which is currently higher for households than for businesses (see Figure 25).

The outlook

A number of possible alternatives to the TGAP have been suggested. One would be to abandon the fiscal approach in favour of voluntary emission reduction agreements, with the possibility of subsequent trading of emission credits. Another would be to combine this voluntary approach for energy-intensive firms with taxation for other firms. A third alternative would be to establish an energy/carbon tax for each branch of industry, the net proceeds of which could be zero. Using pollution and energy consumption indices by branch of activity, firms that pollute at levels below the "norms" would be lightly taxed, or could even benefit

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from a tax credit, while the heavier polluters would be more severely taxed. The latter could however reduce their tax burden by undertaking to cut their emissions. The government is currently studying alternatives to the greentax project struck down by the Constitutional Council and notably the negotiation of voluntary commitments to reduce greenhouse gas emissions with industrial firms.

International experience reveals a number of important drawbacks to the voluntary approach (Box 9). In the case of GHG emissions, in particular, which concern a great number of businesses and where abatement costs are potentially significant, economic instruments (taxes or cap-and-trade) are recognised to be superior and the voluntary approach would be less effective. Negotiated agreements most often deliver unambitious objectives, due to the difficulty of defining a reference scenario resulting from the asymmetry of information between public authorities and business. To avoid this, agreements should be negotiated based on a reference scenario established independently in accordance with a properly defined framework. Moreover, the negotiation process required to agree on sufficiently ambitious goals would be very costly. Introducing a market for emissions credits resulting from these objectives would lower the costs of reducing emissions for the firms concerned by equalising marginal abatement costs; it would not solve the problem of setting objectives inherent to voluntary agreements.

Combining a tax with a cap-and-trade scheme for energy-intensive firms (with "grandfathering" of emission credits, *i.e.* distribution for free on the basis of past emissions, if the objective is to cushion the cost to firms) that are those most likely to enter an eventual international market, could be an effective approach.[137] Comparing the market-established price against the tax would make it possible to check that the abatement cost gap is not too great and would allow the burden to be rebalanced if necessary. Ongoing EU work on the establishment of an allowance trading market for highly energy-intensive European industries is a move in this direction. It would be more effective, however, to extend the trading scheme to all emitters. To the extent that, for reasons of management cost, it can concern only firms of a certain size (current systems in the United States apply only to electric power stations, for example), such a system should be established upstream, at the level of oil product distributors. Emission credits should be auctioned[138] and a system of compensation could be established for certain sectors (in particular energy-intensive firms) to redistribute a portion or all of the revenues from sale of emissions credits in the form of a tax credit or subsidy, linked for example to the level of past emissions or the level of production. Such a system would in principle be equivalent to a carbon tax where the revenues are refunded to energy-intensive firms in the form of a tax credit or subsidy, not tied to current emission levels. It might take more time to implement than an environmental tax but it would have the advantage of being integrated more readily into the future international market for emission credits.

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Box 9. Voluntary agreements: an assessment

The OECD has conducted an evaluation of voluntary approaches to environmental policy (OECD, 1999*c*). In general, the environmental effectiveness of voluntary approaches and, in particular, negotiated agreements (involving commitments negotiated between an industry and the government) is modest. First, the evidence points to the key role of the industry in the target-setting process, the scope for a deficient participation of firms (free riding) and uncertainty over regulatory threats, leading to generally unambitious goals. Second, at the implementation stage, negotiated agreements perform poorly due to non-enforceable commitments, poor monitoring and lack of transparency. In addition, as far as economic efficiency is concerned, it seems that the burden-sharing between firms is driven more by equity considerations than by cost-effectiveness concerns. Finally, the usual claim that negotiated agreements tend to reduce administrative burdens is not confirmed either by empirical evidence or analytical arguments. In fact, when the number of potential participants is large, the costs of administration, control and implementation can be very high. Establishing a reference scenario for emissions, in particular, requires a great deal of time and effort.

An evaluation by Krarup and Ramesohl (2000) of voluntary agreements in energy policy in some European countries (Denmark, Germany, France, Netherlands and Sweden) provides similar conclusions. While the agreements reviewed had clear targets, no business-as-usual estimation had carried out before the schemes were implemented. In the end, targets were achieved, but they were found to correspond to outcomes that required no behavioural changes beyond what would have occurred anyway, if only to cut back on energy costs. Some agreements resulted in substantial cleanup, but at a high cost, while others produced modest improvements at lower cost. The effectiveness of voluntary approaches adopted for reducing carbon dioxide emissions, such as in Germany, Australia and Canada,* has also proven to be weak.

The OECD (1999*c*) offered some recommendations for action. Used as a complement to traditional command-and-control, voluntary approaches can play some beneficial role as they allow some flexibility in meeting targets. While rarely implemented, voluntary agreements could be used in association with economic instruments. Nevertheless, the target itself should be set outside the negotiated agreement, as in some of the Dutch agreements. Krarup and Ramesohl (2000) conclude as well that the most important thing is to set a sufficiently ambitious target on the basis of an in-depth analysis of the reference situation. Once this condition is fulfilled, possible safeguards against the major pitfalls in voluntary approaches could be: to set individual targets for each firm, establish reliable and clear surveillance procedures before the agreement is implemented, provide a credible regulatory threat, penalties for failure to respect commitments, and involvement of third parties in the definition of targets and monitoring of results. It should also be noted that if firms expect future introduction of a market for emissions credits (or tax credits) with grandfathering of credits based on emissions observed immediately prior to this introduction, the fear of being penalised in the future for emission reductions achieved now will be a sharp disincentive to participating in a voluntary agreement.

* *Cf.* OECD (2001*f*) for Australia and IEA (2000*b*) for Canada.

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A tax by branch of industry as described above would in fact be equivalent to the Swedish tax on nitrogen oxide emissions for the electricity sector[139] except that it would apply to GHG emissions and to all branches of activity taken separately. This could be interpreted as superimposing two tax credit mechanisms: an "inter-branch" mechanism that avoids penalising energy-intensive industries in comparison with other industries and another (intra-branch) mechanism that avoids penalising firms through retroactive taxation, while minimising adjustment costs. It would provide the same marginal incentive to all sectors but, in comparison with an energy/carbon tax with tax allowances, it would provide no incentive for capital to shift from more polluting to less polluting sectors and would not encourage structural adjustment towards a less GHG-intensive economy.

Urban air quality

Atmospheric pollution problems are particularly important in cities, given the high population densities and the concentration of economic activity. This is especially so in the Ile-de-France region (which includes Paris and where nearly one fifth of France's population lives on two per cent of its land), on which this chapter will primarily focus. Recent research confirms the harmful effects on health – effects that are not merely short-term[140] – and the sensitivity of public opinion to these issues is increasing. Now that industrial emissions have been largely brought under control,[141] the principal source of pollution is the transportation sector, and in particular road traffic. Until now, policy in this area has relied primarily on implementing Europe-wide emissions standards. Besides, public transport has been strongly developed and most French cities now have public transit systems (buses, streetcars, subways, trains) that are of high international standards. Measures to reduce road traffic, in particular through economic instruments, and to reduce vehicle parking have been less used, reflecting the difficulty of co-ordinating the players responsible, incomplete knowledge of their effects and the fact that mobility is often perceived as an inalienable right for which people should not have to pay. The Air Law of 1996 and the Solidarity and Urban Renewal Law of 2000 appear, however, to open new perspectives.

Objectives

The policy objectives of France's air pollution policy are derived from various European directives, in particular that of 1996, imposing concentration limits for several pollutants and setting air quality objectives expressed in terms of average concentration (annual and daily). Generally speaking, it is difficult to judge how air quality has changed, since, except in the Paris region, reliable measurements have been taken only since 1997.[142] Certain trends are emerging, however. Emissions of pollutants like sulphur dioxide and lead have been significantly reduced and the pollutants that pose a problem today, and which are likely to become of even greater concern as European standards are tightened

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between 2005 and 2010, are nitrogen dioxide and, to a lesser extent, ozone particles.[143] Projections indicate that the measures currently in place will not be sufficient to meet the new targets for 2010.

Reducing specific emissions from vehicles

The introduction of European regulations on engines and fuels has already resulted in a sharp reduction in emissions per kilometre travelled (known as "specific emissions") for private vehicles. This trend is bound to continue as the fleet is renewed, emission standards are tightened and the efficiency gains to which automakers are committed under the Auto-Oil Program become apparent.[144] It is estimated that between 1995 and 2020, specific emissions will be reduced by a factor of 5. The "dieselisation" of the automotive fleet, on the other hand, has significantly lessened the effect of specific emission reductions, since the very fine particles emitted by diesel vehicles are very harmful to human health (Figure 26). For the longer run, the recent development of the particle filter should make it possible to reduce these emissions, as the filter becomes more widely adopted and the fleet is renewed.

In order to meet the CO_2 emission reduction targets negotiated with automakers[145] and the "Euro IV" emission standards that will apply to vehicles in 2005, the European Commission is now proposing mandatory introduction of gasoline and diesel fuel with a very low sulphur content in 2005 and a complete switch to these fuels by 2011.[146] If this proposal is adopted, the most cost-effective way of doing so would be to offer a tax break on this kind of fuel, as was done in the case of unleaded gasoline.

At the same time as specific emissions have been declining, road traffic has continued to grow, reflecting both increased mobility and the growing market share of automobiles and heavy trucks in the transport sector. The number of vehicles and the number of kilometres travelled per vehicle have grown steadily with continuing urban sprawl, significantly diminishing the impact on the sector's emissions, which have nevertheless shown a downward trend in recent years (Figure 30).[147] In fact, urban sprawl and traffic growth have been encouraged by gains in engine efficiency, which reduce the cost of vehicle use, thereby demonstrating the limitations of a policy focused on efficiency and emissions standards. More recently, the authorities have been trying to introduce a better-integrated policy of transport development, notably through urban travel plans.

Traffic management

Co-ordination problems

Measures have also been taken to reduce road traffic. Relying primarily on public transit and parking fees, they have been motivated by concerns over the

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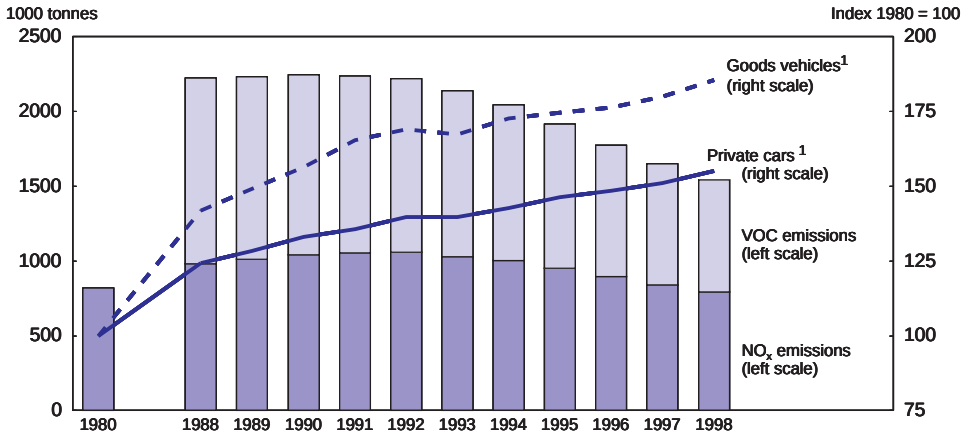
Figure 30. **Road-freight traffic and emissions**



1. Volume index of road traffic, expressed in billion kilometres travelled on the national road network by national vehicles. In 1980, it represented 239 billion kilometres for private cars and 55 billion kilometres for goods vehicles.

Source: OECD, *OECD Environmental Data*, Compendium 1999.

environment and the quality of life and also by the need to reduce congestion, which itself represents a significant cost. The impact of these measures, however, appears to be less well understood and appreciated than for those aimed at specific emission reductions and a comprehensive approach to these issues seems to be lacking. In part, the situation reflects the dispersal of responsibility for transport and urban development among different players, which makes it difficult to harmonise policies. Given their policing powers, mayors are responsible for regulating traffic and parking, except in Paris, where this duty falls to the prefect. The Ile-de-France region has a single authority responsible for public transport, the Syndicat des Transports d'Ile-de-France (STIF). There are many problems of co-ordination among these players and also among the communes or municipalities, in particular when it comes to promoting intermodal transport, such as by creating commuter parking lots ("park-and-ride"). The difficulty here is that the parking facility is dependent on the commune in which it exists, while the principal beneficiaries come from elsewhere.

The Air Law of 1996 made it compulsory for cities of more than 100 000 to adopt five-year urban travel plans (*plans de déplacement urbains*, PDU) as a means of reinforcing these measures and strengthening their coherence. The Solidarity and Urban Renewal Law (SRU) of 2000 also reinforces the technical competence of the inter-communal authorities responsible for the parking and circulation aspects of transport policy. In order to balance mobility needs and ease of access, on one

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hand, and environmental protection and health concerns on the other, the PDU aim at reducing automobile traffic, expanding collective transportation and organising parking lots and fees. Currently, their intermediate objectives and the quantifiable effects of planned measures on traffic, emissions and noise reduction are unclear, which makes it difficult to assess them. The PDU for Ile-de-France is the only one, so far,[148] to have established quantitative objectives for reducing automobile traffic (by five per cent over five years for Paris and the *petite couronne* or near suburbs, and by two per cent for the remainder). However, it has yet to establish quantitative environmental goals. Nevertheless, the government has provided grants to ADEME and to CERTU (a study centre attached to the ministry of transport) so that they can help local authorities to evaluate and take account of environmental aspects in implementing their PDUs.[149]

Regulatory measures and the supply of transport infrastructure

In the past, few of the country's larger urban centres have adopted the goal of reducing road traffic. Indeed some of the measures taken, such as prohibiting parking on the major thoroughfares of Paris and developing peripheral highways as ways of preventing congestion and facilitating movement and access to the city centre actually encourage automobile use over the longer term.

The only traffic-restricting regulation currently in force in France is an emergency measure. When pollution alert thresholds are exceeded, an "alternating car use" rule is activated whereby only vehicles with even-numbered license plates may travel on even-numbered days and only those with odd-numbered plates on odd-numbered days. This rule has been evoked on only two occasions. Moreover, vehicles bearing a green sticker (indicating that they are relatively less polluting), which account for a significant portion of all cars, are not subject to the rule. As a long-term traffic regulating instrument, measures of this kind are unlikely to be very effective: they are unable to take proper account of different people's travel needs and experience in other cities shows that they can be circumvented (for example, by buying a second car).[150] Municipalities are more inclined to physical traffic control measures, such as widening sidewalks, creating bicycle lanes, and reserving lanes to public transit vehicles. By making roads less accessible to vehicles, these measures exacerbate congestion over the short term. In the case of sidewalks and bicycle lanes, the additional short-term congestion cost falls on public transport as well. By contrast, the extension of dedicated bus-only corridors recently undertaken by the city of Paris places congestion costs squarely on private vehicles, and enhances the efficiency of bus travel.

The supply of transport infrastructure also conditions traffic growth. Currently, transportation receives more public investment than any other commercial sector, a fact that reflects the scale of the non-market advantages that this offers. In Ile-de-France, investment during the 1990s was distributed in roughly equal

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proportions between highway infrastructure and public transport.[151] Since 1994, in light of the conclusions of a working group of the *Commissariat Général du Plan* (General Planning Commission), the evaluation of transportation projects (road, rail, or urban with exclusive right-of-way in the Paris region) has included effects such as atmospheric pollution, road safety problems, noise and urban congestion, on the basis of standardised criteria.[152] These criteria have been recently revised (Commissariat Général du Plan, 2001). In particular, the value of time, which had been criticised as too high and as overshadowing other values, was reduced. To take account of the greenhouse effect, a value of 100 euros per tonne of carbon has been set, consistent with the values produced by carbon price models for the year 2010 assuming implementation of the flexibility mechanisms allowed by the Kyoto Protocol (the favourable treatment recently accorded to carbon sinks during the Bonn negotiations in July 2001, however, suggests a substantially lower international price for carbon). The value of human life has been increased, as has the assessment of the health impact of atmospheric pollution, to reflect the conclusions of the recent studies referred to above. Overall, the new criteria should allow to better take into account external effects when evaluating transport projects.

The central government helps to finance urban public transport, notably in Ile-de-France. The purpose of its aid is to rectify the conditions of competition between transport modes due to under-pricing of road transport and to help the local authorities respond to environmental concerns, like the greenhouse effect, that exceed their geographic area of competence. A reform of state assistance given to provincial public transport is under way that aims to increase their environmental rating. Under this programme, an aid from ADEME would be provided to quantify the environmental impacts. The allocation of subsidies will be made conditional on a socio-economic evaluation taking into account the environmental effects, based on the above mentioned criteria.

Parking fees

Parking fees are one of the instruments used in attempting to reduce automobile traffic in major cities. Surveys have shown that the availability of a free parking space at one's destination is a strong incentive to automobile use. In Paris, for example, the use of parking meters was greatly expanded between 1989 and 1997 in all neighbourhoods, and now nearly 70 per cent of street parking spaces are metered. This policy seems to be directed primarily at reducing use of the automobile to get to work, since parking is limited to two hours, except (in general) for local residents.

The Parisian system would appear to provide inadequate incentives, however. Since 1993, three different hourly rates have been in force (initially FF 15 in the central *arrondissements*, FF 10 in the intermediate *arrondissments* and five francs

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in the outlying *arrondissements*) except for residents, who pay FF 15 per day. However, with the fine for non-payment set at FF 75 and for illegal parking at FF 230, the incentive to respect the rules is weak, in contrast to other cities such as London where the fine can be as much as 40 times the hourly parking rate. Given infrequent surveillance, a driver who wants to park for the whole day in Paris will nearly always find it worthwhile to pay the fine rather than go to a parking lot or feed a meter. Moreover, compared with real estate prices, parking rates, as a charge for land use, would seem to be both too low and inadequately differentiated. Estimates suggest that if parking rates were based on market prices for land, office rents and housing, as an approximation of the value of the space occupied by a vehicle, the average hourly rate would be FF 21, with sharply higher differentiation among neighbourhoods.[153] In relative terms, it may also be noted that since 1993, parking rates have gone up only in the central areas of Paris and, even there, the increase has been significantly less than that for public transit passes. On the other hand, the recent decision of the city of Paris to cut the resident parking rate to FF 3 (0.5 euro) a day, by increasing the gap between the cost of parking at home and at work, should significantly diminish the incentive for residents to use their car to go to work.

Another problem in terms of parking management policy is that the elasticity of vehicle use to the price of street parking depends heavily on the existence of alternative parking options. Private parking lots (garages attached to housing, employers' parking lots, commercial parking lots) have multiplied. The offer of parking, for the most part free, by employers is particularly harmful because it encourages workers to commute by car. Although some countries and some foreign cities are attempting to restrict employer-sponsored parking, in France it has been strongly favoured by building construction codes that impose minimum parking space requirements for new projects throughout a municipality, regardless of the status of public transport facilities. Since 2000, however, the new Solidarity and Urban Renewal Law authorises the PDU to cut back or eliminate these requirements, depending on the availability of public transport, or even to set a ceiling on the number of spaces offered. Thought should also be given to having employees pay for such parking.[154]

The urban tolls option

In any case, while parking fees may help reduce automobile traffic and take greater account of the social costs of automobile use, they are not a well-targeted instrument for addressing the externalities of congestion, pollution and noise nuisances. The marginal costs that vehicles generate vary greatly depending on their technical characteristics and their time and place of use (Figure 31) and only a system of urban tolls on individual vehicles can allow for real internalisation. There are currently three systems of urban tolls in France[155] but they relate

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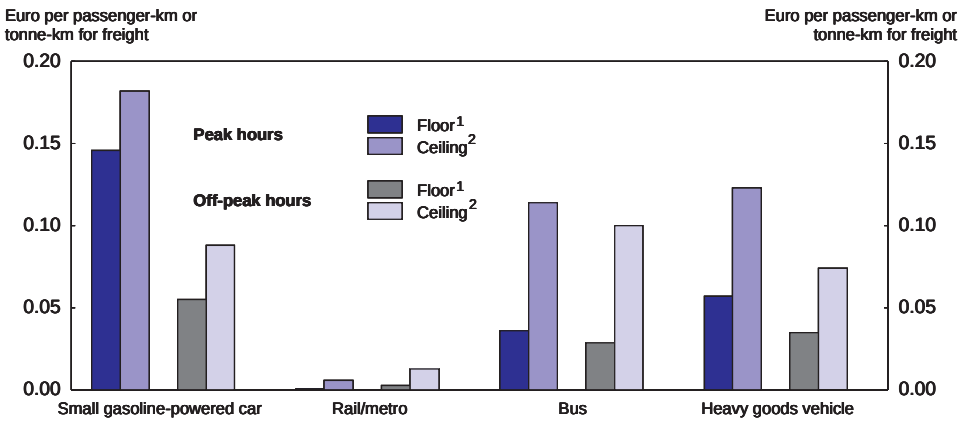
Figure 31. **Marginal external cost of various transport modes in the Ile-de-France region**



1. The pollution values used are those of the ExternE project for urban areas outside the Ile-de-France.
2. The pollution values used are those of the ExternE project for the Ile-de-France, which according to the authors are higher than obtained by other studies of the area.

Source: Quinet, E. and J.P. Taroux (2000).

only to the use of recently built facilities and were instituted primarily as a financing device. Norwegian cities have toll beltways but toll rates are generally not differentiated by type of vehicle or hour of use. A new project in Singapore has tolls that are differentiated by time of day in order to minimise congestion.[156] In the Netherlands, an electronic toll road is to come into service in 2002, aimed primarily at reducing peak-hour congestion on the motorways linking the four cities of Amsterdam, The Hague, Rotterdam and Utrecht. If such a system were to be effective as an incentive in the Paris region, it would have to apply to the entire area where congestion, pollution and noise are considered problems, *i.e.* at least Paris proper and the near suburbs.

The introduction of such a system is now technically feasible. Equipping vehicles with electronic transponders can avoid congestion at toll booths and can differentiate toll rates according to selected criteria. GPS-type systems such as the one now operating in Washington DC (on a voluntary basis) for automobile insurance can also be considered. Compared with a "belt" system, they would reduce the possibility of induced effects, such as on real estate prices, which might be sensitive to the placement of the belt's limits. The cost of implementing either of these systems is still high[157] but it should decline in the future. The social acceptability of measures of this kind is very low, partly because mobility is considered as a fundamental right that should not depend on the ability to pay.[158] Yet this argument needs to be viewed in light of the fact that tolls will merely shift to

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vehicles the social costs that they generate. In any case, the problem could be lessened by ensuring the availability of an efficient and high-quality public transit system (remembering as well that the performance of public transit, especially buses, will be improved by the reduced congestion resulting from tolls).

Water quality

The management of water resources is an area in which France has traditionally been in the lead and watershed or river basin management agencies have been in place since the 1960s. After several decades of policies to reduce water pollution, significant progress has been made in controlling pollution of industrial and household origin but agriculture-induced pollution and, in particular, nitrate pollution continues to grow. Reflecting intensive livestock operations and the use of fertilisers and irrigation,[159] this problem is of primary concern in north-western France, especially Brittany, where water-borne nitrate levels are high and steadily rising. Ever greater numbers of drinking water extraction points exceed European standards, even though pollution charges paid by consumers are increasing. The costs to the environment and to economic activities dependent on it, such as tourism, are also significant, especially in Brittany, where "green tides" or algae infestations are frequent.[160]

Having made a good start, the water agencies are now the focus of debate. Although founded with the intention of implementing economic pricing that would force users to recognise water as a scarce resource by paying a water extraction fee and applying the polluter-pays principle in the form of a pollution charge (which required creation of institutional arrangements for each river basin), the system of water agencies seems to be operating essentially under a logic of financing water management expenses.[161] Moreover, farmers have in effect been exempted from the pollution charge, so it is the final consumers of water, those who are least represented on the river basin committees, who have had to bear the financial burden associated with agricultural pollution. Generally speaking, water pollution control policy (charges, regulations, subsidies) has failed to take account of agricultural pollution issues. Moreover, undercharging farmers for water and the provision of farm subsidies act as incentives to pollution. The draft Water Law adopted by the Council of Ministers in June 2001 aims to remedy some of these problems, and, in particular, to reform the system of water charges so as to base water-management activities on new principles.

Operation of the water agencies: an approach far removed from the polluter-pays principle

The water management framework in France is fairly complicated, because it is shared among many players. Water distribution and treatment are the responsibility of the municipalities, as is the fight against pollution and the

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maintenance and restoration of watercourses, since the 1992 Water Law. Yet the municipalities often resort to private operators for managing the distribution network. The price of water is set by the municipality or the delegated group. Moreover, with the 1964 Water Law, France is the first OECD country to have adopted a decentralised system for integrated river basin management, a system that lay behind the recent European Union water directive.[162] The river basin committees, consisting of representatives of the Central Government, local authorities and water users (industries, farmers, consumers, associations), define the objectives for their river basin, consistent with national water policy and European directives, which are applied through both regulatory and economic instruments. Regulatory action relies on the regional offices of the Central Government, such as the regional directorates of industry, research and environment in the case of industrial wastes, or the departmental directorates of agriculture and forestry for farm charges and pollution control. On the other hand, the economic incentive aspect, in the form of subsidies for reducing pollution and charges for water use and pollution, is the responsibility of the water agencies,[163] which thus play a key role.

Pollution charges

In principle, each agency sets a base rate for nine different types of pollutants, which may then be revised as a function of the geographic zone to take account of the state of the environment. These rates are then applied to various categories of users, through different methods. Pollution charges are rather hybrid in nature, somewhere between an earmarked tax and a service charge, but their payment is not directly linked to the service provided. For industries, the calculation base is estimated or measured pollution:[164] in contrast to the system prevailing in some other OECD countries, where the calculation base is the volume of water consumed or discharged, this should in theory give firms a real incentive to reduce pollution.[165] However, certain industrial activities or types of polluting discharges are virtually exempt from charges while others, although subject to payment, benefit from significant discounts.[166]

It is difficult to design an instrument to take proper account of agriculture-sourced pollution. Whether produced by livestock or crops, nitrate pollution is hard to measure. First of all, in the case of crops, it is diffuse in nature. Secondly, the impact of fertiliser and manure runoff on the quality of water varies as a function of the climate, hydrological characteristics, the nature of the soil, the type of crop and farming practices. Measuring these pollutants individually, which would be necessary to develop an incentive instrument, is therefore very costly. In principle, pollution charges would have been applied to large livestock undertakings as of 1975; technically, they would have paid the charge not on pollution as such but on output of nitrogenous matter. In practice, they were entirely exempt until 1992, and have been *de facto* exempt since then.[167]

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For household users, it is nearly impossible to implement an incentive-type pollution charge. Because it would be so expensive to measure the volume of household pollutant discharges, pollution charges are usually calculated on the basis of water consumption, on the assumption that the cost of treating wastewater is proportional. In France, the basic rates are applied to a flat estimate of pollution produced per inhabitant per cubic meter of water consumed.[168] This calculation is complex and involves a number of multiplier coefficients, including the so-called "collection" coefficient, instituted to finance wastewater systems.[169] Yet the "collection" coefficient applies only to final consumers of water, who are relatively underrepresented on the river basin committees and hence on the agencies' boards of directors and not to industries connected to the sewage system.

Earmarking of revenues and lack of economic evaluation in allocating subsidies

Earmarked taxes are problematical. They stand in contradiction to the polluter-pays principle, according to which the polluter should, at least, pay the (marginal) cost of cleanup and, at best, the (marginal) cost of all externalities associated with the pollution (which would mean, for example, in the case of farm-sourced pollution, the cost of denitrification and the costs borne by the tourism sector, local municipalities, etc.). Pollution charges should not be viewed as creating a right to receive subsidies in return but rather as a cost of production like any other. Most importantly, to the extent that they fix the amount of expenditures to be made in the area concerned (equal to revenue from charges), they do not allow the optimal amount of these expenditures to be assessed in light of economic and environmental criteria. Historically, the earmarking of water charges to finance subsidies for investment in this area, often cited as the principle that "water pays for water", and the fact that charge-payers are involved in the decisions taken in this area, in particular through the river basin committees, have certainly played an important role in making these charges acceptable, particularly to businesses. Yet as was shown in the 1997 evaluation report on agencies conducted by the *Commissariat Général du Plan*, the system has shifted towards essentially a "financing" focus, far removed from the principle of economically efficient pricing. If cleanup efforts are to be subsidised, this should be done independently without linking the amount of subsidies to the proceeds from water charges.

The agencies are under no obligation to conduct an economic analysis of the projects they subsidise and no such analysis is undertaken. In general, subsidies are awarded on the basis of pre-established criteria set forth in the five-year plan, which favour certain categories of beneficiaries and projects (for example, municipal rather than individual water treatment works and large projects). There are special programmes for municipalities or their planning and water treatment

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associations (*syndicats*); private contractors in charge of distribution and network management receive subsidies for investments that they would normally have to finance themselves without any guarantee that they will pass reduced costs on to consumers. Any project within the eligible categories is entitled to a subsidy and the only limit is the size of the agency's budget. The distribution of subsidies is also heavily influenced by the makeup of the river basin committees and in practice ensures industry a certain "rate of return". Thus, industry has received subsidies under the various programmes roughly equal to the charges it has paid (14.5 per cent of subsidies against 13 per cent of pollution charges paid[170] for all agencies under the 1997-2001 programme), something that would probably not have occurred if the system were designed to produce the best economic and environmental results. On the other hand, agriculture has been receiving a growing share of the subsidies (6 per cent for the 1997-2001 programme), yet has paid virtually no charges.

Finally, the water agencies still have few quantified objectives, so that European directives remain more constraining than the French plans. Subsidies are often provided for actions to implement these regulations, in particular the urban wastewater treatment directive, which establishes obligations in terms of wastewater collection and treatment facilities, rather than in terms of water quality. The scope of subsidies granted for certain projects by the agencies and other public bodies, by reducing the burden on the municipalities or *syndicats*, has probably led to accepting policies that have not been sufficiently evaluated. Thus, the development of rural sewer systems has been criticised by environmental groups who believe that individual sanitation facilities are generally more effective.[171] Moreover, local environmental trends are not sufficiently monitored and so it is impossible to assess the impact of the investments subsidised by the agencies (Commissariat Général du Plan, 1997).

Agriculture and water: deficient policies

The deterioration of water quality as a result of nitrate leaching is explained in part by the fact that the farm sector is under no real pressure to take account of its effects on the environment, either through pollution charges or other environmental measures. If these deficiencies are difficult to justify at the economic and environmental levels, they result from the willingness to preserve economical and social balances in the agricultural world. Moreover, the current farm subsidies system (including Community subsidies) provides strong incentives to intensive farming, and hence to pollution. Indeed, a recent evaluation of public policy for water quality – undertaken at the request of the National Evaluation Council by the Commissariat Général du Plan – notes that environmental concerns still play a minor role in agricultural policy despite the importance of the problem and stresses the necessity of an in-depth reform.

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Very little internalisation

In light of the 1991 European directive on nitrates of organic origin, two major measures were taken. First, maximum limits were set on the land application of organic nitrogen-containing substances in areas classed as nitrate-vulnerable ("in structural excess"),[172] *i.e.* those where the levels of livestock effluents and the size of the available farming area are such that the ability of land and crops to absorb more manure is exhausted. In France, these limits were recently tightened.[173] Inasmuch as they relate to gross nitrogen use, and take no account of the absorptive capacity of the local setting, these limits are not very effective from the environmental viewpoint. Moreover, there is no regional differentiation in terms of the associated environmental damage. Finally, these regulations are poorly implemented, reflecting inadequate supervision efforts (many agencies are responsible for enforcing them, and the subnational monitoring and policing systems are fragmented),[174] even if the resources devoted to the water police have recently been increased.

Regulations governing expansion permits for livestock operations have also been poorly implemented and manuring plans are not well supervised. In April 2001, following legal action taken by the society contracted to provide water services in the Guiguamp area in Brittany, the administrative court of Rennes judged the State responsible for not applying the regulations concerning agricultural classified facilities.[175] The ministries of agriculture and environment undertook an inquiry in July 2001 into the operations of the regional health councils in Brittany, for example, after a member of an ecological association[176] reported that permits for the expansion of intensive indoor livestock operations were being granted automatically, even when those operations were non-compliant with the regulations.

In 1993, when large-scale livestock operations became in principle subject to the pollution charge, the farm-source pollution control programme (PMPOA) was launched to deal with the problem of effluents by upgrading buildings to be used for manure storage pursuant to the regulations on classified facilities. Since the degree of nitrogen pollution from manure depends on when it is spread, the idea was to store the substance until the right moment.[177] A recent interministerial assessment found, however, that the programme is not very cost-effective.[178] In fact, it subsidies 65 per cent of the cost to bring buildings in line with environmental regulations but also allows a farm modernisation that is likely profitable for the farmers concerned. Moreover, the programme appears to be costing more than it should.[179] As a national programme, it does not focus on high risk zones, where the environmental benefit of reducing nitrate leaching would be most significant. The programme was suspended in December 2000 following a decision by the European Commission that it was inconsistent with state-aid rules, and the government plans to reform it by focusing aid on high-priority zones in respect of

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water quality. When it comes to crops, no pollution charge is being collected or is collectable. In addition to the earlier-mentioned regulations on manure spreading, there are voluntary agreements for improving farming practices, such as the "Ferti-mieux" programme launched in 1991, which covers only a small portion (about 11 per cent) of arable lands and is difficult to assess in terms of cost-effectiveness.[180]

Environmentally harmful subsidies

The agricultural demand for water has grown sharply with the expansion of irrigated farming. From the 1960s to the mid-1990s, the area under irrigation in France increased by a factor of 4.5, compared to a doubling in the European Union as a whole. This expansion in irrigated farming can be explained in part by under-charging for water and subsidies for irrigation projects. Some water extractions are not recorded, there is little metering (half of extractions are paid at a flat rate) and the charge applied to metered withdrawals is low. All of these factors mean that, as in other OECD countries, the farm sector pays significantly less for its water than do other sectors (Figure 32). All told, metered withdrawals pay an average charge that is only one-fifteenth of the rate paid by municipalities,[181] a discrepancy that is certainly too great to be explained by differences in water quality.[182] Irrigation works are also directly subsidised through public grants and through Community grants specific to the irrigated area (see below), to as much as 65 per cent.[183] Apart from the impact on resource allocation efficiency, this also has consequences for the environment, since irrigated farming is input-intensive and generally more polluting.

A more fundamental problem and one that goes well beyond the efficacy of farm-sourced pollution-control policies is the issue of agricultural subsidies themselves. Subsidies to agriculture are high:[184] direct grants to agriculture (*i.e.* outside price support programmes) represented about 60 per cent of French farm income in 1997. The reform launched in 1992 in the EU as a whole has begun to redirect these subsidies[185] but they still go primarily to support production and in particular the more intensive and more polluting forms of production.[186] Moreover, French subsidies make a distinction between irrigated and dryland farming that has greatly favoured the former. Since 2000, however, subsidies to irrigated farming (and not dryland farming) are conditional on compliance with certain environmental requirements, which should lessen their negative impact.

The reform also includes environmental measures, offering financial compensation for cost overruns and income loss to farmers adopting environmentally friendly techniques developed and implemented in Member States. However, these grants are not distributed on the basis of environmental priorities. In France they are distributed among the regions on the basis of exclusively agronomic criteria (number of operators and hectares under cultivation) and not concentrated on environment priority zones or on the agricultural activities most harmful

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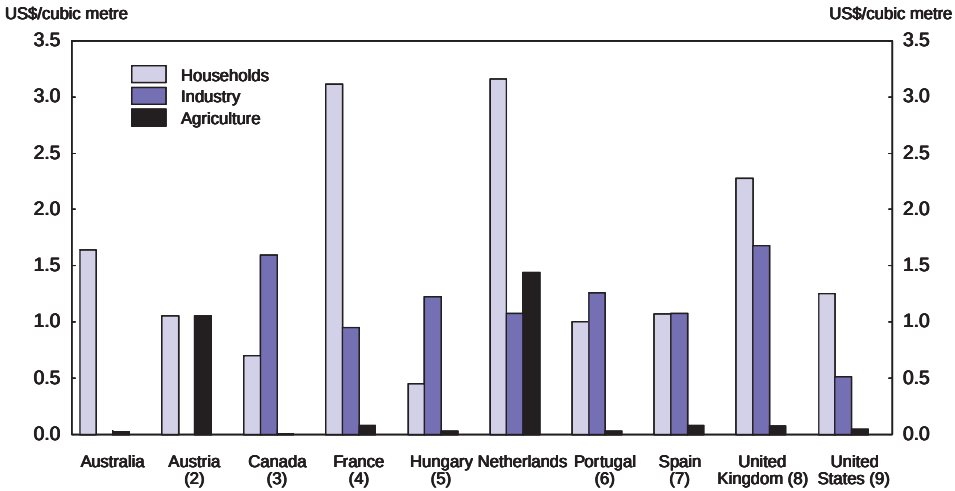
Figure 32. **Water prices in selected OECD countries**[1]
Mid-1990s



1. Prices are the median values for the range of prices for each category. Industrial water prices in Australia and Austria are not available.
2. Water used for livestock activities is obtained from municipal systems and priced at household rates.
3. Industry: rates apply to commercial establishments only. These may include small industries but the rates do not apply to major industrial operations.
4. Agriculture: data refer to the Adour-Garonne and Côteaux de Gascogne regions. Industry: the values refer to 1990-93 and exclude taxes, pollution and abstraction fees.
5. Agriculture: the values to 1998 water abstraction charges. Households and industry: the values to 1998 maximum and minimum user charges for public water supply.
6. Agriculture: data refer to the Sorria and Vigia regions. In the case of two-tier rates, the values were based on the estimated water volumes and the price of the cubic metre.
7. Agriculture: data refer to Andalucia, Castille and Valencia. Industry: data refer to 1992-94.
8. Agriculture: data refer to Northumbria and Wales.
9. Agriculture: data refer to the Sacramento River and Tehama regions.

Source: OECD, *Environmental Indicators for Agriculture* – Volume 3: Methods and Results, and *The Price of Water*.

to the environment (Special Report of the Court of Audit of the European Communities on the CAP and the environment, 2000). Generally speaking, most of the subsidies appear to be used to support extensive farming operations, rather than reducing environmentally harmful intensive practices, without any evaluation as to whether they would cease in the absence of funding. Since 2000, subsidies in France are provided under territorial farming contracts that aim, in particular, to finance the environment-friendly production methods approved by the European Commission.[187] In any case, subsidies targeted more or less directly at the environment or beneficial to the environment are still marginal in comparison to those that foster intensive production. In 2000, they accounted for only three per cent of the total of subsidies in the European Union.

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Recent and planned reforms

In 1999, two ecological taxes aimed at reducing water pollution were created and included in the General Tax on Polluting Activities (TGAP), the revenues from which go not to the agencies but to the Fund to Finance the Reform of Social Security (FOREC) (see earlier section). One of these taxes is a charge on detergent phosphates and the other is a charge on agricultural pesticides (grouped under eight categories of harmfulness). On the other hand, an attempt at that time to introduce taxes on excess nitrogen surplus, on thermal waste and on radioactive elements, which were also to be paid to the Fund encountered sharp resistance. The tax on radioactive waste was abandoned, while those on nitrogen surplus and thermal waste are included in the government's draft legislation on water. If passed, their revenues will be paid to the budget of the water agencies if the legislation goes through.

A major goal of this proposed legislation is to reform the system of water charges so as to better reflect the polluter-pays principle, while bringing it into conformity with the Constitution.[188] In 1982, the Constitutional Council ruled that water charges belong to the category of "taxes of all kinds", for which Parliament must set the base and the rates, implying that the current system, where the agencies collect the charges and determine the calculation base and the rate, was unconstitutional. The draft bill would at the same time transpose into French legislation the Community Water Framework Directive, which promotes the principle of cost recovery, including environmental costs, and the polluter-pays principle.[189] The major changes with respect to charges would be, on one hand, having Parliament set the calculation base and a range of rates, and on the other hand introducing a charges on excess nitrates. It is also planned to set a ceiling on the total amount of charges and to stabilise the amounts paid by households. On the other hand, in contrast to what was initially proposed, the current draft bill does not challenge the earmarking of charges to the budget of the agencies.

As was discussed earlier, the diffuse nature of agricultural pollution and the impact of various natural and agronomic factors on the way nitrates end up in the water make it expensive and difficult to measure pollutants at the individual level and hence to devise an economic instrument that will take them properly into account. Some OECD countries, such as Sweden, have introduced taxes on the nitrate and phosphate contents of fertilisers, which are relatively easy to implement. Such taxes are not well targeted, however, since the link between fertiliser usage and leaching depends on the various factors cited. Moreover, they address only part of the problem, since they leave aside nitrates of animal origin, which in the case of France would be a serious oversight. Other countries such as Belgium and Denmark have introduced systems amounting to a fine on "excess" use of nitrogen (in the sense of the European nitrates directive), and these have the advantage of covering all sources of nitrates. Yet the systems are sharply nonlinear and pose the same targeting problem as a tax on fertilisers.

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The only way to resolve this problem is to measure the flows of nitrogen entering the farm and leaving it (nitrogen outputs incorporated into harvests, livestock and manure) and to tax the difference between the two measures (the nitrogen balance) as representing the amount that is left in the soil and that might end up in the water. The administrative and information collection costs of establishing and controlling the nitrogen balance and taxing it are potentially significant and the net benefits associated with such an instrument need to be compared to those of a system that combines a simple fertiliser tax and a pollution charge on livestock operations. The nutrient balances would be established by the management centres that currently examine farm accounts and that therefore already have a substantial part of the necessary information, which could reduce the cost of drawing up the balance sheets. However, to the extent that they are subject to taxation, these balances will have to be controlled. A nutrient-balance tax of this kind has been in place in the Netherlands since 1998 and seems to be producing good results (Box 10).

Another difficulty lies in setting the tax rate. In principle, it should be equal to the marginal cost of the damage caused by excess nitrates. In practice, however, the latter is difficult to evaluate. A fallback alternative is to estimate the

Box 10. The Dutch system of taxes on excess nitrogen*

The Netherlands introduced a tax on excess nitrogen in 1998 in the wake of findings that the measures taken hitherto (improvements to feeds, geographic dispersal of manure, reprocessing and export) had failed to achieve environmental objectives in a country where the livestock sector is particularly intensive. Nationally, the nutrient surplus amounted to some 220 kg of nitrates per hectare in the late 1990s, and 70 kg per hectare for phosphates.

Farmers are supposed to keep an account on nitrates and phosphates and if the balance between inputs and outputs exceeds certain limits, they must pay a tax. The system is intended to bring inputs and outputs into balance over the medium/long-term. To this end, the thresholds are progressively reduced over time (for nitrates, from 175 kg/ha in 1999 to 60 kg/ha in 2003 on arable land, and from 300 to 140 kg/ha for meadows). Similarly, the rate of taxation on the surpluses, initially set 0.68 euro per kg for nitrates exceeding the thresholds, will rise to 2.27 euros per kg in 2001. The system thus allows farmers to prepare for the transition towards equilibrium. The initial effects of the system are already being felt, and farmers who have high surpluses, and are thus liable to the tax, have reduced their fertiliser consumption and their excess nitrates and phosphates far more than those who are not subject to the tax.

* The information for this box is taken primarily from Oltmer and Brouwer (2001).

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cost of denitrification. In the case of France, an estimate of the tax on surplus nitrates that would be needed to cover the average cost of denitrification yielded a rate of five francs per kg of excess nitrogen.[190] However, it is difficult to assess whether the average cost used in this calculation is lower or higher than the marginal cost, which is what should be used to set the tax rate.[191] If the objective is to reduce nitrogen loss, the level of the tax should be sufficient to offset the forgone yields to the farmer associated with reducing the use of fertilisers.[192] The rate proposed in the draft Water Law falls between 1.3 and 1.5 franc per kg of nitrogen (€ 0.20 to 0.23), which is roughly the average rate planned for industrial municipal wastes. The proposed rate is similar to that charged on fertilisers in Sweden, but it is only one-third the rate applied to excess nitrates in the Netherlands. It is difficult to tell whether the level established will offer the appropriate incentive. In contrast to pollution charges paid by other users, the plan does not make any rate differentiation by zone in terms of environmental risk, although such an approach would probably be quite effective.

The tax would initially be applied, beginning in 2003, only to large-scale farming operations, which are estimated to generate about 80 per cent of excess nitrates; after 2008, it would also apply to medium-sized operations.[193] In calculating the tax base, there will be a flat allowance of 25 kg per hectare of useful farmland, which is supposed to represent the threshold after which nitrogen may reasonably be considered in surplus.[194] This allowance probably preserves the incentive factor[195] and could also be viewed as a way of limiting the impact of the tax on farm incomes, which have already been seriously affected by a series of crises, in particular mad cow disease. The threshold is planned to decrease between 2003 and 2007 and a secondary but still significant impact of this tax would be to reduce greenhouse gas emissions by an estimated half million tonnes of carbon equivalent.

The draft legislation also calls for reforming the water withdrawal charge in order to improve the efficiency of resource allocation and help preserve the water supply. The charge would be imposed on water consumption and would therefore be based on the difference between amounts extracted and those returned; the base and the rate would be identical regardless of the use to which the water is put, with a possible modulation of the rate as a function of resource availability. However, an escape clause has been introduced that would allow users to sign a water resource management protocol (to distribute consumption among users concerned with respect for the aquatic environment) and thereby be eligible for a charge of between ½ and ¼ of the normal rate. Management protocols of this kind are already in force in certain regions such as the Beauce and involve farmers exclusively. The preferential rate is intended to encourage farmers to protect water resources, and also to move towards the metering of water in the agriculture sector. In principle, these protocols should be open to all users but in practice households will not be able to benefit from the preferential rates.[196]

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The pollution charge rates, based on net pollution, would be established by parliament. There would be a possible variation in the reference rate of 20 per cent between agencies, which would be allowed to differentiate the rates in accordance with local sensitivities and the state of the local environment. The principal modification proposed for the calculation base of the pollution charge is to replace the collection coefficient used in calculating the charge for households by a new charge (called the "collection fee") based on the volume of water used. Contrary to the current system, this charge would apply to all users of the public sewage system, including industries, which would however receive preferential treatment. Industries would benefit initially from a rate only 20 per cent of that paid by household users, but the rates would gradually converge until 2008 towards a level amounting to 80 per cent of the household rate; this charge, aimed at financing the renovation of waste water networks would be abolished in 2015. A new charge is also introduced on warm water discharges, applicable to thermal power stations. On the other hand, the idea of introducing a charge on radioactive discharges from nuclear power stations has been abandoned.

Conclusions and recommendations

This chapter has examined some key questions in the environmental field. While far from exhaustive, it has highlighted several features of French policy on the environment. The principal recommendations flowing from this analysis are set forth in Box 11.

While environment policy is still largely based on regulatory-type instruments, France has considerable experience in using economic instruments, in the form of taxes and charges. With the exception of transport taxes, however, these taxes have traditionally been earmarked for financing depollution subsidies. The French authorities have begun to reform these taxes: those now classified under the general tax on polluting activities (TGAP) will no longer be earmarked for environmental expenditures and this constitutes a step in the right direction. The issue now is to get away from the financing focus. As has been done for recently introduced taxes, such as that on pesticides, there is also a need to align the taxes more closely to the external social costs associated with the production of emissions or wastes, so that businesses and consumers can take them into account in their production and consumption decisions, as they do with other costs. This will require more systematic economic analysis and efforts to give value to environmental externalities, which are still little used in France. The recent creation of an economic and environmental evaluation directorate within the Ministry of Environment should help in this respect. It will also be important to reinforce the capacities of the local authorities in this area.

The Constitutional Council has rejected the proposed "TGAP-énergie" on the grounds that the tax allowances and exemptions included in the plan would introduce inequality among taxpayers. That plan constituted a major part of the

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Box 11. Principal recommendations

Ecotaxes and the greenhouse effect

Existing environmental taxes

- Existing environmental taxes should be more closely aligned to the cost of externalities and economic evaluations should be undertaken to this end. In order to provide adequate incentives, the level of these taxes should be rethought and should be better differentiated according to the harmfulness of products or emissions, especially in the case of atmospheric emissions and wastes. Sectoral regulations and taxes should be better designed, so that taxes as price signals can influence decisions (for example, quantitative recycling goals or waste-to-energy incineration targets for treating packaging wastes should be eliminated).
- Certain features of existing fuel taxation that are harmful from an environmental viewpoint should be reformed. The narrowing of the tax differential between gasoline and diesel, suspended in autumn 2000, should be revived: this gap cannot be justified either from the viewpoint of emissions (greenhouse gases and other pollutants) or of other externalities and it should be totally eliminated in time. The various exemptions and tax reductions should also be abolished, in particular those for road and air carriers.

Greenhouse effect

The French government, as part of the European Union, has taken an active position in international bodies for combating climate change: projections show that greenhouse gas emissions in France will rise and measures need to be taken.

- The introduction of an economic instrument, a tax or a cap-and-trade scheme, would by itself allow for significant and cost-effective reductions in greenhouse gas emissions. To avoid an increase in overall fiscal pressure, other taxes could be reduced, and there are mechanisms (tax credits and refunds in the case of a tax, or grandfathering of emissions credits in the case of a market) for moderating the competitiveness losses in highly energy-intensive sectors that such an instrument might entail.
- Voluntary agreements with businesses would not significantly reduce their greenhouse gas emissions at least cost. If these voluntary agreements are to have any real impact on greenhouse gas emissions levels, they would have to meet several conditions:
  - In the first place, the reference scenarios must be established independently, in accordance with a predefined framework common to all enterprises, so as to ensure that the asymmetry of information to the prejudice of government does not lead to the setting of objectives that are insufficiently ambitious (and which would have been achieved in any case without such agreements); targets should be set at the individual enterprise level, even if an initial sectoral objective has to be established and they should be specified in terms of an absolute level of emissions.
  - Next, transparent criteria, rules and surveillance procedures need to be set before the agreements come into effect and the penalties for non-compliance need to be established.

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Box 11. **Principal recommendations** (*cont.*)

Air pollution

- On the institutional front, greater co-ordination among the various local players (municipalities, regional governments, public transport authorities, associations, employers) will be required if urban vehicle traffic is to be effectively reduced: the transfer of responsibilities in this area to the municipal authorities that was begun by the Urban Solidarity and Renewal Act should be pursued; quantitative environmental and traffic objectives should be set (on the basis of an economic evaluation) in urban transport plans (PDU).
- There is a need at the local level for more economic evaluation, both *ex post* and *ex ante*, of measures and projects implemented, both individually and overall.
- Consideration should be given to introducing toll systems for an entire urban zone, taking into account the type of vehicle and the time of day. In the short term, tolls could be instituted on certain arterial roads, which would provide incentives for carpooling.
- If the introduction of sulphur-free fuels is adopted at the European level, as the Commission envisages, the most effective means for doing so would be through differentiated taxation, as was successfully done in the case of unleaded gasoline.

Water quality

- The proposed tax on excess nitrogen should be implemented because it constitutes a first step toward internalising the costs of pollution generated by agriculture. The tax should, however, be differentiated by zone as a function of environmental risk. If experience shows that the incentive is inadequate, the rate and the incidence thresholds could be adjusted.
- Reducing pollution of agricultural origin will also mean continuing to reform farm subsidies. In order to reduce the incentives to intensive production and hence to pollution, the eco-conditionality of subsidies should be strengthened and the differentiation in bonuses between irrigated and non-irrigated lands should be abolished.
- The programme for controlling agricultural pollutants should be reformed to avoid financing investments that are already profitable and to target subsidies more clearly in terms of environmental priorities.
- The "narrow" financing focus encouraged by the present earmarking of pollution charges for the budget of the agencies to finance subsidies for depollution projects needs to be called in question. Tax rates should be more closely aligned with the cost of environmental externalities (or at least with the cost of treating them) and preferential rates for certain categories of users should be avoided (particularly for industries, in the case of the collection fee).

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Box 11. **Principal recommendations** (*cont.*)

– Procedures for allocating subsidies should be reformed so as to enhance their economic and environmental effectiveness. Subsidies should not be allocated automatically as an entitlement and an economic analysis of projects should be a mandatory requirement. Different projects should also be compared among themselves to ensure better cost-effectiveness at the river basin level. Finally, the efforts begun in 1999 to rebalance representation on the river basin committees should be pursued, so as to take greater account of the interests of final users of water.

programme for combating climate change. It was far from perfect – it excluded several categories of polluters, notably agriculture and transport – but, in contrast to the carbon taxes in force in other European countries, it represented an attempt to provide the same marginal incentive to all sectors involved, including those that are highly energy intensive. A degree of inequality is in fact difficult to avoid if the intention is to provide the same incentive to all sectors while attenuating the redistributive effects of the tax, primarily on firm competitiveness.

The advantage of economic instruments (tax or cap-and-trade scheme) has been clearly established when it comes to reducing carbon dioxide emissions. In France as elsewhere, such measures have run up against concerns over the loss of competitiveness but these concerns should not be overstated.[197] In the first place, an emissions reduction policy based on non-economic instruments (such as regulation or voluntary agreements) would also have effects on the competitiveness of particular sectors, even if they were less obvious. Moreover, the cost of emission reduction shortfall to the economy as a whole would be higher under a regulating approach as compared with economic instruments. In the second place, if the intent is to avoid unnecessary adjustment costs linked to the fact that France would be acting ahead of other countries party to the Protocol or if it is merely to smooth the way for enterprises during a period of transition, it would always be possible to negotiate mechanisms that would reduce the financial burden of the tax without changing the rate. This was the purpose behind the tax allowances in the "TGAP-énergie" proposal; it deviated from the polluter-pays principle but by imposing at the margin a single price for pollution it would have minimised the total cost of reducing emissions in the sectors concerned.

There were perhaps some possible solutions for resolving the constitutional obstacle, such as tax credits or refunds independent of current emission levels. A cap-and-trade scheme in which emission credits could be grandfathered, at least in part, could also have been envisioned. Negotiation of voluntary

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agreements with enterprises, advocated by some parties, is clearly not the most cost-effective solution. International evaluations show that administrative costs of establishing the reference scenarios, monitoring and auditing the implementation of voluntary agreements are very high, as is the risk of manipulation and appropriation. If voluntary agreements are really to reduce emissions, the reference scenarios will have to be established independently according to a predefined framework common to all enterprises and the quantitative objectives for emission reduction that are set must be sufficiently ambitious. As well, there must be financial or regulatory penalties provided in case of failure to respect the agreement.

Allowance trading between firms having negotiated a voluntary agreement would reduce the cost of emission reduction for them but would not alter the intrinsic problem of setting the target for the voluntary agreements. Moreover, to design voluntary agreements as a transition to a cap-and-trade system (now under discussion at European level for energy-intensive industries) would pose certain problems. If the total emission reduction targets and the rules for allocation of emission allowances among firms (notably the base year) are not set in advance and credibly, there will be a strong incentive for firms not to enter voluntary agreements or simply to minimise their efforts, for fear of being subsequently penalised for the emission reductions they have made under those agreements.

As in many other OECD countries, certain features of French taxation on transport are harmful to the environment and should be reformed. This is the case with the tax differential between gasoline and diesel (which is particularly high in France), the narrowing of which was suspended in the autumn of 2000 a move that is symbolic of a lack of continuity that is undermining the credibility of the policy. Diesel is still more polluting than gasoline and total elimination of the differential would be one of the most cost-effective measures that could be taken to reduce urban air pollution. Moreover, the various exemptions from fuel taxes, in particular the exemption for road and air carriers (which will constitute the most important source of growth in greenhouse gas emissions to 2010) and the for the farming sector, even though they result largely from current EU regulations, are inappropriate from both the environmental and economic standpoints. In the end, given the sharp variation in social costs associated with fuel consumption as a function of place of use and type of vehicle (for example, the level of gasoline taxes is probably lower than the cost of the externalities that a vehicle generates in a city, but perhaps higher than this cost for a long-distance trip through the countryside), it would seem wise to overhaul completely the system of transportation taxes and charges, and rely on a system of tolls. Such a reform would be even more effective if it were co-ordinated at the European level.[198]

In urban areas, and particularly in the Paris region, automobiles do not pay the costs that they generate (congestion, pollution, noise and damage to infrastructure). This indirect subsidy has had a cumulative impact by promoting

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urban sprawl and thus encouraging car use. The problem is complex, since addressing it requires co-ordination among different institutional levels (municipalities, region, Central Government) in different areas (urban development, transport) and the effects of traffic reduction measures are not well understood – some indeed could significantly increase short-term congestion and pollution. Urban transport plans could help achieve greater co-ordination but at this stage, with the exception of the plan for Ile-de-France, they are still very vague when it comes to their environment and traffic objectives. Quantifying these objectives on the basis of an economic, environmental and social evaluation would provide all players with more solid ground for considering the measures to be taken. Also, the work being done to develop methods and tools of evaluation and monitoring of urban travel plans should be strengthened. It would also be appropriate to better analyse the reasons for popular resistance to the idea of "fewer cars". A campaign to sensitise public opinion to the costs generated by the automobile would help to reduce opposition to the measures.

Socio-economic analyses of public transportation investment programmes are supposed to assess environmental degradation. The recent revision of evaluation criteria should help improve such assessments. Debate among the various areas of government on valuing these costs has also helped to enhance public understanding in this area. Yet more evaluation (both *ex ante* and *ex post*) of policies, and not only projects, is required. The creation of a body for public policy evaluation partly answers this need and it would be useful if it were to review the policy for air quality. It would also be important to ensure that its recommendations are put into practice.[199] The central government will also need to help disseminate available information and environmental assessment tools among municipalities and thereby reduce their costs.

Economic instruments are best suited to the purpose of reconciling the objectives of economic efficiency and minimisation of congestion, on the one hand, and health and environmental protection, on the other. In the first place, parking policy should be used more actively. In Paris, for example, non-residential parking fees are still low and relatively undifferentiated, while the level of fines provide little incentive. Regulations requiring employers to build parking lots reduce the effectiveness of public-parking policy and should be reformed as a function of public transport services. Having workers pay their employer for parking privileges should also be considered. In the second place, urban toll systems are one of the most promising instruments. An electronic toll system applied to an entire zone is still a costly undertaking but the cost will decline and this remains the only way of truly taking the various externalities into account. Over the short term, alternative but partial solutions could be envisaged, such as electronic systems for imposing rush-hour tolls on highways and express ways that go into, across or around major cities, as is now done in the United States and is about to be introduced in the Netherlands. Such systems provide an incentive for

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carpooling, which could eventually be strengthened further through preferential toll rates. In every case it is important that these measures be taken coherently and that their effects be well gauged in advance, so as to prevent undesirable outcomes from harming policy credibility.

With respect to water, the current draft bill represents only a first move, given the objective of introducing more economic pricing and having the water agencies take better account of externalities. It fails to challenge the earmarking of fees and the lack of economic evaluation in allocating investment subsidies. Water charges should be more clearly aligned with externalities, in order to provide a proper price signal concerning water usage and pollution. The economic and environmental effectiveness of subsidies should also be enhanced. This could be achieved through mandatory publication of the economic analysis of the projects to be subsidised, consistent with the new EU framework directive on water, and by continuing to rebalance representation on the water boards in favour of end-users of water, a process begun in 1999. The present allocation of charge revenues to water agency budgets has led to the notion of a "rate of return" for the various sectors and its continuance is making it more difficult to introduce the necessary reforms.

The proposed reform of the method for calculating pollution charges would limit the ability of the water agencies to set those charges solely in terms of the financing needs of their five-year investment programmes. Yet the framework established by the draft bill for setting water charges should allow those rates to be more closely linked to the costs generated by externalities. This would probably require greater differentiation as a function of the state of the environment, beyond the 20 per cent margin currently being proposed. Extending the collection charge to all users of the sewage system, not only to households, would represent a step forward but the slow alignment of industry rates with the rates paid by households is difficult to justify from an economic or environmental standpoint. The highly preferential rates for water withdrawal charges allowed to parties to a collective management agreement (primarily farmers) are also hard to justify from an economic standpoint. But if those arrangements can offer a practical incentive to adopt practices more respectful of the environment, this can be a step towards progressive integration of external costs.

The proposed tax on excess nitrogen included in the draft Water Law, on the other hand, constitutes an important step towards taking account of the externalities generated by farming and livestock activity, which have hitherto been *de facto* exempt from any pollution taxes. A typical example of an environmental fiscal instrument, it could perhaps be somewhat complicated to implement but it should help to reduce the public spending needed to control nitrate pollution. Apart from its water quality impact, this tax would also help to reduce agricultural emissions of nitrous oxide, an important greenhouse gas. The proposed level of

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the tax has been significantly reduced and incidence thresholds have been raised compared to the initial plan, primarily in order to soften the impact on farm incomes. These provisions should be adjusted in the future if the tax is found to provide inadequate incentives for reducing excess nitrates. It would also be effective to differentiate the tax in light of the environmental risk, so as to take better account of local disparities.

The control plan for pollutants of agricultural origin, the major measure undertaken in terms of farming, proved to be very costly and of limited effectiveness environmentally. It financed investments of a type that often seems profitable to the farmer and subsidies were not allocated in accordance with environmental priorities. The current suspension of the programme for inconsistency with the European Union rules on state aid provides an opportunity to conduct an in-depth economic and environmental analysis to decide on the next steps. It is important that the programme's reform proposed by the government and now under review by the European Commission lead to an effective integration of environmental priorities.

In any case, a policy of farm-source pollution control by itself will not suffice. A process of farm assistance reform has been in progress since 1992 but most of the subsidies still incorporate incentives to intensive farming and hence to pollution. In order to make policy in this area more consistent, those subsidies should continue to be reduced. The ecological conditionality of subsidies should be strengthened and agro-environmental measures should aim to reduce practices harmful to the environment rather than only maintaining those that are environment-friendly. Continuing reform of CAP subsidies for Europe as a whole, in order to delink them more from output and pollution, would further these aims.

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Notes

1. Other major European countries also have a relatively low level of ICT investment.
2. Definition used for the OECD's Analytical Data Bank, civilian employment, *i.e.* excluding conscripts (national service).
3. The longer periods of plant utilisation may reflect a medium-term trend towards more intensive use of plants.
4. As explained later on, the analysis of recent wage growth has been made difficult by the introduction of the 35-hour week. In particular, it would seem that the manner of booking hours worked has changed without the official statistics being adjusted, thus causing a break in the statistical series.
5. The calculation of hourly wage growth became somewhat distorted in 2000-2001 because of the measurement difficulties created by changes in the definition of working hours (see Chapter III). As a result, the rate of hourly wage growth in 2000 and 2001 has been overestimated. While certain estimates by DARES suggest that effective working time decreased by only 1.5-2 per cent, the reduction of working time as measured by the ACEMO survey – which is used for calculating the hourly wage – was much steeper.
6. The agreements already concluded show that employees generally have accepted less rapid wage growth for a limited period (according to DARES, 2001, wage moderation is of the order of 1 percentage point over two years in firms that have moved over to the 35-hour week, relative to firms still on the 39-hour week).
7. The most recent presentations of this strategy are contained in the R*apport* É*conomique et* F*inancier du* P*rojet de* L*oi de* F*inances pour* 2002 (September 2001, Paris) and in the R*apport sur l'*É*volution de l'*É*conomie* N*ationale et des* F*inances* P*ubliques* (2001), a report produced by the government for the parliamentary debate on the fiscal stance, May 2001, Paris.
8. Since the bulk of public expenditure is independent of the economic cycle, the authorities consider that this norm is inflexible and must not vary with cyclical fluctuations. However, the expenditure of UNEDIC on unemployment insurance benefits is linked to the cycle. For consistency's sake, the growth norm for public expenditure should exclude this category.
9. To determine real expenditure, the authorities deflate nominal expenditure by the consumer price index excluding tobacco.
10. On a budgetary accounts basis and with unchanged perimeter.
11. The sum of taxes and effective social security contributions, as defined in the national accounts.
12. The elasticity of direct taxation with respect to GDP is more than 1 in a period of rapid growth of activity.

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13. This strategy is described in the *Rapport sur l'Évolution de l'Économie Nationale et des Finances Publiques* (2001) (cited earlier).

14. The multiyear public finance programme for 2002-2004 contains two scenarios. In the first, growth of activity is 3 per cent a year and the general government account is in balance in 2004. In the second, growth of activity is 2½ per cent a year and public finance is in deficit by 0.5 per cent of GDP in 2004.

15. The official projection is based on growth of world trade, at constant prices, of 2 per cent in 2001 and 5.8 per cent in 2002. In the OECD countries real GDP is assumed to grow by 1.4 per cent in 2001 and by 1.8 per cent in 2002. The authorities recognise that the risks overhanging the macroeconomic outlook, which were already pronounced prior to the September 11 terrorist attacks, have been made even more so by those events. They consider that the biggest risk – aside from any adverse developments in the military and geopolitical situation – is loss of confidence on the part of American and European households. Drawing on the precedent of the Persian Gulf crisis, the authorities are assuming that the index of household confidence will fall by 20 points in the United States and by 10 points in Europe. On this basis they estimate, *ceteris paribus*, the downside risk at about ½ percentage point of real GDP growth in 2002. In their view, the sharper slowdown of activity would reduce inflationary pressures and allow more latitude for monetary policy in the euro area. Basing their judgement on simulations of macroeconomic models (Dalsgaard, André and Richardson), the authorities consider that a cut of 50 basis points in ECB key rates would make it possible to increase activity by 0.25 per cent of GDP in 2002. All told, assuming a responsive monetary policy, the authorities put the downside risk at only ¼ point of GDP. Accordingly, they have decided to keep their economic forecasts unchanged, while stressing the downside risks. The draft budget is therefore built on an assumption of real GDP growth of 2.3 per cent in 2001 (with a bottom figure of 2.1 per cent) and 2½ per cent in 2002 (with a bottom figure of 2¼ per cent).

16. These measures include an extension of the "employment bonus" tax credit, a special depreciation rate of 30 per cent for investments made between 27 October 2001 and 31 March 2002, accelerated refund of VAT and the creation of two-off budget funds to finance small business. Additionally, it has been decided to suspend payments for third-generation mobile phone licences over and above the € 1 238 million already paid and to introduce a new system based on the prospective revenues from this activity. The immediate impact of these measures on the central government budget would be limited. Extension of the employment bonus would increase the 2001 budget deficit by 0.1 per cent of GDP, whilst the accelerated depreciation of investments would be partly offset by reduced future depreciation and should cost the central government € 0.8 billion in 2003. The other measures have no impact in national accounting terms because, even though they bring the expenditure schedule forward, they do not change the deficit for the year as a whole or are classified as financing transactions. Nonetheless, the new system of UMTS licence payments is liable to deprive the Pensions Reserve Fund of much of its envisaged resources.

17. That is, the general government borrowing requirement adjusted for cyclical variations.

18. *Rapport sur l'Évolution de l'Économie Nationale et des Finances Publiques*, report produced by the government for the parliamentary debate on the fiscal stance, May 2001, Paris.

19. European Commission, 2001.

20. OECD *Economic Outlook*, No. 69.

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21. The estimates of the structural public deficit in the year 2000 are fairly similar. The French authorities put it at 1.5 per cent of GDP, the European Commission at 1.3 per cent and the OECD Secretariat at 1.6 per cent.
22. The ratio of current public receipts to GDP is higher than in France in Denmark, Germany, Norway and Sweden.
23. These different priorities are fully described in the report produced by the government for the parliamentary debate on the fiscal stance (May 2001).
24. The *Conseil des impôts* considers that the targeted VAT cuts are less effective in terms of job creation than cuts in social security contributions (Conseil des impôts, 2001).
25. The breakdown made by the OECD (2001*b*) shows that this increase in pension expenditure results from a rising old age dependency ratio (ratio of the number of pensioners to the population of working age). On the other hand, the lowering of average pension levels under the Balladur Reform of 1993 made it possible to limit the growth of borrowing requirements.
26. Speech by Prime Minister Lionel Jospin on 21 March 2000.
27. These measures had to be phased in so as not to penalise persons nearing retirement age.
28. The simulations show that for a typical OECD country an increase in the primary surplus of 1 per cent of GDP would be sufficient to maintain an unchanged debt to GDP ratio. Alternatively, an average rise of one year in the age of retirement would suffice. On the other hand, a reduction of average pension benefit would be more painful, since it would have to be 17 per cent.
29. *Le Figaro* of 5 July 2001.
30. Bac and Cornilleau (2001) note that demographic factors will have the same impact on health expenditure in the period 1998-2005 as they did in 1990-1997, since population ageing is offset by the slowdown in population increase. However, the effect of population ageing will probably be felt towards the end of the present decade.
31. Bodies benefiting from these parafiscal taxes include a technical centre for meat-curing and canning and pigmeat products, a cider fruit committee, an office for the Calvados apple-brandy trade, regional hunters' federations, an association for the support of private theatre and an association for the support of song, popular music and jazz.
32. Certain *fonds de concours* are intended to introduce flexibility in government accounting procedures where several cofinancers are involved (central government, regions, European Union). Others have a more debatable purpose and it would be better if they were brought into the budget. Examples are the financing of watercourse study units, financing of temporary museum staff, receipts from catalogue sales, management expenses of the audiovisual tax service, financing of ASFNE early-retirement pensions, financing of surveillance of horse and greyhound racetracks and receipts from exhibitions at the national sports museum.
33. Expenditure from the special treasury accounts does not require parliamentary approval. These special accounts include the special appropriation accounts, accounts for advances, the loan accounts, the business accounts and the coin-issue account.
34. Speech by Mr. Michel Sapin, Minister for the Civil Service, 19 September 2001.
35. A report on public service pay and pensions is supplied with the finance bill.

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36. The debate on "social modernisation" was initiated by the *Mouvement des Entreprises de France* (MEDEF) and concerns the future of pension schemes, unemployment compensation and Social Security financing, *inter alia* (see *www.medef.fr* for further information).
37. Speech by Mr. Laurent Fabius at the 16 January 2001 press conference announcing the creation of the PPE.
38. For a couple earning the SMIC and with two children, the amounts will be FF 3 400 in 2001, 6 400 in 2002 and 9 400 in 2003.
39. Source: OECD, *Employment Outlook* (2001), figures updated for 2000 on the basis of DARES estimates.
40. I.*e*. in excess of which, hours worked are considered to be overtime.
41. Ministry for Employment and Solidarity, "L'impact macroéconomique d'une politique de RTT: l'approche par les modèles macroéconomiques", DARES-*Mission Analyse Économique*, No. 17, 1998.
42. That unemployment is characterised by hysteresis implies that its equilibrium level at any point of time depends on its cyclical level.
43. By way of comparison, the annual cost of a job paid at the SMIC is about FF 100 000.
44. There are to be exceptional arrangements concerning overtime for 2002-2003, identical to those extended to firms with more than 20 employees in 2000-2001. In the first year, the overtime rate for hours between the 35th and 39th hour will be +10 per cent, instead of +25 per cent, and this premium will have to be granted in the form of leave, unless an industry-wide agreement or a collective agreement provides for it to be paid in cash. The overtime quota will only be counted beyond the 37th hour in 2002 and the 36th hour in 2003, and beyond 35 hours only in 2004.
45. The monthly remuneration guarantee is equal to the monthly SMIC calculated on the basis of 169 hours (*i.e*. 39 hours per week) at the time when the reduction in working time takes effect.
46. The hourly SMIC will be adjusted on the basis of production workers' hourly pay and will therefore increase faster than monthly wages because of the reduction in working time. In other words, the amount equivalent to 169 hourly SMICs will mechanically increase faster than the monthly guarantee.
47. The later the shift to the 35-hour week, the higher the level of the guarantees: on 1 July 2000, the guarantees were, respectively: FF 7 092.27 for moves to the 35-hour week before 30 June 1999; FF 7 180.43 for moves between 01/07/99 and 30/06/00; FF 7 307.77 for moves between 01/07/00 and 30/06/01 and FF 7 388.68 for moves after 01/07/01.
48. Let S2005 be the hourly SMIC in 2005. The monthly guarantee will by then be [S2005 × (39 × 52/12)], while 35 times the hourly SMIC will give a monthly wage of [S2005 × (35 × 52/12)]. S2005 will therefore have to be adjusted by (39/35), *i.e*. 11.4 per cent, for the monthly wage for 35 hours to be equal to the old monthly wage for 39 hours, and for the monthly guarantee system to be abolished.
49. The hourly SMIC is indexed on half the purchasing power of the hourly wage of production workers. With the hourly wage rising faster because of the reduction in work time, once all production workers are working 35 hours a week the hourly SMIC will be adjusted as follows: 0.5 × 39/35.
50. See Henry (1997); Cohen and Henry (1997); Ilzkovitz *et al*. (1999); Bergougnoux *et al*. (2000).

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51. See the special issue of OECD *Economic Studies* (No. 2, 2000/1) on "Regulatory Reform", which includes an overview and several sectoral studies on the impact of regulatory reform in network industries.
52. Network industries were traditionally organised as "national public services": uniform services at standard prices were offered by public monopolies that extended their networks throughout the territory, creating cross-subsidies between regions: Air France and Air Inter in the transport sector, EDF in electricity, GDF in gas, SNCF and RATP in rail service, France Telecom in telecommunications and La Poste in postal services. These establishments had significant financial and human resources. In 1995 they represented about 45 per cent of the capital stock of all industrial groups and in 1999 they employed 1.2 million people, including a significant portion of high-level professionals trained in prestigious specialised schools.
53. France Telecom, Cégétel, Télé 2 and 9 Télécom are the major competitors in the long-distance non-business market.
54. Comparison of the average per-minute prices of national calls (taking into account the average duration of calls, the effect of the minimum call charge (crédit-temps) and the breakdown of charge rate periods) shows an average difference of 30 per cent between the rates charged by France Télécom and those of its new competitors. However, the discounts offered by the incumbent operator (25 per cent on calls to numbers of subscriber's choice, up to 35 per cent according to the duration of the call, and between 20 and 30 per cent for all-in packages) mean that in practice the differences are smaller. The price differences of 5 to 15 per cent are based on the average profile of a consumer benefiting from these discounts.
55. France Telecom (Orange), SFR and Bouygues Telecommunications.
56. A broader definition of the local area could help to explain part of these price differentials.
57. Including for DOM-TOM service. Portugal insisted on an additional length of time for service to the Azores and Greece did the same for the Aegean islands.
58. This does not amount to a full "open skies" agreement, such as those signed by the United Kingdom and the Netherlands with the United States. In contrast to "open skies" agreements, additional capacity creation across the Atlantic is still subject to authorisation and flights originating in either country are not allowed to take on new passengers at their point of landing in the other.
59. Air transport deregulation in the United States caused prices to collapse, reflecting the lack of congestion at most airports and the great number of new entrants. In Europe, air traffic constraints and the power of the major incumbent airlines did not allow the same degree of "contestation".
60. These companies are established and licensed in European Union countries and are subject to the same safety standards as the large national airlines.
61. These rates represent the strict minimum called for under the directive. They designate the portion of domestic demand represented by bulk consumers authorised to choose their supplier, as of January 1, 2000 and 2003. This eligibility threshold is currently set at 100 million kilowatt hours of annual consumption.
62. A limited portion of national output (about six per cent) is beyond the control of EDF: Compagnie Nationale du Rhône, SNET and the 6 000 MW of supplementary blocks of power that EDF has agreed to make available to its competitors, under requests from the European Commission, in return for an authorisation to penetrate into other European markets.

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63. These prices fell by 13.3 per cent between 1997 and 2000.

64. In 2000, EDF recorded operating profits of FF 3.5 billion, sharply down from the 14.9 billion in profits for 1999. About half of this decline reflected damage from the storm of December 2000. RTE contributed FF 5.6 billion to the parent company's profits during the year. EDF expects total profits of FF 7 billion for 2001.

65. Three European Directives require that the main rail tracks in Member states be liberalised for international freight by 15 March 2003 and that a toll be paid to the network manager. The entire freight network will opened to competition by 15 March 2008.

66. The theory of marginal cost pricing applied to France can partly explain this fact, since maintenance costs include a fixed component and a marginal-cost toll does not cover all maintenance costs.

67. In June 2000, a new proposal for a postal directive was rejected by the European Parliament, after having obtained the grudging consent of governments to reduce the threshold for services reserved to the historic monopolies.

68. See Senate (1988).

69. The price in euros of a stamp for a domestic letter weighing less than 20 g was in the summer of 2001: 0.59 in Finland, 0.56 in Germany, 0.56 in Sweden, 0.46 in France, 0.43 in the United Kingdom, 0.38 in Ireland, 0.36 in the Netherlands and 0.24 in Spain.

70. By a law of 1990, pension costs are borne by the enterprise, though the burden is capped by a contrat de plan.

71. France has the largest postal network in Europe, with 17 000 service points, this being related to the country's low population density. According to one survey, activity at 20 per cent of these wickets amounts to less than two hours a day.

72. Twenty per cent of consumption was to be open to competition as of August 2000, 28 per cent in 2003 and 33 per cent in 2008. In France, these thresholds correspond to users consuming respectively 25, 15 and 5 million cubic meters of gas per year, as a minimum.

73. Compagnie Française du Méthane (CFM) and Gaz du Sud-Ouest (GSE), of which it is co-owner with TotalFinaElf.

74. The European Commission brought an appeal before the Court of Justice of the European Communities for failure to apply the gas directive, in May 2001.

75. According to the French authorities, several other countries that have legally deregulated a greater proportion, or even the entirety, of their gas market are in fact less open to competition because of practical obstacles to the use of their distribution networks.

76. Operating profits declined by 20 per cent in 2000 to EUR 0.87 million, from 1.09 billion in 1999 but this drop reflected primarily the increases in gas supply prices. However, part of the decline due to competitive pressure was offset by the registration of 200 000 new customers in 2000. GDF also publishes its financial statements for a "constant environment" (with the price of gas unchanged): under this scenario, earnings would have risen by 32 per cent during 1999.

77. The United Kingdom in particular. A spot market for gas has started operating in the UK (National Balancing Point), with flows that are interconnected with the continent (the Interconnector System). Such an integration approach is not recommended in the current directive.

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78. The fact that French markets are relatively closed creates a perception of asymmetry in European countries open to French public network enterprises. Public ownership of these enterprises, while compatible with European rules, also sparks concerns about asymmetry in the conditions of competition between public and private firms, notably the strategic asymmetry in terms of the cost of capital. Indeed, French public undertakings can make acquisitions in foreign markets without the reciprocal possibility that their competitors take controlling stakes in France. Thus, the governments of Spain and Italy, where EDF's market share is growing, are seeking to slow its rate of penetration, by restricting the voting rights associated with its equity shares.
79. French administrative law establishes the principles of *continuity*, *equality* and *adaptability* and makes the government legally responsible for providing services. *Continuity* relates to "a social imperative that justifies the exclusion of any risk of disruption or interruption in the service"; *equality* requires "equal status for all users of the service, without discrimination or particular advantage for any"; and *adaptability* means that "services provided to the public are always adapted to the public's needs". The legal commitment of the central government to ensuring such availability of services appears unique among OECD countries. See Chevallier (1997).
80. France Telecom's competitors, some of whom also question the method used to estimate the cost, regularly complain to the ART about the high cost of their public service obligations, which increase their contributions to the universal service fund. The size of the country, the dispersal of the population and the low overall population density are also factors that increase the cost of the universal service compared to more densely populated countries.
81. Local authorities indicate they are willing to cofinance such costs.
82. In 2000, out of FIDA's total available resources of FF 279 million, only FF 64 million were disbursed, largely for airlines operating in French Guyana and those providing links between St.-Pierre-et-Miquelon and Canada.
83. Moreover, invoicing for high-voltage transmission services by RTE, which is standardised throughout the country, seems to imply some price cross-subsidisation but it has never been necessary to make this explicit because it is financed by transfers within a natural monopoly. As long as transmission remains a monopoly, this equalisation mechanism appears sustainable. However, the transparency of transmission costs might suffer as a result.
84. In the short term, the costs of maintaining reserve capacity are covered explicitly, rather than treated as a charge paid solely by the historic operator. Assuring energy security over the longer term, whether viewed from a national or European perspective, will mean diversifying electricity production sources in ways that a competitive market may not by itself generate. The question, then, is to define the desirable portfolio and capacity of alternative generating modes and if the spontaneous results of the market diverge from that target, to call for public tenders to undertake the needed investments. Such a procedure looks cumbersome but it could be managed in a pro-competitive manner. This is the approach that the 2000 Law and the consultations between the French government and the European Commission have opted for. How this will be done remains to be worked out, however.
85. The impossibility of maintaining uniform service became clear with the TGV. The service is available only in areas and on routes of commercial interest. The TGV applies the fare-setting practices of airlines, including computerised management, to maximise business revenues and passengers on any one train will find themselves paying very different fares, just as on an airplane.

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86. In the 2000 budget, out of a total Central Government contribution of FF 5.9 billion for regional passenger services, 3 billion was paid directly to the 7 experimental regions and the rest disbursed through the SNCF to the remaining regions. The seven participating regions are: Alsace, Centre, Limousin, Nord-Pas-de-Calais, Pays-de-la-Loire, Provence-Alpes-Côte-d'Azur and Rhône-Alpes.

87. This enhanced competition between rail and road reflects the choices that the regional authorities can now make between the two modes of intraregional connections. However, entry licensing ("*conventionnement*") for interurban bus companies is generally stymied by the Central Government's preference for rail. The Ministry of Transport refuses, as a matter of principle, to authorise bus services over routes already served by the train, in order to maintain the integrity of the rail network.

88. Every country has its own definition of universal postal service and the French definition is particularly ambitious in its range of services, its geographic coverage and the frequency of service. For example, few other countries guarantee residential delivery of parcels up to 20 kg throughout the country, at a standard price. This service has played an important role in the growth of mail-order selling in France (preservation of which is actively supported by the French Association of Mail-Order Enterprises).

89. Banks are legally prohibited from discriminating among clients on the basis of the size of their accounts. In practice, however, partly because of the uneven accessibility of branch services (the banking system is more polarised then the postal network), the post office is the sole source of some services, such as providing banking ID, chequebooks and payment services, for a significant portion of the population.

90. Between 5 and 10 million account holders are prohibited from writing checks.

91. Most of the financial customers of La Poste in fact appreciate the availability of the vast network spanning the entire country. This activity is not classified as "banking", since the post office does not make loans apart from those provided with home savings (*épargne logement*) plans and accounts. The Post Office has nevertheless established distribution agreements with certain establishments. In June 2001, it announced a major deal with the *Caisse des Dépôts et Consignations* and the *Caisse d'Epargne* to create a large quasi-banking network. The pending regulatory and competition implications of this agreement, if it is confirmed, may be significant.

92. Personnel can retire very early (as of age 50 for some categories of staff) and in-house supplementary health insurance schemes refund care expenditure in full.

93. In the 1990s the rate of unionisation averaged 26 per cent in the public services, compared with 8 per cent in private enterprises. It averaged 50 per cent in EDF-GDF and 25 per cent in the SNCF and the PTT (the then post office and telephone service). Of the 143 000 EDF employees, 1 200 are seconded as permanent union officers.

94. The loss of earnings for days of strike is very often made up by the enterprises through return-to-work agreements, eliminating any financial deterrent to work stoppages.

95. The three richest employee committees (*comités d'entreprise*) in France are those of the EDF, SNCF and Air France. The EDF central fund for social activities has an annual budget of FF 2 billion and employs 3 600 full-time staff and 12 000 seasonal workers in its recreation and vacation centres.

96. From just the demographic standpoint, the employees and pensioners of the French public network enterprises constitute a large social group, with very considerable political and electoral weight. (The employees and pensioners of these enterprises, together with those of general government, in fact form the largest group of French households. In 1996 more than half of all households did not have a member working

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in the private sector.) Similarity of status in public network enterprises and common concerns over privatisation have created a conjunction of interests that is not so strong in other countries. Moreover, unionisation is very advanced in these sectors, where national confederations have established their bastions and their main officer-recruiting bases. This explains why prospects of any changes in labour standards arouses vivid collective irritability.

97. The pension issue is particularly acute in EDF and La Poste, where the pensions of former employees are overhead expenses for the enterprise and entered as an annual charge in its operating account. The SNCF has its own pension funds but these are in deficit, since employment and contributions are not increasing proportionately with pension claims. The deficit of the SNCF retirement scheme was running at annual FF 19 billion in the late 1990s.

98. A background document on the challenges facing Air France was sent to all employees in 1993. Eighty per cent of the employees questioned said they thought that the diagnosis put forward in the paper was fair (Corcos *et al.*, 2001).

99. That role and its institutional mechanisms are described by Cohen (1992).

100. The distribution and treatment of drinking water is a local network activity that is not examined in this survey. The way it is organised around local authorities follows the national monopoly model but supply is ensured by a private duopoly which is recognised as a world technological leader in this sector: the *Générale des Eaux* group (now *Vivendi*) and the *Lyonnaise des Eaux* group (now *Suez*).

101. The economies of scale, scope and learning afforded by the electro-nuclear programme are reviewed by the International Energy Agency (2000). These economies could explain why the returns on nuclear investment have withstood the vacillations of the international market.

102. The pragmatic approach of national enterprises has helped them to withstand the temptation of technological autarky. Technical options adopted in nuclear and telecommunications industries, and the aircraft choices made in air transport, were decided on the basis of the technologies available in the world market.

103. The "private" and "social" profitability of investment in individual TGV lines is estimated by the Ministry of Transport. In principle, projects with a socio-economic return of less than 8 per cent are ruled out. If the return is over 8 per cent, the studies of the project can continue but subject to the constraint that subsidies will have to be paid to RFF so that the project shows a private return of at least 8 per cent to RFF, the minimum return below which RFF is not legally authorised to invest. Thus, among the TGV projects under way, the private return on the Franco-Spanish Perpignan-Figueras high-speed line was estimated at 2 per cent, and the socio-economic return at 10.6 per cent, while the private return on the Rhine-Rhone line was estimated at 2.4 per cent and the socio-economic return at 11.3 per cent. These estimates take into account TGV development costs insofar as those costs are incorporated in the purchase price of the rolling stock. Nevertheless, part of the R&D expenditure on TGV technology was subsidised by the State and the SNCF in the form of basic research and the profitability calculations do not include these research costs.

104. Examples being the *Superphenix* project in the energy sector and the plan to automatise entirely the handling of mail in post offices. See Cohen (1992) and Fauroux and Spitz (2001).

105. European directives do not specify how competition surveillance should be organised.

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106. See OECD (2001*c*) for a comparison of the approaches of various OECD countries to the vertical separation of monopolistic infrastructures and competitive activities. It led to the adoption by the Council in 2001 of a Recommendation on structural separation in regulated sectors.

107. The regulatory authorities' role in "proposing" prices or "approving" them (on basis of the prices proposed by enterprises), and giving "opinions" (on the prices proposed by the government) varies according to areas. Regulated prices are always the result of a conflation of the positions of the three parties. In the electricity sector, the CRE "proposes" network access tariffs and "gives an opinion" on the consumer tariffs suggested by the government.

108. See the Round Table of the OECD Committee on Competition Law and Policy of June 1998, *www.oecd.org/daf/clp*, for information on the relations between the competition authorities and sectoral regulators, notably regarding the promotion and protection of competition in specific sectors.

109. The CRE approves the Electricity Transport Network's annual investment plan.

110. Thus, the ART monitors the inter-penetration of telecommunications, media and Internet services.

111. Usually headed by a State Secretary, it was attached to several ministries in turn: Culture, Quality of Life, Tourism, Public Works and the Prime Minister.

112. Also created were the National Institute for Industrial Environment and Risks (INERIS) and the French Environment Institute (IFEN), with statistical monitoring duties.

113. Lévêque *et al.* (1999) show that a tax at the same rate as that on fuel oil would produce revenues of FF 2.8 billion.

114. This variation is even more important if European emissions standards prior to 1993 and after 2000 are taken into account, as shown for Belgium in OECD (2001*e*).

115. C*f*. ECMT (2000)

116. The relative fuel economy of diesel versus petrol engines is irrelevant as far as an environmental tax is concerned. It is true that greater fuel economy reduces emissions per kilometre, but it will reduce tax per kilometre in the same proportion. Such a tax should be set in direct proportion to the environmental damage per litre consumed, which generally implies a higher tax, per litre, on diesel. If diesel is more efficient, it may still be used even if it is more expensive – but the consumer would only do this if the lower fuel consumption is enough to make the cost (including tax) per kilometre lower.

117. At the end of August 2001 the government announced that the alignment process would not be resumed in 2002.

118. This tax refund is capped at 25 000 litres of diesel per vehicle per half-year. Since the average truck is estimated by the National Road Transport Federation to consume 38 400 litres a year, the cap is not constraining on average.

119. The stabilisation mechanism, introduced on 1 October 2000, operates symmetrically when oil prices fall. It was activated in an upward direction for the first time in March 2001. On the other hand, the "fiscal bonus", which was supposed to be suspended when the average oil price fell below its January 2000 level, was not suspended in March 2001when this happened.

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120. In theory, environmental taxes are defined as unrequited levies imposed on a tax base of environmental relevance, while charges are levied in proportion to the service provided.

121. It has since been amended and expanded. A charge of between FF 250 and FF 375 per tonne is now levied on emissions of sulphur dioxide, hydrochloric acid, nitrous oxide, nitrogen oxides and volatile organic compounds.

122. The tax on storage of household wastes has been increased but this was offset by a reduction in VAT on garbage collection and sorting.

123. The tax on nitrous oxide, which is a greenhouse gas, amounts to about three francs per carbon-equivalent tonne, well below international estimates of the carbon price under the Kyoto Protocol.

124. *Cf.* IEA (2000*a*). Nuclear energy is profitable in France for electricity generation for the basic market but nuclear power generation is in a situation of overcapacity relative to this market. In the semi-basic market, other types of production, notably gas-based, are more competitive. For technological reasons, nuclear energy cannot supply the high-tech market.

125. *Cf.* Charpin *et al.* (2000).

126. In 1999, GHG emissions were five per cent lower than those of 1990. This is explained primarily by the elimination of nitrous oxide output from industrial processes and the weak economic growth that characterised the early part of the decade.

127. The price that the French authorities have selected for their programme is in line with those produced by several international models, where only carbon emissions are considered. The GREEN model, for example, yielded a price of $90 per tonne of carbon (OECD, 1999*b*).

128. See "Programme national de lutte contre le changement climatique", *www.effet-de-serre.gouv.fr/pd/effetserre/effet_d e_serre.pdf.*

129. The measures already implemented are estimated to restrain emissions between 1990 and 2010 by 14 percentage points. In quantitative terms, the most important measures would be the regulation of industrial nitrous oxide emissions, Europe-wide voluntary agreements with automakers to reduce specific consumption by private vehicles and efforts to restrain urban vehicle traffic (see the following section).

130. According to the PNLCCC these should provide about 45 per cent of the emission reductions required to achieve the stabilisation objective.

131. Except in the case of natural gas.

132. After the Bonn negotiations, the carbon tax would probably have to be lower in order to match the international carbon price.

133. For a description of the proposed tax, see Bureau (2000).

134. *Cf.* Maire (2000). The industry section of the Ministry of Economy, Finance and Industry is still responsible both for promoting nuclear energy, as shareholder of the major firms in the sector (EDF and Cogema) and for economic regulation of the system, as well as for nuclear safety. In recent years, however, institutional reforms (in particular the move to make the nuclear protection and safety institute independent of the atomic energy commission and to place this new public entity under the budget of the Ministry of Environment) have tended to enhance the transparency of policy in the sector.

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135. Since the technology is relatively inflexible, the nuclear industry is precluded from the "peak" power market, which must be supplied by other types of electricity.

136. The tax payable would have been reduced by an amount equal to the difference between the quantities of energy products as evaluated in the reference situation for the voluntary agreement and the amounts actually used, multiplied by a rate of FF 520 per tonne of carbon for fossil fuels and FF 26 for electricity (with the tax rate at FF 13).

137. See for example Cournède and Gastaldo (2001).

138. Handing out emission credits free of charge to petroleum product distributors would be tantamount to granting them a rent, since they would pass on the going market value of the emission credits to consumers.

139. Tax credits received by electricity firms are determined on the basis of the quantity of energy produced and in fact exhaust the revenues produced by the tax, which amounts to paying firms that pollute less than the average (or the "reference") and taxing the others.

140. Scientific understanding of the health effects of pollution has made significant progress in recent years. While it was generally thought that exposure to pollution peaks could cause death or respiratory disease, recent research has also revealed long-term effects (prolonged exposure to pollution can provoke or aggravate chronic diseases and cancers). Epidemiological studies have quantified and assessed these effects (the most significant are presented in Commissariat General du Plan, 2001). There remain some uncertainties about the nature of the pollutants in question and the scope of the effects remains a subject of debate.

141. These are no longer a real problem except in the region of Le Havre and Rouen and in the Bouches du Rhône, particularly at Fos-Berre, the site of the only ozone alert in 1997.

142. Cf. *Le bilan de la qualité de l'air en France* 1991-2000, Ministry of Regional Planning and Environment, 18 May 2001. Measurement networks have been greatly extended and modernised since 1997. Since air quality varies sharply according to meteorological conditions, there is not as yet enough consistent evidence to form an objective judgement.

143. Concentrations of sulphur dioxide, primarily from stationary sources, are clearly declining, as are those of lead, thanks primarily to the use of unleaded gasoline which benefitted from tax incentives. On the other hand, air quality objectives with respect to nitrogen dioxide have not been met in areas with heavy road traffic, in particular the Paris region, where this compound has been the sole cause of pollution alerts. The average daily concentration of particulates on certain days of the year also exceeds the lower limit of the prohibited range. Finally, while the alert thresholds have never been exceeded for ozone (a secondary pollutant formed by the chemical reaction of nitrogen dioxide with volatile organic compounds) in metropolitan Paris, the "information thresholds" have been exceeded on certain days of the year and the health protection limit is regularly exceeded across the entire region. Overall, annual averages in central Paris have risen sharply since 1992.

144. This programme establishes vehicle emission standards for conventional pollutants (nitrogen dioxide, particles, carbon monoxide, VOCs, benzene) as well as fuel quality standards.

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145. European automakers have undertaken to reduce average CO_2 emissions for their new models to 140 g of CO_2 per kilometre by 2008; this will involve sharp reductions in fuel consumption.

146. The cost-benefit study for this measure concludes that the current value of the net benefits will be 1.1 billion euros [*cf. http://europa.eu.int/eur-lex/fr/com.dat*/2001/*fr*_501PC0141.*html*].

147. Nitrogen dioxide emissions apparently declined by 10 to 20 per cent between 1994 and 1998 in the Ile-de-France region and those of VOCs, another precursor of ozone, by about 25 per cent, while emissions of particles remained unchanged.

148. It is the only one in France to have been prepared by the regional prefect (*i.e.* by the central government) rather than by local officials.

149. The CERTU is also supposed to monitor implementation of the PDU against performance indicators.

150. *Cf.* May (1999).

151. The most important projects were the construction of the A86 and the Francilienne bypasses, for road transport, and the EOLE and METEOR radial connectors, for public transit. A start has also been made at the construction of ring lines linking various suburbs of the *petite couronne*.

152. A socioeconomic evaluation based on these criteria is mandatory for projects exceeding FF 500 million, and those subject to a public hearing.

153. Ministry of Economy and Finance, Directorate of Forecasting, "L*a tarification du stationnement à Paris*", September 1999. Rates would range from FF 4 in the outer *arrondissements* of the Northeast to FF 40 in the seventh *arrondissement*.

154. *Cf.* Dron and Cohen de Lara (2000), pp. 215-216.

155. The Prado-Carénage Tunnel in Marseille, the East-West Transversal (TEO) in Lyon, and the A14 near Paris.

156. Willoughby (2001).

157. In a city such as Paris and its immediate surroundings, there are a great number of possible entry points that would have to be electronically equipped. Outfitting every vehicle with a GPS system is also a costly option. Moreover, in either case a system of paper payment and surveillance would have to be included for occasional users without such equipment.

158. The CERTU has made a number of studies on the acceptability of urban road transport pricing. Some studies on introducing road-pricing schemes in the Paris region have been conducted in universities but not yet at the governments level.

159. Pollution from pesticides and phytosanitary chemicals is also a problem in some regions but probably to a lesser extent.

160. One-third of French watercourses suffer from eutrophication and groundwaters are also affected. In Brittany, 49 per cent of drinking water extraction points exceeded the European standard of 50 mg of nitrates per litre, at least once during the year. The European Court of Justice ruled in March 2001 that France had failed in its obligations to enforce the standards.

161. While this is not one of the agencies' explicit objectives, it does make possible a certain degree of cross-subsidisation between different localities.

162. Directive 2000/219/CE, establishing a framework for Community action in the field of water policy.

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163. The Agencies of Adour-Garonne, Artois-Picardie, Loire-Bretagne, Rhin-Meuse, Rhône-Méditerranée-Corse and Seine-Normandie.

164. The calculation of the charge paid by industries, whether or not connected to the community sewer system, can be based either on actually measured pollution or on pollution estimated using technical coefficients for specific emissions (expressed in terms of the pollution produced on a normal day of the month of maximum discharge). The gross charge is then diminished by a treatment bonus that takes into account depollution efforts, which are also measured or estimated on a flat basis.

165. The charge should offer an even greater incentive when the portion that is geared to an accurate measurement of polluting discharges (for which there are at present no estimates) is high.

166. *Cf.* Committee on Finance, the General Economy and Planning (1999). The mining and base chemicals industries, for example, enjoy special and highly favourable treatment.

167. The PMPOA and integration of the principle of large livestock undertakings into the pollution charge are a response to the 1991 European directive on nitrates. Since then, under the PMPOA programme for control of agricultural pollution, large livestock farms, covered by the 1976 law on classified facilities, are supposed to pay the pollution charge. In practice, the many discounts and exemptions have meant that pollution charges paid by the farming sector have been virtually nil, even though that sector constitues a growing source of pollution. In 1993, farmers' organisations won a moratorium on payment of charges: this was supposed to be temporary but has been extended to 2002. Treatment bonuses, which serve to reduce the charge, have been granted liberally, without any relationship to actual pollution reduction. Finally, high thresholds of incidence have been set by the water agencies, whereas other users must pay as of the first unit of pollution.

168. For determining the calculation base, a fixed daily amount of pollution per inhabitant is set by interministerial decree for a maximum term of five years.

169. The so-called "counter value" flat rate is calculated as follows: the first step is to estimate the pollution charge that should be paid by the community as a whole. This requires calculating the number of inhabitant-equivalents by multiplying the population (corrected for seasonal variations) by an urbanisation coefficient (since large cities pollute more than smaller cities, because of their activities). This is then multiplied by a zone coefficient that is supposed to represent the sensitivity of the immediate environment, and the "collection" coefficient, which may vary considerably from one municipality to the next. To these "corrected" inhabitant-equivalents, the pollution rates per inhabitant-equivalent are applied (set by interministerial decree for a maximum term of five years) and then the basic charge rates per pollutant established by the water agencies (the same as for industry). In the second step, this total estimated charge is divided by the expected number of cubic meters consumed annually in the municipality, which provides the "counter value" rate that appears on the water bill.

170. Pollution charges represent more than 80 per cent of total charges.

171. *Cf.* Commission on Finance, the General Economy and Planning (1999).

172. These zones are located in Brittany, Poitou-Charentes, the Loire Valley, the Paris basin, and in certain parts of the Rhone Valley, Aquitaine and Midi-Pyrénées.

173. The maximum limit was reduced from 240 to 210 kg of nitrogen per hectare in 2001 and will be reduced to 170 kg per hectare in 2002, in specified zones.

174. *Cf.* OECD (1997).

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175. The firm Suez Lyonnaise des eaux, which was condemned in 1997 to indemnify a consumer group for having provided water not respecting the nitrate norms, has in turn taken action against the State (*www.seaus.org/jurisprudence/tribunal/rennes/ta*02052001.*html*).

176. Eau et Rivières de Bretagne.

177. The runoff risk, for example, is higher in winter, when the land is bare. The risk also depends on precipitation levels.

178. R*apport d'évaluation sur la gestion et le bilan du programme de maîtrise des pollutions d'origine agricole*, Ministry of Economy, Finance and Industry, Ministry of Regional Planning and Environment, Ministry of Agriculture, No. 99-RI-TERM-023-R, July 1999.

179. C*f. ibid*. Financed to the amount of 35 per cent by the central and local governments, 30 per cent by the water agencies and 35 per cent by the operator, the cost of the programme has doubled from its initial estimate of FF 7 billion, reflecting a poor cost estimation and higher-than-expected participation rates. The incorporation of livestock operations by decreasing size favours the most important ones and by this fact alone generates strong pressure to extend the programme to smaller-scale operations.

180. C*f*. Finance, General Economy and Planning Committee (1999).

181. Gastaldo and Kosciusko-Morizet (1999).

182. This is particularly true of the differential with the price of water for industry.

183. Rainelli and Vermersch (1997). The subsidy rate is particularly high in the Midi-Pyrénées region.

184. Producer support was estimated at 49 per cent in 1999 for Europe as a whole.

185. See OECD (2000).

186. For an evaluation of the environmental impact of the CAP reform, see the special report of the Court of Audit on "Greening the CAP" (14/2000), O*fficial* J*ournal of the* E*uropean Communities* C353, 8 December 2000. Even if proportionally they show a marked decline from the previous decade, the farm subsidies provided in the European Union by way of price support, production support and input use support still represented 71 per cent of the assistance given to agricultural producers in 2000 (source: OECD).

187. As of 25 September 2001 about 12 000 territorial farming contracts had been subscribed and more than 16 000 validated by the departmental agricultural boards, on which representatives of the environmental protection associations and consumers' associations now sit. Financing provided under these contracts is also used to promote economic performance, employment and regional development.

188. A further objective of the draft bill is to establish a framework for public water services and provide greater transparency in the relationships between municipalities and their agencies, which enjoy a situation of natural monopoly, by setting rules for water rates and creating a new regulatory body that will conduct comparative studies of water pricing and quality practices and will advise local governments in negotiating contracts with the water enterprises.

189. The directive also requires the use of economic analysis in evaluating measures taken in order to achieve the fixed targets at the least cost. Compared to previous directives, this one establishes obligations in terms of water quality that relate to outcomes and not only to the means employed.

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190. Gastaldo and Kosciusko-Morizet (1999). These calculations are based on an estimated total final cost of denitrification of FF 3.2 billion, of which two-thirds is of agricultural origin (one-third from field crops and one-third from livestock).

191. In theory, it is likely that the costs of denitrification, strictly speaking, decrease with the quantity processed and the nitrate content. On the other hand, if what is sought is an estimate of external marginal cost, the calculation would also have to include other external costs (such as environmental costs), which probably exhibit threshold effects and are therefore likely to be increasing.

192. There is considerable uncertainty about the quantity of fertiliser that plants require and these relate primarily to climatic conditions. To avoid a decline in yields, a farmer will tend to use more fertiliser than may be necessary.

193. It will apply as of 2003 to operations with annual turnover exceeding FF 500 000, and in 2008 to those exceeding FF 300 000.

194. Moreover, as in the Netherlands case, supplementary allowances are granted for meadows and for intermediate nitrogen catch crops. These allowances are justified on the grounds that more nitrates will be absorbed by meadows than by croplands, and that this will not be reflected in the nutrient balances, which include only inputs and outputs.

195. Brouwer *et al.* (1999) indicates that the average nitrates surplus in France is estimated at 50 kg per hectare, with regional variations ranging from 6 kg/ha in the Limousin to 120 kg/ha in parts of Brittany.

196. The local governments or the contracted companies could in principle participate in these protocols but they have no incentive to do so, nor to pass possible preferential rates on to households.

197. For a discussion of competitiveness issues, see OECD (2001*b*) and O'Brien and Vourc'h (2001).

198. See the September 2001 White Paper of the Commision of the European Communities entitled: "European transport policy for 2010: time to decide" [*http://europa.eu.int/comm/energy transport/library/lb texte complet fr.pdf*].

199. This body for public policies evaluation, whose secretariat is provided by the Commissariat Général du Plan, can review all types of public policies. In the environmental area, it has already evaluated the policy for water quality, it is reviewing the policy for public service in waste management and is just starting to review the policy for combined rail-road transport.

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Annex I

Are the national accounts forgetting the new economy?

There is lively debate in the OECD concerning the capacity of the national accounts to give an accurate picture of the development of the new economy (OECD, 2001*d*; Landefeld and Fraumeni, 2001; Schreyer, 1998). The three topics traditionally referred to have recently been discussed by French statisticians (Lequiller, 2000):

1) The first question relates to the capacity of national accounting to **encompass the whole of output**. Some doubts arise because of the exceptional level of company start-ups and bankruptcies in these activities and the intangible nature of the output. It is feared that these characteristics may result in output being "lost" in the national accounts. That risk has been deemed slight. The accepted practices in measuring intellectual property and software production are on the whole well-tried and tested, even if there is room for improvement when it comes to better distinguishing between definitive sales and transfers of user rights. Concerning aggregated measures, the present system is based on a comprehensive list of firms, including unincorporated ones (the *Sirène* database), which in principle does not "lose" any activity, old or new. Within-year statistics, on the other hand, are based on a sample of firms taken from a population of 550 000 minimum-sized firms liable to the normal VAT system and filling in monthly statements. Here, the rotation of new economy firms can result in two distortions: on the one hand, by "losing sight" of extremely young firms not included in the reference population for a given month, which only contains firms already in operation from the previous month; and on the other hand, by recording the activity of small firms that have just been "absorbed" in the sample by merger or acquisition as internal growth of the members of the sample. The two distortions pull in opposite directions and their net effect is deemed insignificant. Also, they only concern within-year accounts and do not affect the final annual accounts.

2) The second question concerns the **volume-price breakdown** between the output and value added of new activities, according to the detailed measures of the telecommunications regulations authority (ART). The new economy being based on continuous technical progress in the information and communications technologies (ICT), there is a danger of overestimating price indices if quality improvements are not taken into account, hence underestimating volume growth. This problem has been known about for a long time and gave birth to hedonic indices,[1] a technique used mainly in the United States to revalue the output of the electronic and computer components industries. Recently it was extended to software packages but has not been applied very much to other activities. France is one of the most advanced countries in Europe when it comes to applying hedonic indices to its computer industry output. National accounts experts recognise, however, that where mobile telephony is concerned, deflating turnover by the fixed line price index (as INSEE does at present) seriously

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underestimates the growth of that activity. INSEE puts its annual growth at 7 to 8 per cent whereas the volume of minutes of mobile phone communication is growing twice as fast (14.5 per cent according to the more detailed measuring done by the Telecommunications Regulation Authority). INSEE has decided to carry out an hoc adjustment in this area, but admits that this formula creates an artificial difference between the price index and the private consumption deflator. Another aspect of the same problem has to do with the volume-price breakdown in ICT-user activities, whose product quality improvement is not fully taken into account. The problem is more significant than in ICTs themselves since ICT -user activities account for a larger proportion of GDP. The main example is in financial services, where ICT-induced improvements in quality are not measured, though this drawback comes on top of well-known traditional measurement difficulties in this sector.[2] Other ICT-user activities affected by this phenomenon include education and health.

3) The last question concerns the measurement of firms' **investment in software**. How this expenditure breaks down between intermediate consumption and capital formation remains unclear. Different countries apply different conventions, which tends to prevent their capital formation and GDP indicators from being perfectly comparable. The United States, for example, has decided to view a large proportion of the software industry's output as a component of capital formation, whereas the European method agreed in Eurostat aims to adopt to the accounting conventions used by the firms themselves, which tend to look upon the bulk of their software purchases as intermediate consumption. The stakes involved in these methodological choices are high because only one-third of software expenditure is included in investment and GDP in France, whereas in the United States the proportion is two-thirds. According to INSEE estimates, the difference in method could account for about 0.3 per cent of the roughly 2 per cent growth differential between the United States and France over the recent period.

To meet some of these challenges, France will be introducing some methodological innovations in its national accounts when it moves from base 1995 to base 2000. A special price index for mobile telephony will be introduced, a breakdown of investment in computer, software and telecommunications equipment will be added to the annual business surveys and a more detailed nomenclature of the markets for electronic publishing, computer games and on-line databases will be used.

Notes

1. Hedonic price indices are used to measure price trends after adjusting for the effected improvements in product quality.
2. Financial services output is measured on the basis of production costs, a technique which does not take account of improvements in productivity or quality.

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Annex II

A breakdown of growth factors

Faster growth in some OECD countries in the 1990s has prompted renewed interest in growth accounting exercises. This is because a sustained increase of output growth reflects a fundamental shift in the dynamics of production factors and not a cyclical phenomenon. Faster growth can result from an expansion in labour supply, more intensive capital accumulation, or a more rapid increase in the productivity of these factors. It is the latter aspect that is of course the most interesting as new information and communications technologies are though to make labour and capital more productive. Recent work has therefore focused on the diffusion of these technologies to the economy as a whole. Scarpetta *et al.* (2000), for example, discuss these various aspects in the context of the OECD countries, while Mairesse *et al.* (2000) look at the contribution of information and communications technologies to French growth.

This annex has a more narrow focus: it only considers the factors that may have caused the slowdown of French economic growth during the 1990s. Covering an entire decade eliminates cyclical fluctuations in demand, as it may be assumed that they cancel each other out over a period of that length. The traditional approach is to consider that production potential depends on the quantity of labour and capital utilised, as well as on the technical and organisational progress made by firms. A growth accounting exercise can help identify the roles played by each of these components. The results of that exercise are shown in Table A1, on the assumption of constant returns to scale.

This exercise suggests that the slowdown in output growth in the 1990s was caused by weaker labour productivity growth. The average annual rate of growth in hourly labour productivity in the business sector of the economy fell from 2.2 per cent between 1980 and 1989 to 1.1 per cent between 1990 and 1999. That result is consistent with the findings of other authors (Duchène and Jacquot, 1999). The break in the trend of labour productivity occurred at the start of the decade, between 1990 and 1993. Lerais (2001) estimates that average annual growth of per capita productivity in the 1990s was only 1.2 per cent in the non-agricultural business sector. Gubian and Ponthieux (2000) calculate that the average growth in output per worker was only 0.5 per cent over the period 1994-98. The slowdown in labour productivity in France contrasts with the acceleration observed in other OECD countries, especially the United States, which many observers attribute to capital deepening and the diffusion of new information and communications technologies (OECD, 2001*d*). In the case of France, existing studies point to technological innovation having a positive impact in the sectors producing high-technologies but do not show any diffusion to sectors using the technologies over the period 1997-98 (Crépon and Heckel, 2001).

The slowdown in hourly labour productivity is usually explained by labour cost moderation, which has encouraged firms to substitute labour for capital. Wage claims did in fact ease against a background of rising unemployment. In addition, as of 1994 the authorities

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Table A1. **Average annual growth of activity and the factors of production**

	1970-79	1980-89	1990-99
Total GDP	**3.5**	**2.3**	**1.6**
Business sector GDP	3.7	2.4	1.7
Contributions by:			
1. Capital stock	1.5	1.1	0.9
2. Labour	–0.4	–0.6	0.0
2.*a* *Population*	0.4	0.3	0.3
2.*b* *Participation rate*	0.1	0.0	0.1
2.*c* NAIRU	–0.2	–0.3	0.0
2.*d* *Hours worked*	–0.7	–0.5	–0.3
2.*e* *Other*	0.0	–0.1	–0.1
3. TFP, of which contribution by	2.6	1.9	0.8
3.*a* *Labour productivity*	3.0	2.2	1.1
3.*b* *Capital productivity*	–0.4	–0.3	–0.3

Source: OECD.

implemented programmes to reduce social contributions for low wage workers and this encouraged firms to substitute unskilled labour for skilled labour. Recent econometric studies on individual data do show that targeted reductions in employers' social insurance contributions had a large impact on unskilled labour (INSEE, 2001). Lastly, the pattern of activity has changed in favour of less productive sectors such as services (Pisani-Ferry, 2000).

Lower labour cost for unskilled workers is beneficial to employment growth. In so doing, however, it lowers the average skill level of the labour force and so has a negative impact on output per worker. The risk is that the downturn in per capita productivity may be not entirely offset by an increase in the labour force. Although the statistical estimates are extremely flimsy, it is a scenario that cannot be ruled out in the case of France. It is not impossible that the decline in labour productivity growth may have had a negative impact of potential growth. As Carnot and Quinet (2000) observe, however, the downturn in per capita labour productivity is a passing phenomenon, linked to the transition from a growth pattern based on "capital deepening" to a growth pattern based on "labour deepening". Once the transition has been made, hourly labour productivity should move back to its earlier growth path.

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Annex III

Follow-up to the recommendations on policies concerning the health sector

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OECD Recommendations[1]	Measures taken[2]	Comments by the OECD
Hospitals		
Reform the method of payment for hospital services (payment by pathology, with the necessary adjustments).	A number of working parties, set up by the Ministry for Solidarity and bringing together health professionals, are making progress in defining the conditions for preparing, simulating and then testing payment by pathology. The private-practice activities of hospitals-based doctors was the subject of a decree in 2001 that sought to provide a better legal framework for such activities (2001-367 of 25 April 2001).	A good first step. To be continued.
Strengthening of the role of the *Agences régionales d'hospitalisation* (ARH): made independent, purchase of care, tendering procedure for the purchase of standard forms of care.	No measures taken (the time needed to introduce new regulatory arrangements and the re-sizing of administrative services are being discussed).	Prepare the introduction of new regulatory arrangements for the reform of the ARHs.
The status of hospitals changed to *Établissements publics autonomes*, with increased management independence. Introduction of cost accounting and depreciation of fixed assets.	Cost accounting is gradually being introduced in all hospitals.	Change the status of hospitals.
Increased mobility of hospital staff; strengthening of performance appraisal procedures for managers and hospital doctors. Reorganisation of working hours with the introduction of the 35 hour week. Abolition of private beds in public hospitals.	The reorganisation of work time with the introduction of the 35-hour week is being negotiated with the unions.	Continue with the implementation of the other recommendations.
Increased role of governing boards, plus increased transparency with regard to financial results and medical performance.	Under consideration.	Pursue implementation.
Private medicine and private clinics		
Joint discussions with complementary insurance schemes on ways of covering health risks, the balance between preventive and curative medicine, and defining a basket of reimbursable types of care.	The discussions with complementary bodies are being conducted primarily in the framework of the implementation of and developments in universal sickness cover (CMU – *couverture maladie universelle*).	Publicly announced what the main lines of these joint discussions are and publicise the results.

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OECD Recommendations[1]	Measures taken[2]	Comments by the OECD
Incentives designed to encourage a system of "subscribing" to general practitioners so as ensure greater continuity of care.	No action.	The "wise men's committee" on future regulations governing health services ("Grenelle de la santé"), to report in July 2001, can make new proposals in this area.
Filtering by general practitioners of access to certain categories of specialist.	Wide-ranging (including inter-ministerial) discussions are in progress on the medical population, including the way jobs and relations between health professionals are evolving.	This wide-ranging debate can include the possibility of "gentle" filtering by the referring doctor.
Changes in the way doctors are paid, the object being to discourage excessive amounts of separate treatment.	No action.	Wait for any proposals on this point by the "wise men".
Introduction of a new nomenclature of medical treatment, which more accurately reflects technical progress.	The preparation of a new "common classification of categories of medical treatment" for the ambulatory and hospital sectors has just been completed. The economic valuation of this nomenclature is under way, on the basis in particular of a review of doctors' costs.	A step in the right direction. Publish the effective economic utilisation of this nomenclature.
Creation by the CNAMTS of a telephone call centre to guide patients as to whether or not to consult a doctor.	Envisaged.	Pursue implementation.
Periodic re-accreditation of doctors.	A system of individual and collective assessment of doctors in private practice by their peers (on a voluntary basis) is in the process of being implemented (on the basis of the decree of 28 December 1999).	A compulsory framework has to be adopted.
Follow-up to the impact of CMU on patient behaviour. Presentation to Parliament of a periodic assessment of CMU.	Many studies on the impact of CMU have been undertaken by the DREES, the CREDES and the CNAMTS, with a particular focus on health care consumption. Some initial findings have already been published, while a global assessment will be presented in autumn 2001.	Wait for the results of the global assessments.

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OECD Recommendations[1]	Measures taken[2]	Comments by the OECD
Medication		
Drugs whose therapeutic effectiveness is not proven to cease being reimbursed.	Launched in 1999, the review of medical services rendered (SMR) by all reimbursed drugs has just been completed. For the 800 drugs with insufficient SMR, there are no immediate or scheduled plans to cease reimbursement; instead it is planned to reduce their price (by some 20 per cent) and possibly lower the rate of reimbursement (alignment on the 35 per cent rate).	Pursue implementation of the recommendation.
Gradual abolition of government pricing of medicines.	The possibility of experimenting with reimbursing certain classes of drug at the reference price was again considered by the government in spring, and was again rejected.	New initiatives are needed.
Additional incentives to use generic drugs.	Following the "medication plan" announced by the Health Ministry in June 2001, ways of promoting generic drugs are being negotiated with pharmacists' representatives.	Introduce new incentives.
Develop the market for drugs that can be purchased without a prescription.	No action.	Such a market could encourage fewer medical consultations. A step in this direction should be considered.
Insurance schemes		
Develop unified management of treatment forms (*feuilles de maladie*) and payments (CNAM: state health insurance fund, and complementary insurance).	Central funds to some extent streamlining the procedure for settling expenditure in clinics (co-ordination between sickness insurance schemes) were set up in February 2001. Difficulties with co-ordination persist, however, between basic schemes and complementary bodies as regards direct payment by insurers for ambulatory care, particularly in the context of CMU.	Pursue implementation.

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OECD Recommendations[1]	Measures taken[2]	Comments by the OECD
Stimulate competition between complementary insurance schemes (transparency of rates).	The Community directives on complementary insurance bodies have just been transposed. This will result, in particular, in the activities of these bodies being more strictly separated and in increased financial transparency (stricter separation of accounts relating to sickness risk management, in particular).	A step forward. The encouragement of competition will, however, depend on the progress made in reducing to a minimum the excess consumption prompted by the existence of complementary schemes which eliminate the disincentive effect provoked by co-payments.
CNAM business plan aimed at cutting management costs.	This is gradually being achieved (agreement on objectives and management between the government and the CNAMTS and its annual additional clauses).	Pursue implementation with renewed vigour.
Develop computer tools in doctors' offices (help with prescriptions, patient follow-up, auditing).	Doctors are making real progress in the use of computers; there was a big increase in the remote transmission of treatment forms and in the use of *Sésam-Vitale* in 2001. Developing help with prescriptions is built into the government's "medication plan" announced in June 2001 (notably via the *Fonds de promotion de l'information médicale et médico-economique*, created by the social security financing Act (LFSS) for 2001. The effective implementation of the "good practice agreements" between the CNAMTS and doctors, provided for under the LFSS 2000, is still pending.	Satisfactory progress.
Develop public health policy: anti-smoking measures, screening for certain cancers, frequent and comprehensive medical check-ups.	These priorities are being pursued assiduously within the context of French public health policy. Other avenues that have been added include dental care (improved cover) and the safety of health care processes.	Satisfactory progress.

1. Recommendations made in OECD (2000).
2. Source: Ministère de l'Économie, des Finances et de l'Industrie.

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Annex IV

The reduction in working time: a post-mortem on the 1999 Interlink simulations

This annex presents several Interlink model simulations relating to the 35-hour workweek. Simulations have already been run with this model for the 1999 *Economic Survey*. They were based on a range of assumptions deemed at the time to be plausible, given what was known about government measures to reduce working time – *i.e.* mainly the Aubry Law of 13 June 1998 (hereinafter called Aubry Law I). In the meantime, a second law has been adopted, adding to and clarifying the legislative mechanisms (hereinafter called Aubry Law II) and there is now a great deal of literature available, which monitors the concrete results of the reduction in working time (see, in particular, Rouilleaut, 2001). The present annex does not go back over the legislative details, which were described in detail in the July 2000 *Survey*, nor over the impacts of the 35-hour workweek, which are described in this *Survey*. It summarises the assumptions put forward in the 1999 analysis, explaining in what way they need to be changed and, on that basis, provides the results of new simulations and finally compares them with the initial ones. The Interlink model hardly changed between 1999 and 2001 (apart from the move to ESA95), so the differences in the macroeconomic results likely derive mainly from the new set of assumptions.

On the whole, the 1999 results are confirmed: a reduction in working time imposed on firms can indeed result in a short-term increase in the numbers employed; however, this result incorporates a cost for firms and the public purse and a potentially negative impact on the growth of GDP.

The 1999 simulations

The 1999 *Economic Survey of France* provided an opportunity for an initial review of the government's measures to reduce working time. To this end, a series of simulations with the Interlink model were run on the basis of assumptions deemed plausible at the time. Basically, the assumptions were the following:

– *Rate of introduction*: the main assumption was that the process would be gradual, but a number of variants were considered. In the most rapid scenario, 60 per cent of employees in firms with more than 20 employees and 25 per cent of employees in small firms would be on a 35-hour week by the end of 2001; once introduced, that proportion was expected to stabilise at 80 and 75 per cent, respectively, by the end of 2003. In the slowest scenario, the corresponding figures were 40 and 12 per cent by the end of 2001, and 50 and 33 per cent by the end of 2003.

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- *Aid to firms*: the simulations were based on the assumption of long-term assistance of FF 5 000 per year and per worker for firms that reduced their working time. That amount had been calculated, within the framework of Aubry Law I, as being compatible with public finance equilibrium, bearing in mind feedback effects.
- *Wage restraint*: the assumption made in 1999 was for considerable wage moderation. It was assumed in the different scenarios that negotiations would result in an increase in hourly wage rates, but a fall in monthly wages in nominal terms. Various degrees of wage moderation were assumed, depending on what employees were prepared to accept, but in every case monthly wages diminished.
- *Job creation*: the "offensive" provision of Aubry Law I made the granting of government assistance conditional on the creation of a minimum number of jobs. In order to qualify for State aid, firms had to undertake to reduce working time by at least 10 per cent and to hire the equivalent of at least 6 per cent of the number of workers concerned by the reduction. The range adopted for the simulations was that hirings would be between 5 and 7½ per cent of the workforce.

With the set of assumptions adopted in 1999, it was possible to cover the full range of possible macroeconomic implications, and the results of the simulations thus produced a varied set of possibilities. The general conclusion to emerge from the simulations over a five-year time horizon was that the impact on employment would be neutral or positive, while the impact on growth would be negative. With regard to employment, the ultimate impact would either be neutral or markedly positive (up to 380 000 additional jobs), depending on the assumptions taken. Assuming rapid implementation and strict application of the coercive nature of hirings provided for in the "offensive" part of Aubry Law I, the impact on employment would be in the upper part of the range. Compared with the simulations carried out by other bodies using other models, that sort of impact on employment was rather moderate, the simulations conducted by the official bodies and by economic research institutes reckoned on an impact of between 450 000 and 700 000 jobs.

The new simulations

Two years on from that analysis, government mechanisms for introducing the 35-hour week are more clearly established. The assumptions adopted in 1999 can now be revised and updated in the light of the first available data.

- *Rate of introduction*: according to the official statistics based on the ACEMO survey, the move to the 35-hour week in firms with more than 20 employees has been even quicker than in the simulation's "rapid" scenario. The figure of 60 per cent of the employees of large firms being on a 35-hour week, which was supposed to be reached by the end of 2001 in the rapid scenario, would in fact seem to have been reached as of the first quarter of 2001. On the other hand, small firms have introduced the 35-hour week more slowly than expected, only 8 per cent of the employees concerned having reduced their working time, which more or less matches the "slow" scenario drawn up in 1999.
- *Actual reduction in working time*: although large firms have brought the measures in more rapidly than expected, the fall in actual working time is not so great. This is because it was not foreseen in 1999 that Aubry Law II would allow new methods of calculating working time. Thanks to this legislation, firms agreed, within the framework of the negotiations, to methods of calculation excluding certain periods of time at work when the employee is not actually working (breaks and time spent changing clothes). This meant that some firms were able to arrive at 35 hours with a reduction of only 2 hours, *i.e.* half of what was reckoned to be necessary (Heyer and Timbeau, 2000). All told, the reduction in working time is less than forecast in 1999-2000 (2 per cent on average for

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employees in the market sector, according to the DARES composite indicator, as against a reduction of 3.4 per cent assumed in the Interlink simulations in the rapid introduction scenario). Looking forward to 2003, the reduction is also expected to be less marked (3.6 per cent, instead of the 5.2 per cent assumed initially).

– *Aid to firms*: the aid package finally adopted is more generous than expected. Long-term aid under Aubry Law II is in the region of FF 7 000 per year and per worker (Heyer and Timbeau, 2001), compared with the FF 5 000 assumption used in the 1999 Interlink simulation.

– *Wage restraint*: the assumption formulated in 1999 postulated an increase in hourly wages, but a reduction in monthly wages per capita. The agreements negotiated within firms appear to have dismissed that assumption. In most cases, nominal monthly wages seem to have been maintained, even if some agreements provide for wages to be frozen for between a year and eighteen months, or increased by less than inflation. In this exercise, two types of wage behaviour have been simulated: on one hand, wage equations have been left free to react to the shocks imposed on the model; on the other, wages have been frozen in real terms so as to simulate a form of lasting restraint.

– *Impact on potential growth*: the 1999 simulation postulated that the 35-hour workweek would have no impact on potential growth. However, the neutrality assumption is debatable in that, by cutting the number of hours worked, the reduction in working time can have a negative impact on output potential. Two factors can offset the effect of the fall in working time on potential growth: an increase in underlying labour productivity and higher participation rates. Both are uncertain. Three scenarios have been simulated in order to reflect these uncertainties: no fall in output potential by comparison with the reference scenario; fall in potential output limited to 1.3 per cent after 5 years; 2.7 per cent fall in potential output after 5 years. The NAIRU is assumed to remain unchanged (see Chapter III).

The model is simulated on the basis of a "shock" affecting working time (maximum differential of 3.9 per cent over time for the market sector, by comparison with the reference scenario) and revenue from social insurance contributions (maximum differential of 6½ per cent). Six scenarios are constructed on this basis, postulating three different impacts for potential growth and two for wages. The model is simulated up until 2006. The reference scenario is the medium-term projection in the OECD's *Economic Outlook No. 69*. Figure A1 shows the new and old results of the simulations in terms of differences compared to the reference scenario.

To understand the results of the simulations, it is important to distinguish between the long-term and the short-term effects. Long-term equilibrium following the 35-hour "shock" reflects the assumptions made concerning the economy's output potential: *i*) as in most other studies, the impact of the 35-hour week on the NAIRU is assumed to be neutral, so the unemployment rate deviates only temporarily from the level in the reference scenario; *ii*) the impact on GDP over time depends on the capacity of the economy to offset the reduction in working time with an increase in labour productivity or an upward shift in participation rates; and *iii*) the long-term impact on employment depends on the upward shift in participation rates. In the long term, the impact on GDP, on unemployment and on employment is either neutral or negative.

In the short term, the "shock" can be interpreted as a change in fiscal policy stance, the reduction in working time and government aid in the form of lower social insurance contributions resulting in a rise in employment which boosts household income and increases consumption. However, inflation gathers pace and the public finance balance worsens.

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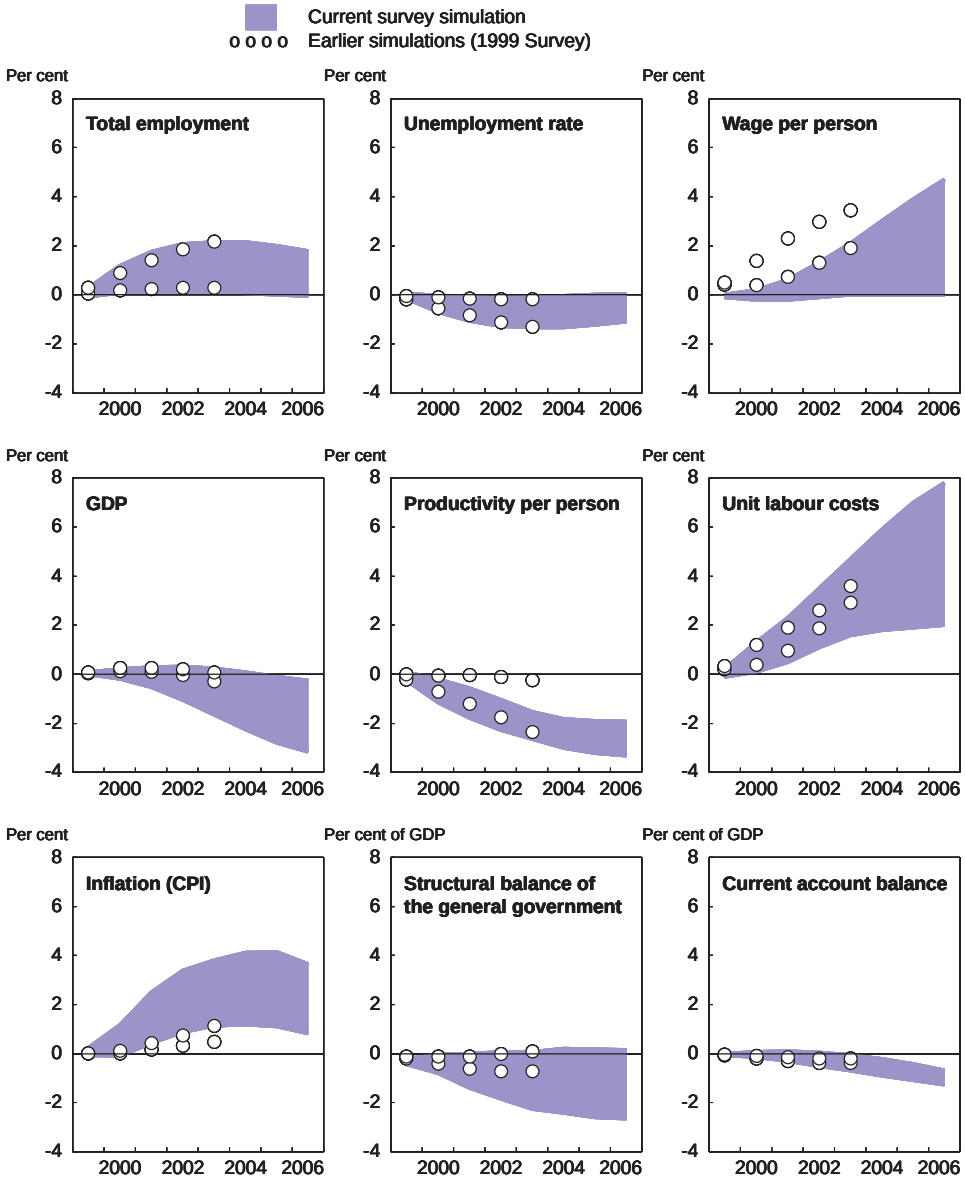
Figure A1. **Simulation of the effects of the reduction in the statutory working week**

Deviation from a baseline scenario of a 39 hour statutory work week



Source: OECD.

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The transition from the short term to long-term equilibrium involves a number of channels: *i*) wages costs go up, *ii*) corporate margins are reduced, *iii*) investment declines, *iv*) external competitiveness worsens, and *v*) foreign trade deteriorates. The precise breakdown of this move to long-term equilibrium depends on the assumptions made about the behaviour of wages (free play of the Interlink equations or fixed real wages).

The new simulations only partially bear out those carried out in 1999. As with the old simulations, the impact on employment appears to be neutral or positive: in the best scenario, the numbers employed increase by 2 per cent (some 350 000 jobs) in 2003. In contrast with the old simulations, however, it transpires here that the long-term impact is either neutral or negative. The gains in terms of jobs and unemployment can disappear in the long term and GDP can be lower.

The results of this exercise need to be viewed with caution. Like all macroeconomic models, the Interlink model is not well suited to simulating complex supply-side shocks and to long-term analyses. If the requisite caution is exercised, however, three useful lessons can be learnt from these simulations, the application of which would prevent the introduction of the 35-hour week turning into a major handicap for the French economy in the medium term. F*irst*, economic growth potential is being jeopardized by the increased rigidities induced by the scheme. Flexibility is therefore required when it comes to the actual application thereof: flexibility in applying the main parameters – for example, the overtime quota which may have to be raised if hiring difficulties persist and flexibility in applying it to SMEs which are not large enough and do not have necessary administrative resources. S*econd*, wage restraint is under threat and with it the competitive disinflation measures in force for the past ten years. In this connection, it will be important to make sure that decisions concerning the SMIC do not contribute to a further inflationary spiral. T*hird*, the feedback effects on public finance that can neutralise the fiscal cost of government aid are very uncertain. Measures need to be taken to prevent FOREC expenditure handicapping France's budgetary situation over the long term. Reductions in social insurance contributions are necessary but they need to be offset by cuts in public spending.

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Annex V

Calendar of main economic events

2000

January

Universal health insurance comes into force, giving the 10 per cent of the most deprived members of the population non-contributory basic and supplementary health insurance.

The *Conseil constitutionnel* approves the second 35-hour week law except as regards the payment of overtime. The agreements already signed by the social parters will continue to apply.

As part of the Stability and Growth Pact, the Prime Minister forwards to the European Commission the triennial finance plan, under which taxation is to be cut by FF 120 billion between now and end-2003.

February

The bill opening up the electricity market to competition is voted: 30 per cent immediately and 34 per cent in 2003.

Entry into force of the second Aubry law setting out the provisions for the reduction of the working week from 39 to 35 hours in all companies employing more than 20 people.

The Minister for the Economy, Finance and Industry and the Secretary of State for the Budget announce a tax receipts surplus of FF 30.7 billion in 1999, bringing the budget deficit down to FF 206 billion. Announcement of an amending budget to be submitted to parliament in spring, setting the deficit target for the end of 2000 and a introducing a reform of the property tax.

March

The Prime Minister announces the main measures included in the forthcoming revised Budget Law: tax cuts amounting to FF 40 billion (lower property tax, income tax cut, and reduced normal VAT rate), and new public spending of FF 10 billion.

The Prime Minister presents broad guidelines to reform the pension system: financing of pensions through the reserve fund, with funding reaching FF 1 000 billion in 2020; "pension treaty" for pensions in the public sector, including lengthening to 40 years of the contribution period.

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April

The standard VAT rate is lowered from 20.6 per cent to 19.6 per cent.

The Prime Minister announces measures to support enterprise creation: all state taxes on enterprise creation are eliminated; new credits will be provided to enterprises working in the new technology and proximity areas; social security contribution cuts will be pursued for young enterprise creators and, in due time, it will be possible to declare an enterprise on the Internet.

May

The government sent a framework letter for next year's budget: central government spending will be increased by 0.3 per cent in real terms, taking into account an inflation rate expected to be 0.9 per cent; the central government deficit will be reduced to FF 195 billion.

The Minister of Economy, Finance, and Industry announces that the proceeds from the sale of licenses for third generation mobile telephones will be mainly earmarked for the pension reserve fund.

Negotiations between social partners start with a view to establishing a new unemployment insurance convention.

June

The Minister for the Economy, Finance and Industry announces that four UMTS third-generation mobile telephone licences will be put up for sale between now and June 2001, each at the price of FF 32.5 billion, and that the proceeds will be allocated to the pensions reserve fund.

July

The SMIC is raised by 3.2 per cent for employees still on 39 hours and by 1.45 per cent for those who have switched to a 35-hour week. Unemployment benefit increased by 2 per cent.

The interest rate on the *Livret* A, *Livret bleu*, *Codevi* and *Compte d'épargne logement* (CEL) raised from 2.25 per cent à 3 per cent; the rate on the *Livret d'épargne populaire* increased from 4 per cent to 4.25 per cent, and that on the *Plan d'épargne logement* (PEL) raised from 3.6 per cent to 4.5 per cent.

The Minister for the Economy, Finance and Industry announces supplementary receipts of FF 30 billion, in addition to the FF 50 billion already announced in February.

August

The Minister for the Economy, Finance and Industry announces further tax cuts of FF 120 billion over the period 2001-2003: reduction in income tax (FF 45 billion), reduction of the *Contribution sociale généralisée* (CSG) on low wages, abolition of road tax and the surtax on companies, the latter being offset by a widening of the tax base.

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September

The draft Finance Act for 2001 (growth assumption: 3.3 per cent) is submitted to the Council of Ministers. The main measures are:

- On the revenue side: modification of the TIPP (*Taxe intérieure de consommation sur les produits pétroliers*); tax reductions of FF 57 billion.
- On the expenditure side: stabilisation of central government expenditure (–0.3 per cent); increases in four main items: environment (+8.2 per cent), policing (+4.9 per cent), justice (+3 per cent) and education (+2.7 per cent); public deficits to be cut to 1 per cent of GDP and income tax and social insurance contributions to 44.7 per cent of GDP.

October

Presentation of the Social Security financing bill setting a new national target for health insurance spending of FF 693 billion (up by 3.5 per cent); pensions increased by 2.2 per cent; housing allowances to be awarded using a single scale; FF 85 billion allocated to finance the 35-hour week (reductions in charges).

Agreement between the government and employers on the reform of the UNEDIC.

November

Gas tariff raised by 13 per cent.

The government decides to suspend the use of animal meal in pig and poultry feed.

December

France Télécom announces an overall reduction of 5.8 per cent in the price of local calls.

The three basic social benefits raised by 2.2 per cent on 1 January 2001; an exceptional end-of-year payment (of between FF 705 and FF 2 500) will be made to the most needy.

Adoption by Parliament of the Budget Law: a budget deficit of FF 210 billion (on the basis of growth of 3.2 per cent or more) – reducing the deficit to 1.4 per cent du GDP, total tax cuts representing nearly FF 100 billion.

The *Conseil constitutionnel* vetos several provisions of the Social Security Financing Act for 2001, in particular the degressive reduction in the CSG and the abolition of the Loi Thomas which created retirement savings plans, and rejects the the extension of the ecotax and the transfer of tobacco duty (two measures intended to finance the 35-hour week).

2001

January

The government announces the broad lines of the tax credit (*primes pour l'emploi*) for low wage-earners, which will concern 10 million people.

Orange, the subsidiary of France Télécom, is listed on the stock exchange.

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February

The Prime Minister revises downwards the growth forecasts for 2001; the slowdown of the US economy could knock 0.5 point off GDP.

France Télécom launches a single Internet connection charge of 14 centimes per minute after the first minute, which is billed at 70 centimes.

March

The telecommunications regulatory authority (ART) obliges France to cut its wholesale tariffs for the resale of high-speed Internet access: FF 210 per month per access and FF 1 330 per megabit/second per month for use.

April

The Prime Minister announces several measures :

– The *prime pour l'emploi* to be doubled in 2002 and the 2001 *prime* to be paid on 15 September.

– The period during which a wage can be combined with benefit doubled to six months from September 2001.

– Redundancy plans to be made more expensive for firms that lay off workers.

– Publication of a decree at the end of April doubling the appropriations for the transition to the 35-hour week, which is to become effective at the start of 2002 in small and medium-sized entreprises.

– Spending increase target of 0.5 per cent in real terms for the preparation of the 2002 budget (inflation: 1.7 per cent).

May

The management of AOM-Air Liberté announces 1 338 redundancies and the reduction of its fleet to 27 aircraft.

France Télécom and SFR obtain the first UMTS mobile telephone licences.

June

The government presents a plan to make youth jobs (*emplois jeunes*) permanent: FF 40 billion over five years, of which FF 28 billion for education and the police, FF 8 billion for associations and FF 4 billion for local authorities.

The government presents a "medecines plan" to the Social Security Accounting Commission designed to achieve savings of between "FF 4 and 5 billion" on spending on medecines.

The Prime Minister announces a raft of measures to help families: two weeks' paid paternity leave (100 per cent of gross salary), an extra billion francs for the F*onds d'investissement pour la petite enfance*, a change in the method of means testing used by the C*aisse nationale des allocations familiales* for 800 000 young people, reform of the special education allowance for the parents of handicapped children.

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July

The new unemployment insurance agreement and the plan for the return to employment (PARE) takes effect.

The minimum wage (SMIC) is raised by 4.05 per cent for those working 39 hours and by 2.85 per cent for those working 35 hours. Unemployment insurance benefits raised by 2.4 per cent. The floor used in calculating social welfare benefits is raised by 1.6 per cent.

The government presents the expenditure side of the 2002 draft budget bill. State spending to grow by 2 per cent: increases for four large chapters of: 5.5 per cent for justice; 4.3 per cent for interior; 7 per cent for environment and 4 per cent for education.

August

The Prime Minister announces an easing of the manner in which the 35-hour work week legislation will be applied to small and medium enterprises with fewer than 20 employees, allowing them, by decree, to more overtime hours than larger firms.

September

The Minister of Employment and Solidarity announces that the number of net new jobs in the hospital-based civil service would rise from 40 000 to 45 000 as of 1 January 2002.

October

The government revises its growth forecast for 2002 from 2.5 to 2.25 per cent.

The Ministry of Employment and Solidarity announces the creation of 30 000 solidarity employment-contracts (CES) and the creation of insertion internships, costing a total of 180 million euros.

The government presents a FF 35 billion, five-year programme to improve the quality of life in the suburbs.

The 2002 budget bill is presented. It includes the distribution of a new employment premium (PPE) in January, measures to support investment of enterprises in various sectors (air transport, tourism and insurance) and a very large reduction in the price of UMTS licences.

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OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 2001 13 1 P) ISBN 92-64-19685-4 – No. 52243 2001
ISSN 0376-6438